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Patent Public Search | Text View

United States Patent Application Publication

20250275768

Kind Code

A1

Publication Date

September 04, 2025

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END TOOL OF SURGICAL INSTRUMENT, AND SURGICAL INSTRUMENT COMPRISING SAME

Abstract

The present disclosure relates to an end tool of a surgical instrument and a surgical instrument including the same, and more particularly, to an end tool of a surgical instrument that may be mounted on a robotic arm or operable manually to be used in laparoscopic surgery or other various surgeries, wherein the end tool is rotatable in two or more directions and is moved in a way that intuitively matches a motion of a manipulation part, and a surgical instrument including the same.

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Family ID: 83723025

Appl. No.: 19/210381

Filed: May 16, 2025

Foreign Application Priority Data

KR 10-2021-0053353

Apr. 23, 2021

KR 10-2021-0055322

Apr. 28, 2021

Related U.S. Application Data

parent US continuation 18288002 20231023 parent-grant-document US 12349903 WO
continuation PCT/KR2022/005795 20220422 child US 19210381

Publication Classification

Int. Cl.: A61B17/072 (20060101); A61B17/00 (20060101); A61B34/30 (20160101)

U.S. Cl.:

CPC A61B17/07207 (20130101); A61B2017/00367 (20130101); A61B34/30 (20160201)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation application of the U.S. Utility patent application Ser. No. 18/288,002 filed on Oct. 23, 2023, which is a U.S. national-stage application of PCT/KR2022/005795 filed on Apr. 22, 2022, and claims priority to Korean patent application Nos. 10-2021-0053353 filed on Apr. 23, 2021, and 10-2021-0055322 filed on Apr. 28, 2021, the entire disclosures of which are incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to an end tool of a surgical instrument and a surgical instrument including the same, and more particularly, to an end tool of a surgical instrument that may be mounted on a robotic arm or operable manually to be used in laparoscopic surgery or other various surgeries, wherein the end tool is rotatable in two or more directions and is moved in a way that intuitively matches a motion of a manipulation part, and a surgical instrument including the same.

BACKGROUND ART

[0003] In recent years, laparoscopic surgery has been actively utilized to reduce postoperative recovery time and complications through small incisions. The laparoscopic surgery is a surgical method in which a plurality of small holes are drilled in the abdomen of a patient and the inside of the abdominal cavity is observed through these holes, and is widely used in general surgery and the like.

[0004] In performing the laparoscopic surgery, a suturing instrument inserted into the body is used to suture a surgical site in the abdominal cavity, and a surgical stapler for suturing the surgical site using medical staples is used as the suturing instrument.

[0005] In general, a surgical stapler is a medical instrument that is often used for cutting and anastomosis of an organ in abdominal and thoracic surgery. The surgical stapler includes an open stapler used in thoracotomy and laparotomy and an endo stapler used in thoracoscopic surgery and celioscopic surgery.

[0006] The surgical stapler has advantages of not only shortening operation time since cutting of a surgical site and anastomosis of an organ are simultaneously performed, but also accurately stapling the surgical site. In addition, the surgical stapler has advantages of a faster recovery and a smaller scar than those when tissue is cut and stapled using a surgical stapling thread, and thus has been widely used in modern surgical operations. In particular, the surgical stapler has been widely used in cancer surgery to cut cancer tissue and suture a cut site.

[0007] The aforementioned background technology is technical information possessed by the inventor for derivation of the present disclosure or acquired by the inventor during the derivation of the present disclosure, and is not necessarily prior art disclosed to the public before the application of the present disclosure.

DESCRIPTION OF EMBODIMENTS

Technical Problem

[0008] The present disclosure is directed to providing a surgical instrument, which may be mounted on a robotic arm or operable manually to be used in laparoscopic surgery or other various surgeries

and includes an end tool rotatable in two or more directions and moved in a way that intuitively matches a motion of a manipulation part, and a surgical instrument including the same.

Technical Solution to Problem

[0009] According to an embodiment of the present disclosure, there is provided a surgical instrument including an end tool which includes a first jaw, a second jaw formed to face the first jaw, a first jaw pulley coupled to the first jaw and formed to be rotatable around a first shaft, a second jaw pulley coupled to the second jaw, formed to be rotatable around a shaft substantially the same as or parallel to the first shaft, and formed to be spaced apart from the first jaw pulley by a certain extent, and a staple drive assembly including one or more staple pulleys at least partially formed between the first jaw pulley and the second jaw pulley, and a cartridge which includes a reciprocating assembly that is connected to the staple drive assembly, and linearly moved when the staple pulley is rotationally moved, and an operation member that is brought into contact with the reciprocating assembly, and is moved in one direction by the reciprocating assembly when the reciprocating assembly is moved in the one direction.

[0010] Other aspects, features, and advantages other than those described above will become apparent from the following drawings, claims, and detailed description of the disclosure.

Advantageous Effects of Disclosure

[0011] According to the present disclosure, a manipulating direction of a manipulation part by a surgical operator and an operating direction of an end tool are intuitively the same direction, so that convenience of the surgical operator can be improved, and accuracy, reliability, and speed of surgery can be improved.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0012] FIG. 1 shows (a) a conceptual diagram of a pitch motion of a conventional surgical instrument, (b) a conceptual diagram of a yaw motion thereof, (c) a conceptual diagram of a pitch motion of another conventional surgical instrument, (d) a conceptual diagram of a yaw motion thereof, (e) a conceptual diagram of a pitch motion of a surgical instrument according to the present disclosure, and (f) a conceptual diagram of a yaw motion thereof.

[0013] FIG. 2 is a perspective view illustrating a surgical instrument according to a first embodiment of the present disclosure.

[0014] FIG. 3 is a side view of the surgical instrument of FIG. 2.

[0015] FIGS. 4 and 5 are perspective views illustrating an end tool of the surgical instrument of FIG. 2.

[0016] FIG. 6 is a perspective view illustrating an end tool hub of the end tool of the surgical instrument of FIG. 2.

[0017] FIGS. 7 and 8 are plan views illustrating the end tool of the surgical instrument of FIG. 2.

[0018] FIG. 9 is a side view illustrating the end tool of the surgical instrument of FIG. 2.

[0019] FIGS. 10 and 11 are exploded perspective views of the end tool of the surgical instrument of FIG. 2.

[0020] FIG. 12 is a perspective view illustrating a first jaw pulley of the surgical instrument of FIG. 2.

[0021] FIG. 13 is an exploded perspective view illustrating a staple pulley and a staple link of the surgical instrument of FIG. 2.

[0022] FIG. 14 is a plan view illustrating a first jaw of the surgical instrument of FIG. 2.

[0023] FIG. 15 is a plan view illustrating a second jaw of the surgical instrument of FIG. 2.

[0024] FIG. 16 is a plan view illustrating opening and closing motions of the first jaw of the surgical instrument of FIG. 2.

[0025] FIG. **17** is a plan view illustrating opening and closing motions of the second jaw of the surgical instrument of FIG. **2**.

[0026] FIG. **18** is a plan view illustrating opening and closing motions of the first jaw and the second jaw of the surgical instrument of FIG. **2**.

[0027] FIG. **19** is a perspective view illustrating opening and closing motions of the end tool of the surgical instrument of FIG. **2**.

[0028] FIG. **20** is a plan view illustrating opening and closing motions of the end tool of the surgical instrument of FIG. **2**.

[0029] FIG. **21** is a perspective view illustrating the first jaw and a cartridge of the surgical instrument of FIG. **2**.

[0030] FIG. **22** is an exploded perspective view illustrating the cartridge of FIG. **21**.

[0031] FIG. **23** is an assembled perspective view illustrating the cartridge of FIG. **21**.

[0032] FIG. **24** is a side view illustrating the cartridge of FIG. **21**.

[0033] FIG. **25** is a perspective cross-sectional view illustrating the cartridge of FIG. **21**.

[0034] FIG. **26** is a side cross-sectional view illustrating the cartridge of FIG. **21**.

[0035] FIGS. **27** and **28** are perspective views illustrating an operation member of the cartridge of FIG. **21**.

[0036] FIG. **29** is a side cross-sectional view illustrating a stapling-related structure of the end tool of the surgical instrument of FIG. **2**.

[0037] FIGS. **30** and **31** are perspective cross-sectional views illustrating a stapling structure of the end tool of the surgical instrument of FIG. **2**.

[0038] FIGS. **32** to **35** are perspective views illustrating a ratchet drive operation of the end tool of FIG. **30**.

[0039] FIGS. **36** and **37** are plan views illustrating a ratchet drive operation of the end tool of FIG. **30**.

[0040] FIG. **38** is a perspective view illustrating an entire ratchet drive operation of the end tool of FIG. **30**.

[0041] FIGS. **39** and **40** are perspective views illustrating an entire stapling motion of the end tool of FIG. **30**.

[0042] FIGS. **41** and **42** are perspective views illustrating a manipulation part of the surgical instrument of FIG. **2**.

[0043] FIG. **43** is a view schematically illustrating only a configuration of pulleys and wires constituting joints of the surgical instrument illustrated in FIG. **2**.

[0044] FIG. **44** is a perspective view illustrating a yaw motion of the surgical instrument of FIG. **2**.

[0045] FIGS. **45** and **46** are diagrams illustrating a configuration of pulleys and wires, which are related to an actuation motion and a yaw motion of the surgical instrument illustrated in FIG. **2**, in detail for each of the first jaw and the second jaw.

[0046] FIGS. **47** to **49** are diagrams illustrating a configuration of pulleys and wires, which are related to stapling and cutting motions of the surgical instrument illustrated in FIG. **2**, in detail for each of the first jaw and the second jaw.

[0047] FIG. **50** is a perspective view illustrating a pitch motion of the surgical instrument of FIG. **2**.

[0048] FIGS. **51** to **53** are diagrams illustrating a configuration of pulleys and wires, which are related to a pitch motion of the surgical instrument illustrated in FIG. **2**, in detail for each of the first jaw and the second jaw.

[0049] FIGS. **54** to **57** are plan views illustrating an actuation motion of the end tool of the surgical instrument of FIG. **2**, and are views illustrating a process of performing an actuation motion in a state in which the jaws are yaw-rotated by -90° .

[0050] FIGS. **58** to **61** are plan views illustrating an actuation motion of the end tool of the surgical instrument of FIG. **2**, and are views illustrating a process of performing an actuation motion in a state in which the jaws are yaw-rotated by $+90^{\circ}$.

[0051] FIG. **62** is a plan view illustrating a stapling motion of the end tool of the surgical instrument of FIG. **2**, and is a view illustrating a process of performing a stapling motion in a state in which the jaws are yaw-rotated by $+90^\circ$.

[0052] FIG. **63** is a plan view illustrating a stapling motion of the end tool of the surgical instrument of FIG. **2**, and is a view illustrating a process of performing a stapling motion in a state in which the jaws are yaw-rotated by -90° .

[0053] FIGS. **64** to **67** are perspective views illustrating a pitch motion of the surgical instrument of FIG. **2**.

[0054] FIGS. **68** to **71** are perspective views illustrating a yaw motion of the surgical instrument of FIG. **2**.

[0055] FIGS. **72** to **75** are plan views illustrating a state in which the end tool of the surgical instrument of FIG. **2** is pitch-rotated and yaw-rotated.

[0056] FIGS. **76** and **77** are views illustrating a cartridge of a surgical instrument according to a first modified example of the first embodiment of the present disclosure.

[0057] FIGS. **78** and **79** are views illustrating a cartridge of a surgical instrument according to a second modified example of the first embodiment of the present disclosure.

[0058] FIGS. **80** and **81** are views illustrating a cartridge of a surgical instrument according to a third modified example of the first embodiment of the present disclosure.

[0059] FIGS. **82** and **83** are views illustrating a cartridge of a surgical instrument according to a fourth modified example of the first embodiment of the present disclosure.

[0060] FIGS. **84** and **85** are views illustrating a cartridge of a surgical instrument according to a fifth modified example of the first embodiment of the present disclosure.

[0061] FIG. **86** is a view illustrating operating states of the cartridge of FIG. **84**.

[0062] FIG. **87** is a perspective view illustrating an end tool of a surgical instrument according to a second embodiment of the present disclosure.

[0063] FIGS. **88** and **89** are magnified views of the end tool of the surgical instrument of FIG. **87**.

[0064] FIGS. **90** and **91** are magnified views illustrating the end tool of the surgical instrument of FIG. **87** from a different angle.

[0065] FIGS. **92** and **93** are views illustrating a state in which jaws of the end tool of the surgical instrument of FIG. **87** are yaw-rotated by 90° in a counterclockwise direction.

[0066] FIG. **94** is a magnified perspective view illustrating an end tool hub of the surgical instrument of FIG. **87**.

[0067] FIGS. **95** and **96** are perspective views illustrating an end tool of a surgical instrument according to a third embodiment of the present disclosure.

[0068] FIG. **97** is a perspective view illustrating a state in which the end tool of the surgical instrument of FIG. **96** is opened.

[0069] FIG. **98** is a perspective view illustrating a state in which the end tool of the surgical instrument of FIG. **96** is closed.

[0070] FIG. **99** is a side view illustrating the end tool of the surgical instrument of FIG. **96**.

[0071] FIGS. **100** and **101** are exploded perspective views of the end tool of the surgical instrument of FIG. **96**.

[0072] FIG. **102** is a plan view illustrating a state in which a first jaw of the end tool of the surgical instrument of FIG. **96** is fully opened.

[0073] FIGS. **103** and **104** are plan views illustrating opening and closing motions of the end tool of the surgical instrument of FIG. **96**.

[0074] FIGS. **105** and **106** are perspective views illustrating an end tool of a surgical instrument according to a fourth embodiment of the present disclosure.

[0075] FIGS. **107** and **108** are perspective views of the end tool of the surgical instrument of FIG. **105** viewed from a different angle.

[0076] FIG. **109** is a side view illustrating the end tool of the surgical instrument of FIG. **105**.

[0077] FIGS. **110** and **111** are exploded perspective views of the end tool of the surgical instrument of FIG. **105**.

[0078] FIG. **112** is an exploded perspective view of a staple link assembly of the surgical instrument of FIG. **105**.

[0079] FIGS. **113** and **114** are a side view and a plan view illustrating respective operating states of the staple link assembly of FIG. **105**.

[0080] FIG. **115** is a perspective view illustrating an internal structure of the end tool of the surgical instrument of FIG. **105**.

[0081] FIGS. **116** and **117** are perspective views illustrating respective operating states of the end tool of FIG. **115**.

[0082] FIGS. **118** and **119** are perspective views illustrating respective operating states of the end tool of FIG. **115**.

[0083] FIGS. **120** and **121** are perspective views illustrating respective operating states of the end tool of FIG. **115**.

[0084] FIG. **122** is a perspective view illustrating an end tool of a surgical instrument according to a fifth embodiment of the present disclosure.

[0085] FIG. **123** is a magnified view of the end tool of the surgical instrument of FIG. **122**.

[0086] FIG. **124** is a side view illustrating the end tool of the surgical instrument of FIG. **122**.

[0087] FIGS. **125** and **126** are exploded perspective views of the end tool of the surgical instrument of FIG. **122**.

[0088] FIG. **127** is an exploded perspective view of a staple pulley assembly of the surgical instrument of FIG. **122**.

[0089] FIG. **128** is a plan view illustrating an operating states of a staple pulley in the end tool of FIG. **122**.

[0090] FIG. **129** is a perspective view illustrating opening and closing motions of the end tool of the surgical instrument of FIG. **122**, and FIG. **130** is a plan view of FIG. **129**.

[0091] FIGS. **131** and **132** are plan views illustrating opening and closing motions of the end tool of the surgical instrument of FIG. **122**.

[0092] FIGS. **133** and **134** are views illustrating a process in which the end tool of the surgical instrument of FIG. **122** is switched from a deactivated state to an activated state.

[0093] FIGS. **135** to **139** are views illustrating a process of performing stapling and cutting motions as the end tool of the surgical instrument of FIG. **122** is switched from the deactivated state to the activated state.

[0094] FIG. **140** is a perspective view illustrating an end tool of a surgical instrument according to a sixth embodiment of the present disclosure.

[0095] FIG. **141** is a magnified view of the end tool of the surgical instrument of FIG. **140**.

[0096] FIG. **142** is a side view illustrating the end tool of the surgical instrument of FIG. **140**.

[0097] FIGS. **143** and **144** are exploded perspective views of the end tool of the surgical instrument of FIG. **140**.

[0098] FIG. **145** is an exploded perspective view of a staple pulley assembly of the surgical instrument of FIG. **140**.

[0099] FIG. **146** is a side view illustrating operating states of a staple pulley in the end tool of FIG. **140**.

[0100] FIGS. **147** and **148** are plan views illustrating opening and closing motions of the end tool of the surgical instrument of FIG. **140**.

[0101] FIG. **149** is a perspective view illustrating a process in which the end tool of the surgical instrument of FIG. **140** is switched from a deactivated state to an activated state.

[0102] FIGS. **150** to **153** are perspective views illustrating an end tool of a surgical instrument according to a seventh embodiment of the present disclosure.

[0103] FIGS. **154** and **155** are side views illustrating the end tool of the surgical instrument of FIG.

150.

[0104] FIGS. **156** and **157** are plan views illustrating the end tool of the surgical instrument of FIG. **150**.

[0105] FIGS. **158** to **160** are plan views illustrating a stapling motion of the end tool of the surgical instrument of FIG. **150**, and is a view illustrating a process of performing a stapling motion in a state in which jaws are yaw-rotated by $+90^\circ$.

[0106] FIGS. **161** to **163** are plan views illustrating a stapling motion of the end tool of the surgical instrument of FIG. **150**, and is a view illustrating a process of performing a stapling motion in a state in which the jaws are yaw-rotated by -90° .

[0107] FIGS. **164** to **166** are perspective views illustrating an end tool of the surgical instrument according to an eighth embodiment of the present disclosure.

[0108] FIGS. **167** and **168** are perspective views illustrating a cartridge of the surgical instrument of FIG. **164**.

[0109] FIG. **169** is a bottom view of the cartridge of FIG. **167**.

[0110] FIGS. **170** and **171** are perspective views illustrating respective operating states of the end tool of FIG. **164**.

[0111] FIG. **172** is a perspective view illustrating an end tool of a surgical instrument according to an embodiment of the present disclosure.

[0112] FIG. **173** is a side cross-sectional view illustrating a cartridge of the surgical instrument of FIG. **172**.

[0113] FIG. **174** is a perspective view illustrating an operation member of the cartridge of the surgical instrument of FIG. **172**.

[0114] FIG. **175** is a magnified perspective view illustrating the cartridge of the surgical instrument of FIG. **172**.

[0115] FIGS. **176** to **179** are views illustrating a process in which the operation member is moved.

[0116] FIGS. **180** to **182** are views illustrating a process in which the operation member is coupled to a second jaw.

[0117] FIGS. **183** to **185** are views illustrating a process in which the operation member is separated from the second jaw.

BEST MODE

[0118] According to an embodiment of the present disclosure, there is provided a surgical instrument including an end tool which includes a first jaw, a second jaw formed to face the first jaw, a first jaw pulley coupled to the first jaw and formed to be rotatable around a first shaft, a second jaw pulley coupled to the second jaw, formed to be rotatable around a shaft substantially the same as or parallel to the first shaft, and formed to be spaced apart from the first jaw pulley by a certain extent, and a staple drive assembly including one or more staple pulleys at least partially formed between the first jaw pulley and the second jaw pulley, and a cartridge which includes a reciprocating assembly that is connected to the staple drive assembly, and linearly moved when the staple pulley is rotationally moved, and an operation member that is brought into contact with the reciprocating assembly, and is moved in one direction by the reciprocating assembly when the reciprocating assembly is moved in the one direction.

[0119] In the present disclosure, when the staple pulley is rotated, the reciprocating assembly connected to the staple drive assembly may be moved toward a distal end or a proximal end of the cartridge.

[0120] In the present disclosure, when the staple pulley is alternately rotated in a clockwise direction and a counterclockwise direction, the reciprocating assembly connected to the staple drive assembly may be alternately moved toward the distal end and the proximal end of the cartridge.

[0121] In the present disclosure, when the reciprocating assembly is moved toward the distal end of the cartridge, the operation member may be moved toward the distal end of the cartridge by the reciprocating assembly.

[0122] In the present disclosure, a bidirectional rotational motion of the staple pulley may be converted into a reciprocating linear motion of the reciprocating assembly, which is connected to the staple drive assembly, by the staple drive assembly.

[0123] In the present disclosure, as the operation member is moved in the one direction, a wedge of the operation member may sequentially push and raise a plurality of staples in the cartridge to perform a stapling motion, and simultaneously, a blade formed on one side of the wedge of the operation member may be moved in the one direction to perform a cutting motion.

[0124] In the present disclosure, the staple drive assembly may include a link member configured to connect the staple pulley and the reciprocating assembly.

[0125] In the present disclosure, the operation member may include a ratchet member having a ratchet formed at least one surface thereon, and the ratchet of the ratchet member may be formed to be in contact with the reciprocating assembly.

[0126] In the present disclosure, the operation member may be moved toward a distal end of the cartridge together with the reciprocating assembly only when the reciprocating assembly is moved toward the distal end of the cartridge.

[0127] In the present disclosure, when the staple pulley is rotated in a first direction between a clockwise direction and a counterclockwise direction, the link member connected to the staple pulley, the reciprocating assembly connected to the link member, and the operation member in contact with the reciprocating assembly may be moved toward a distal end of the cartridge.

[0128] In the present disclosure, when the staple pulley is rotated in a direction opposite to the first direction between the clockwise and counterclockwise directions, the link member connected to the staple pulley and the reciprocating assembly connected to the link member may be moved toward a proximal end of the end tool, and the operation member may remain stationary with respect to the one direction.

[0129] In the present disclosure, the staple drive assembly may include a first link member connected to one region of the staple pulley, and a second link member connected to the other region of the staple pulley, and the reciprocating assembly may include a first reciprocating member coupled to the first link member and a second reciprocating member coupled to the second link member.

[0130] In the present disclosure, the operation member may include a ratchet member having a first ratchet and a second ratchet formed thereon, wherein the first ratchet may be formed to be in contact with the first reciprocating member, and the second ratchet may be formed to be in contact with the second reciprocating member.

[0131] In the present disclosure, when the staple pulley is rotated in one direction, the first ratchet and the first reciprocating member may be in contact with each other, and when the staple pulley is rotated in the other direction, the second ratchet and the second reciprocating member may be in contact with each other.

[0132] In the present disclosure, when the staple pulley is rotated in a first direction between a clockwise direction and a counterclockwise direction, the first link member connected to the staple pulley, the first reciprocating member connected to the first link member, and the operation member in contact with the first reciprocating member may be moved toward a distal end of the cartridge.

[0133] In the present disclosure, when the staple pulley is rotated in a direction opposite to the first direction between the clockwise and counterclockwise directions, the second link member connected to the staple pulley, the second reciprocating member connected to the second link member, and the operation member in contact with the second reciprocating member may be moved toward the distal end of the cartridge.

[0134] In the present disclosure, the surgical instrument may further include a staple wire coupled to the staple pulley and configured to rotate the staple pulley.

[0135] In the present disclosure, each of a deactivated state, in which the staple drive assembly and

the reciprocating assembly are spaced apart from each other, and an activated state, in which the staple drive assembly and the reciprocating assembly are coupled to each other, may exist.

[0136] In the present disclosure, only in the activated state, when the staple pulley is rotationally moved, which may cause the reciprocating assembly to be linearly moved.

[0137] In the present disclosure, the surgical instrument may further include a pair of end tool first jaw pitch main pulleys formed on one side of the first jaw pulley, and formed to be rotatable around a second shaft forming a predetermined angle with the first shaft, and a pair of end tool second jaw pitch main pulleys formed on one side of the second jaw pulley, and formed to be rotatable around a shaft that is substantially the same as or parallel to the second shaft.

[0138] In the present disclosure, the end tool may be formed to be yaw-rotatable around the first shaft and simultaneously pitch-rotatable around the second shaft.

[0139] In the present disclosure, the first jaw pulley, the staple pulley, and the second jaw pulley may be sequentially stacked.

[0140] According to another embodiment of the present disclosure, there is provided an end tool of a surgical instrument, the end tool including a first jaw capable of accommodating a cartridge, a second jaw formed to face the first jaw, a first jaw pulley coupled to the first jaw and formed to be rotatable around a first shaft, a second jaw pulley coupled to the second jaw, formed to be rotatable around a shaft substantially the same as or parallel to the first shaft, and formed to be spaced apart from the first jaw pulley by a certain extent, a staple drive assembly including one or more staple pulleys at least partially formed between the first jaw pulley and the second jaw pulley, and a staple wire at least partially in contact with the staple pulley and configured to transmit, to the staple pulley, a driving force necessary for rotating the staple pulley, wherein the staple drive assembly is connected to a reciprocating assembly of the cartridge and configured to convert a rotational motion of the staple pulley into a linear motion of the reciprocating assembly.

[0141] In the present disclosure, the end tool may further include an end tool hub including a first jaw pulley coupling part and a second jaw pulley coupling part, which are formed to face each other, and a guide part configured to connect the first jaw pulley coupling part and the second jaw pulley coupling part, wherein the first jaw pulley may be disposed adjacent to the first jaw pulley coupling part of the end tool hub, the second jaw pulley may be disposed adjacent to the second jaw pulley coupling part of the end tool hub, and at least a portion of the staple drive assembly may be formed between the first jaw pulley and the second jaw pulley.

[0142] In the present disclosure, the first shaft may be sequentially inserted through the first jaw pulley coupling part, the first jaw pulley, the staple pulley, the second jaw pulley, and the second jaw pulley coupling part.

[0143] In the present disclosure, the first jaw pulley, the staple pulley, and the second jaw pulley may be sequentially stacked in the end tool hub.

[0144] In the present disclosure, the first jaw pulley, the staple pulley, and the second jaw pulley may be formed to be rotatable independently of each other.

[0145] In the present disclosure, the end tool may further include a staple auxiliary pulley disposed between the staple pulley and the guide part.

[0146] In the present disclosure, the staple wire may be located on a common internal tangent of the staple pulley and the staple auxiliary pulley, and a rotation angle of the staple pulley may be increased by the staple auxiliary pulley.

[0147] In the present disclosure, a region of the guide part adjacent to the first jaw pulley, the staple pulley, and the second jaw pulley may be formed such that a cross section thereof is curved with a predetermined curvature.

[0148] In the present disclosure, the staple wire may be located on a common internal tangent of the staple pulley and the guide part, and a rotation angle of the staple pulley may be increased by the guide part.

[0149] In the present disclosure, the staple drive assembly may include a staple link assembly

configured to connect the staple pulley and the reciprocating assembly.

[0150] In the present disclosure, the staple link assembly may include a first link coupled to the staple pulley, and a second link coupled to each of the first link and the reciprocating assembly.

[0151] In the present disclosure, when the staple pulley is alternately rotated in a clockwise direction and a counterclockwise direction, the staple link assembly connected to the staple pulley may be alternately moved toward a distal end and a proximal end of the end tool.

[0152] In the present disclosure, a bidirectional rotational motion of the staple pulley may be converted into a reciprocating linear motion of the reciprocating assembly, which is connected to the staple link assembly, by the staple link assembly.

[0153] In the present disclosure, a guide groove may be formed in a length direction of the first jaw, and the staple link assembly may be moved along the guide groove.

[0154] In the present disclosure, the end tool may further include a jaw rotation shaft inserted through the first jaw and the second jaw and being the center of rotation of the first jaw and the second jaw, wherein the first shaft may be a jaw pulley rotation shaft, which is inserted through the first jaw pulley and the second jaw pulley and being the center of rotation of the first jaw pulley and the second jaw pulley, and the jaw rotation shaft may be moved relative to the jaw pulley rotation shaft when the first jaw pulley and the second jaw pulley are rotated around the jaw pulley rotation shaft.

[0155] In the present disclosure, when the first jaw and the second jaw are closed, the jaw rotation shaft may be moved toward a distal end of the end tool, and when the first jaw and the second jaw are opened, the jaw rotation shaft may be moved toward a proximal end of the end tool.

[0156] In the present disclosure, the surgical instrument may further include a pair of end tool first jaw pitch main pulleys formed on one side of the first jaw pulley, and formed to be rotatable around a second shaft forming a predetermined angle with the first shaft, and a pair of end tool second jaw pitch main pulleys formed on one side of the second jaw pulley, and formed to be rotatable around a shaft that is substantially the same as or parallel to the second shaft.

[0157] In the present disclosure, the end tool may be formed to be yaw-rotatable around the first shaft and simultaneously pitch-rotatable around the second shaft.

[0158] The end tool may further include a first jaw wire of which at least a part is wound on the first jaw pulley and the pair of end tool first jaw pitch main pulleys, and a second jaw wire of which at least a part is wound on the second jaw pulley and the pair of end tool second jaw pitch main pulleys.

[0159] According to another embodiment of the present disclosure, there is provided an end tool of a surgical instrument, the end tool including a first jaw and a second jaw that are rotatable independently of each other, a first jaw pulley coupled to the first jaw and formed to be rotatable around a first shaft, a second jaw pulley coupled to the second jaw and formed to be rotatable around a shaft that is substantially the same as or parallel to the first shaft, a staple pulley formed to be rotatable around a shaft substantially the same as or parallel to the first shaft, and disposed adjacent to the first jaw pulley or the second jaw pulley, and a staple link assembly connected to the staple pulley and reciprocating according to a bidirectional rotation of the staple pulley.

[0160] In the present disclosure, the staple link assembly may be coupled to a reciprocating assembly of a cartridge accommodated in the first jaw and may reciprocate the reciprocating assembly.

[0161] In the present disclosure, the staple link assembly may be moved toward a distal end or a proximal end of the end tool according to a rotation direction of the staple pulley.

[0162] In the present disclosure, a protrusion may be formed on one side of the staple pulley and the staple link assembly, a hole may be formed on the other side of the staple pulley and the staple link assembly, and the protrusion may be axially coupled to the hole.

[0163] In the present disclosure, a protruding member may be formed on the staple pulley, a slot is formed in the staple link assembly, and when the staple pulley is rotated, the protruding member

may be moved in the slot while coming into contact with the slot.

[0164] In the present disclosure, the staple link assembly may include a single link.

[0165] In the present disclosure, the staple link assembly may include a link member.

[0166] In the present disclosure, the link member may include a first link coupled to the staple pulley, and a second link coupled to the first link.

[0167] In the present disclosure, the staple link assembly may include a first link member and a second link member.

[0168] In the present disclosure, a cartridge accommodated in the first jaw may include a first reciprocating member and a second reciprocating member, wherein the first link member may be connected to the first reciprocating member, and the second link member may be connected to the second reciprocating member.

[0169] In the present disclosure, a guide groove may be formed in a length direction of the first jaw, and the staple link assembly may be moved along the guide groove.

[0170] In the present disclosure, the end tool may include a pair of end tool first jaw pitch main pulleys formed on one side of the first jaw pulley, and formed to be rotatable around a second shaft forming a predetermined angle with the first shaft, and a pair of end tool second jaw pitch main pulleys formed on one side of the second jaw pulley, and formed to be rotatable around a shaft that is substantially the same as or parallel to the second shaft.

[0171] In the present disclosure, when the first jaw pulley and the second jaw pulley are rotated in the same direction around the second shaft, the staple pulley may be rotated together with the first jaw pulley and the second jaw pulley.

[0172] In the present disclosure, when the first jaw pulley and the second jaw pulley are rotated in the same direction around the first shaft, the staple pulley may be rotated together with the first jaw pulley and the second jaw pulley.

[0173] In the present disclosure, when the first jaw pulley and the second jaw pulley are rotated in different directions around the first shaft, the staple pulley may be rotated together with either the first jaw pulley or the second jaw pulley.

[0174] In the present disclosure, there may be a case in which the first jaw pulley and the second jaw pulley are not rotated while the staple pulley is rotated around the first shaft by a staple wire.

[0175] In the present disclosure, a cartridge accommodation part in which a cartridge is accommodated may be formed in the first jaw, and an anvil with which a staple of the cartridge is in contact may be formed in the second jaw.

[0176] In the present disclosure, the end tool may further include a first jaw wire of which at least a portion is wound around the first jaw pulley, a second jaw wire of which at least a portion is wound around the second jaw pulley, and a staple wire of which at least a portion is wound around the staple pulley.

[0177] According to another embodiment of the present disclosure, there is provided an end tool of a surgical instrument, the end tool including a first jaw and a second jaw that are rotatable independently of each other, a first jaw pulley coupled to the first jaw and formed to be rotatable around a first shaft, a first jaw wire of which at least a portion is wound around the first jaw pulley, a second jaw pulley coupled to the second jaw and formed to be rotatable around the first shaft, a second jaw wire of which at least a portion is wound around the second jaw pulley, a pair of end tool first jaw pitch main pulleys formed on one side of the first jaw pulley, and formed to be rotatable around a second shaft forming a predetermined angle with the first shaft, a pair of end tool second jaw pitch main pulleys formed on one side of the second jaw pulley, and formed to be rotatable around a shaft that is substantially the same as or parallel to the second shaft, a staple pulley formed to be rotatable around the first shaft and disposed between the first jaw pulley and the second jaw pulley, a staple link assembly connected to the staple pulley and reciprocating according to a bidirectional rotation of the staple pulley, and a staple wire at least partially in contact with the staple pulley and configured to transmit, to the staple pulley, a driving force

necessary for rotating the staple pulley.

[0178] In the present disclosure, a bidirectional rotational motion of the staple pulley may be converted into a reciprocating linear motion of the staple link assembly.

[0179] In the present disclosure, the staple link assembly may be coupled to a reciprocating assembly of a cartridge accommodated in the first jaw, and a rotational motion of the staple pulley may be transmitted to an operation member of the cartridge via the staple link assembly and the reciprocating assembly.

[0180] In the present disclosure, a bidirectional rotational motion of the staple pulley may be converted into a reciprocating linear motion of the reciprocating assembly, which is connected to the staple link assembly, by the staple link assembly.

[0181] In the present disclosure, the end tool may further include a jaw rotation shaft inserted through the first jaw and the second jaw and being the center of rotation of the first jaw and the second jaw, wherein the first shaft may be a jaw pulley rotation shaft, which is inserted through the first jaw pulley and the second jaw pulley and being the center of rotation of the first jaw pulley and the second jaw pulley, and the jaw rotation shaft may be moved relative to the jaw pulley rotation shaft when the first jaw pulley and the second jaw pulley are rotated around the jaw pulley rotation shaft.

[0182] In the present disclosure, when the first jaw and the second jaw are closed, the jaw rotation shaft may be moved toward a distal end of the end tool, and when the first jaw and the second jaw are opened, the jaw rotation shaft may be moved toward a proximal end of the end tool.

[0183] In the present disclosure, a movable-coupling-allowed hole may be formed in the first jaw or the second jaw, a shaft coupling part may be formed in the first jaw pulley or the second jaw pulley, and the shaft coupling part may be movable to a certain extent in the movable-coupling-allowed hole while being fitted into the movable-coupling-allowed hole.

[0184] In the present disclosure, when the staple pulley is alternately rotated in a clockwise direction and a counterclockwise direction, the staple link assembly connected to the staple pulley may be alternately moved toward a distal end and a proximal end of the end tool.

[0185] In the present disclosure, a guide groove may be formed in a length direction of the first jaw, and the staple link assembly may be moved along the guide groove.

[0186] In the present disclosure, the end tool may further include an end tool hub including a first jaw pulley coupling part and a second jaw pulley coupling part, which are formed to face each other, and a guide part configured to connect the first jaw pulley coupling part and the second jaw pulley coupling part, wherein a region of the guide part adjacent to the first jaw pulley, the staple pulley, and the second jaw pulley may be formed such that a cross section thereof is curved with a predetermined curvature.

[0187] In the present disclosure, the staple wire may be located on a common internal tangent of the staple pulley and the guide part, and a rotation angle of the staple pulley may be increased by the guide part.

[0188] In the present disclosure, the first jaw, the first jaw pulley, the second jaw, and the second jaw pulley may be formed to rotate around the same shaft of rotation.

[0189] In the present disclosure, the first jaw pulley, the staple pulley, and the second jaw pulley may be sequentially stacked.

[0190] In the present disclosure, a guide groove may be formed in a length direction of an anvil of the second jaw, and a clamp of a cartridge accommodated in the first jaw may be formed to be movable along the guide groove.

[0191] In the present disclosure, coupling grooves through which the clamp is inserted into and withdrawn from the anvil may be formed at both end portions of the guide groove of the anvil.

[0192] In the present disclosure, the end tool may further include a pair of end tool first jaw pitch main pulleys formed on one side of the first jaw pulley, and formed to be rotatable around a second shaft forming a predetermined angle with the first shaft, a pair of end tool second jaw pitch main

pulleys formed on one side of the second jaw pulley, and formed to be rotatable around a shaft that is substantially the same as or parallel to the second shaft.

[0193] In the present disclosure, the end tool may be formed to be yaw-rotatable around the first shaft and simultaneously pitch-rotatable around the second shaft.

[0194] In the present disclosure, the end tool may further include a first jaw wire of which at least a portion is wound around the first jaw pulley and the pair of end tool first jaw pitch main pulleys, and a second jaw wire of which at least a portion is wound around the second jaw pulley and the pair of end tool second jaw pitch main pulleys.

[0195] In the present disclosure, the end tool may further include a staple second auxiliary pulley disposed between the first jaw pulley and the end tool first jaw pitch main pulley or between the second jaw pulley and the end tool second jaw pitch main pulley, formed to be rotatable around a shaft substantially the same as or parallel to the second shaft, and configured to guide a path of the first jaw wire or the second jaw wire.

[0196] According to another embodiment of the present disclosure, there is provided a surgical instrument including a first jaw capable of accommodating a cartridge, a second jaw formed to face the first jaw, a first jaw pulley coupled to the first jaw and formed to be rotatable around a first shaft, a second jaw pulley coupled to the second jaw, formed to be rotatable around a shaft substantially the same as or parallel to the first shaft, and formed to be spaced apart from the first jaw pulley by a certain extent, one or more staple pulleys at least partially formed between the first jaw pulley and the second jaw pulley, a staple link assembly including a first link member coupled to one region of the staple pulley and a second link member coupled to the other region of the staple pulley, and a staple wire of which at least a portion is wound around the staple pulley.

[0197] In the present disclosure, the first link member may be coupled to a first reciprocating member of the cartridge accommodated in the first jaw, and the second link member may be coupled to a second reciprocating member of the cartridge accommodated in the first jaw.

[0198] In the present disclosure, a first link coupling part and a second link coupling part may be formed in the staple pulley, the first link member may be coupled to the first link coupling part, and the second link member may be coupled to the second link coupling part.

[0199] In the present disclosure, the first link coupling part and the second link coupling part may be disposed on opposite sides of a central axis of the staple pulley.

[0200] In the present disclosure, when the staple pulley is rotated in one direction, the first link member and the second link member may be moved in opposite directions to each other.

[0201] In the present disclosure, when the staple pulley is rotated in one direction, the first link member may be moved toward a distal end of the end tool, and the second link member may be moved toward a proximal end of the end tool.

[0202] In the present disclosure, the cartridge accommodated in the first jaw may include a reciprocating assembly including a first reciprocating member and a second reciprocating member formed to face each other, and a ratchet member formed to be movable along the reciprocating assembly and having a first ratchet and a second ratchet formed thereon.

[0203] In the present disclosure, the first ratchet may be formed to be in contact with the first reciprocating member, and the second ratchet may be formed to be in contact with the second reciprocating member.

[0204] In the present disclosure, when the staple pulley is rotated in one direction, the first ratchet and the first reciprocating member may be in contact with each other, and when the staple pulley is rotated in the other direction, the second ratchet and the second reciprocating member may be in contact with each other.

[0205] In the present disclosure, when the staple pulley is rotated in one direction, the ratchet member is moved toward a distal end of the cartridge as the first reciprocating member is brought into close contact with and pushes the first ratchet, and when the staple pulley is rotated in the other direction, the ratchet member may be moved toward the distal end of the cartridge as the second

reciprocating member is brought into close contact with and pushes the second ratchet.

[0206] In the present disclosure, when the staple pulley is rotated in one direction, the first link member connected to the staple pulley, the first reciprocating member connected to the first link member, and the ratchet member in contact with the first reciprocating member may be moved toward a distal end of the cartridge, and when the staple pulley is rotated in the other direction, the second link member connected to the staple pulley, the second reciprocating member connected to the second link member, and the ratchet member in contact with the second reciprocating member may be moved toward the distal end of the cartridge.

[0207] In the present disclosure, the first link member may include a first link and a second link, and the second link member may include a third link and a fourth link.

[0208] In the present disclosure, a coupling part, which is formed on the first link member and coupled to the first reciprocating member, and a coupling part, which is formed on the second link member and coupled to the second reciprocating member, may have substantially the same height with respect to the first shaft.

[0209] In the present disclosure, the fourth link may be formed in the form of a bar that is bent one or more times.

[0210] In the present disclosure, the first ratchet may be formed on one side surface of the ratchet member, and the second ratchet may be formed on the other side surface of the ratchet member.

[0211] In the present disclosure, when the first reciprocating member is moved toward a proximal end of the end tool, the first reciprocating member may push the first ratchet toward the second reciprocating member, and when the second reciprocating member is moved toward the proximal end of the end tool, the second reciprocating member may push the second ratchet toward the first reciprocating member.

[0212] In the present disclosure, the ratchet member may include a first ratchet member having one surface on which the first ratchet is formed, and a second ratchet member having one surface on which the second ratchet is formed.

[0213] In the present disclosure, when the first reciprocating member is moved toward a proximal end of the end tool, the first reciprocating member may push the first ratchet toward the second reciprocating member, and when the second reciprocating member is moved toward the proximal end of the end tool, the second reciprocating member may push the second ratchet toward the first reciprocating member.

[0214] In the present disclosure, the surgical instrument may further include a first elastic member interposed between the first ratchet member and the first reciprocating member and configured to apply an elastic force in a direction in which the first ratchet member comes into close contact with the first reciprocating member, and a second elastic member interposed between the first ratchet member and the second reciprocating member and configured to apply an elastic force in a direction in which the second ratchet member comes into close contact with the second reciprocating member.

[0215] In the present disclosure, the surgical instrument may further include a pair of end tool first jaw pitch main pulleys formed on one side of the first jaw pulley and formed to be rotatable around a second shaft forming a predetermined angle with the first shaft, and a pair of end tool second jaw pitch main pulleys formed on one side of the second jaw pulley, and formed to be rotatable around a shaft that is substantially the same as or parallel to the second shaft.

[0216] In the present disclosure, the end tool may be formed to be yaw-rotatable around the first shaft and simultaneously pitch-rotatable around the second shaft.

[0217] In the present disclosure, the surgical instrument may further include a first jaw wire of which at least a portion is wound around the first jaw pulley and the pair of end tool first jaw pitch main pulleys, and a second jaw wire of which at least a portion is wound around the second jaw pulley and the pair of end tool second jaw pitch main pulleys.

[0218] According to another embodiment of the present disclosure, there is provided an end tool of

a surgical instrument, the end tool including a first jaw capable of accommodating a cartridge, a second jaw formed to face the first jaw, a first jaw pulley coupled to the first jaw and formed to be rotatable around a first shaft, a second jaw pulley coupled to the second jaw, formed to be rotatable around a shaft substantially the same as or parallel to the first shaft, and formed to be spaced apart from the first jaw pulley by a certain extent, a staple pulley disposed between the first jaw pulley and the second jaw pulley and having a protruding member formed in one region thereof, a link having a slot formed on one region thereof, wherein the protruding member of the staple pulley is inserted into the slot, and a staple wire of which at least a portion is wound around the staple pulley.

[0219] In the present disclosure, when the staple pulley is rotated, the link may be moved as the protruding member is moved in the slot while coming into contact with the slot.

[0220] In the present disclosure, the link may be formed as a single member.

[0221] In the present disclosure, when the staple pulley is rotated, the link is moved in one direction, wherein the link may be moved along a width of the slot in the one direction.

[0222] In the present disclosure, the protruding member may be formed in the form of a pin, and the link may be moved as the protruding member presses the slot of the link while rotating.

[0223] In the present disclosure, the slot may be formed obliquely without being concentric with the staple pulley, and the pin may be moved along the slot.

[0224] In the present disclosure, the protruding member may be formed in the form of a cam, and the link may be moved as the protruding member presses the slot of the link while rotating.

[0225] In the present disclosure, a center of the protruding member may not coincide with a center of the staple pulley, and the protruding member may be formed to be eccentric to a certain extent with respect to the staple pulley.

[0226] In the present disclosure, the cartridge accommodated in the first jaw may include a reciprocating assembly and an operation member, and each of a deactivated state, in which the link and the reciprocating assembly are spaced apart from each other, and an activated state, in which the link and the reciprocating assembly are coupled to each other, may exist.

[0227] In the present disclosure, only in the activated state, when the staple pulley is rotationally moved, which may cause the reciprocating assembly to be linearly moved.

[0228] In the present disclosure, when the first jaw and the second jaw are opened, the reciprocating assembly may be moved toward a proximal end of the end tool.

[0229] In the present disclosure, when the first jaw and the second jaw are closed, the reciprocating assembly and the link may be in contact with each other.

[0230] In the present disclosure, when the staple pulley is rotated in a state in which the first jaw and the second jaw are closed, the end tool may be switched to the activated state in which the link and the reciprocating assembly are coupled to each other.

[0231] In the present disclosure, a bidirectional rotational motion of the staple pulley may be converted into a reciprocating linear motion of the reciprocating assembly, which is connected to the link, by the link.

[0232] In the present disclosure, a bidirectional rotational motion of the staple pulley may be converted into a reciprocating linear motion of the link.

[0233] In the present disclosure, the first jaw pulley, the staple pulley, and the second jaw pulley may be sequentially stacked.

[0234] In the present disclosure, the end tool may further include a pair of end tool first jaw pitch main pulleys formed on one side of the first jaw pulley and formed to be rotatable around a second shaft forming a predetermined angle with the first shaft, and a pair of end tool second jaw pitch main pulleys formed on one side of the second jaw pulley, and formed to be rotatable around a shaft that is substantially the same as or parallel to the second shaft.

[0235] In the present disclosure, the end tool may be formed to be yaw-rotatable around the first shaft and simultaneously pitch-rotatable around the second shaft.

[0236] In the present disclosure, the end tool may further include a first jaw wire of which at least a portion is wound around the first jaw pulley and the pair of end tool first jaw pitch main pulleys, and a second jaw wire of which at least a portion is wound around the second jaw pulley and the pair of end tool second jaw pitch main pulleys.

[0237] According to another embodiment of the present disclosure, there is provided a cartridge of a surgical instrument having an end tool rotatable in at least one direction, the cartridge including a housing, a cover configured to cover one surface of the housing and having a slit formed in a first direction that is a length direction of the housing, a plurality of staples disposed inside the housing, a reciprocating assembly disposed inside the housing, formed to be movable in the first direction with respect to the housing, and having a plurality of recesses formed on at least one surface thereof, and an operation member formed on one side of the reciprocating assembly to be in contact with the reciprocating assembly and to be movable in the first direction by the reciprocating assembly.

[0238] In the present disclosure, the reciprocating assembly may be connected to a staple drive assembly formed on the end tool, and may be moved in the first direction when a staple pulley of the staple drive assembly is rotated.

[0239] In the present disclosure, when the staple pulley is alternately rotated in a clockwise direction and a counterclockwise direction, the reciprocating assembly connected to the staple drive assembly may be alternately moved toward a distal end and a proximal end of the cartridge.

[0240] In the present disclosure, the reciprocating assembly may move the operation member in contact with the reciprocating assembly in the first direction while moving toward the distal end of the cartridge.

[0241] In the present disclosure, the operation member may include a body, one or more wedges formed on one side of the body and each including an inclined surface formed to have a greater height at a proximal end side of the cartridge than a distal end side of the cartridge, a blade formed on one side of the wedge and including a sharp edge, and a ratchet member formed on one side of the body and having one or more ratchets formed thereon to be in contact with the recesses of the reciprocating assembly.

[0242] In the present disclosure, the cartridge may further include an elastic member formed between the body or the wedge and the ratchet member and configured to provide an elastic force to press the ratchet toward the reciprocating assembly.

[0243] In the present disclosure, when the reciprocating assembly is moved toward the distal end of the cartridge, the reciprocating assembly may be brought into close contact with and pushes the ratchet so that the operation member is moved toward the distal end of the cartridge.

[0244] In the present disclosure, when the reciprocating assembly is moved toward the proximal end of the cartridge, the operation member may remain stationary with respect to the one direction.

[0245] In the present disclosure, when the reciprocating assembly is moved toward the proximal end of the cartridge, an inclined surface of the reciprocating assembly may press an inclined surface of the ratchet in a direction in which the ratchet member is away from the reciprocating assembly.

[0246] In the present disclosure, the reciprocating assembly may include a first reciprocating member and a second reciprocating member formed to face each other, and the ratchet member may include a first ratchet formed to be in contact with the first reciprocating member, and a second ratchet formed to be in contact with the second reciprocating member.

[0247] In the present disclosure, when the first reciprocating member is moved toward the distal end of the cartridge, the first reciprocating member may be brought into close contact with and push the first ratchet so that the operation member is moved toward the distal end of the cartridge, and when the second reciprocating member is moved toward the distal end of the cartridge, the second reciprocating member may be brought into close contact with and push the second ratchet so that the operation member is moved toward the distal end of the cartridge.

[0248] In the present disclosure, the first reciprocating member and the second reciprocating member may be alternately moved toward the distal end of the cartridge.

[0249] In the present disclosure, the first reciprocating member may be connected to a first link member connected to one region of a staple pulley of the end tool, the second reciprocating member may be connected to a second link member connected to the other region of the staple pulley, when the staple pulley is rotated in one direction, the first link member connected to the staple pulley, the first reciprocating member connected to the first link member, and the ratchet member in contact with the first reciprocating member may be moved toward the distal end of the cartridge, and when the staple pulley is rotated in the other direction, the second link member connected to the staple pulley, the second reciprocating member connected to the second link member, and the ratchet member in contact with the second reciprocating member may be moved toward the distal end of the cartridge.

[0250] In the present disclosure, the first ratchet may be formed on one side surface of the ratchet member, and the second ratchet may be formed on the other side surface of the ratchet member.

[0251] In the present disclosure, when the first reciprocating member is moved toward the proximal end of the cartridge, the first reciprocating member may push the first ratchet toward the second reciprocating member, and when the second reciprocating member is moved toward the proximal end of the cartridge, the second reciprocating member may push the second ratchet toward the first reciprocating member.

[0252] In the present disclosure, the ratchet member may include a first ratchet member having one surface on which the first ratchet is formed, and a second ratchet member having one surface on which the second ratchet is formed.

[0253] In the present disclosure, when the first reciprocating member is moved toward the proximal end of the cartridge, the first reciprocating member may push the first ratchet toward the second reciprocating member, and when the second reciprocating member is moved toward the proximal end of the cartridge, the second reciprocating member may push the second ratchet toward the first reciprocating member.

[0254] In the present disclosure, the cartridge may further include a first elastic member interposed between the first ratchet member and the first reciprocating member and configured to apply an elastic force in a direction in which the first ratchet member comes into close contact with the first reciprocating member, and a second elastic member interposed between the first ratchet member and the second reciprocating member and configured to apply an elastic force in a direction in which the second ratchet member comes into close contact with the second reciprocating member.

[0255] In the present disclosure, one or more protrusions may be formed in a region of an inner surface of the housing, which is to be in contact with the operation member, and a snap to be in contact with the protrusion may be formed in the operation member.

[0256] In the present disclosure, when the reciprocating assembly is moved toward the proximal end of the cartridge, the snap and the protrusion may be in contact with each other to prevent the operation member from moving toward the proximal end of the cartridge.

[0257] In the present disclosure, one end portion of the snap may be coupled to the operation member and is formed to be elastically deformable to a certain extent.

[0258] In the present disclosure, each of the one or more protrusions may include an inclined surface formed to have a greater height at the proximal end side of the cartridge than the distal end side of the cartridge.

[0259] In the present disclosure, a clamp formed to extend in the first direction may be formed at one side of the blade of the operation member.

[0260] In the present disclosure, a guide groove may be formed in an anvil of a second jaw of the end tool in the first direction, and the clamp may be moved along the guide groove.

[0261] In the present disclosure, coupling grooves through which the clamp is inserted into and withdrawn from the anvil may be formed at both end portions of the anvil.

[0262] According to another embodiment of the present disclosure, there is provided a method of driving a surgical instrument, the method comprising operations (a) in which, when a staple pulley of a staple drive assembly is rotated in a first direction around a first shaft, a staple link assembly connected to the staple pulley and a reciprocating assembly of a cartridge connected to the staple link assembly are moved along a second shaft toward a distal end of the cartridge, (b) in which, when the reciprocating assembly is moved toward the distal end of the cartridge, an operation member in contact with the reciprocating assembly is moved toward the distal end of the cartridge together with the reciprocating assembly, (c) in which, as the operation member is moved toward the distal end of the cartridge, the operation member ejects staples in the cartridge to the outside of the cartridge, and simultaneously, a blade of the operation member is moved toward the distal end of the cartridge, and (d) in which, when the staple pulley is rotated in a second direction opposite to the first direction around the first shaft, the staple link assembly, which is connected to the staple pulley, and the reciprocating assembly of the cartridge, which is connected to the staple link assembly, are moved toward a proximal end of the cartridge.

[0263] In the present disclosure, when the staple pulley is rotated in the first direction or the second direction, the reciprocating assembly may be moved toward the distal end of the cartridge or the proximal end of the cartridge.

[0264] In the present disclosure, a bidirectional rotational motion of the staple pulley around the first shaft may be converted into a reciprocating linear motion of the reciprocating assembly, which is connected to the staple pulley, with respect to the second shaft.

[0265] In the present disclosure, the operation member may be moved toward the distal end of the cartridge by the reciprocating linear motion of the reciprocating assembly.

[0266] In the present disclosure, a rack may be formed on one surface of the reciprocating assembly, the operation member may include a ratchet member having a ratchet formed thereon, and as the rack pushes the ratchet member while being in close contact with the ratchet member, the ratchet member may be moved toward the distal end of the cartridge.

[0267] In the present disclosure, in operation (d), the operation member may remain stationary with respect to a direction of the second shaft.

[0268] in the present disclosure, the operation member may be moved toward the distal end of the cartridge together with the reciprocating assembly only when the reciprocating assembly is moved toward the distal end of the cartridge.

[0269] In the present disclosure, the staple drive assembly may include a first link member connected to one region of the staple pulley, and a second link member connected to the other region of the staple pulley, the reciprocating assembly may include a first reciprocating member coupled to the first link member and a second reciprocating member coupled to the second link member, and the operation member may include a ratchet member on which a first ratchet and a second ratchet are formed.

[0270] In the present disclosure, in operation (a), the first ratchet and the first reciprocating member may be in contact with each other, and in operation (b), the second ratchet and the second reciprocating member may be in contact with each other.

[0271] In the present disclosure, in operation (a), the first link member connected to the staple pulley, the first reciprocating member connected to the first link member, and the operation member in contact with the first reciprocating member may be moved toward the distal end of the cartridge, and the second link member connected to the staple pulley and the second reciprocating member connected to the second link member may be moved toward the proximal end of the cartridge.

[0272] In the present disclosure, in operation (d), the first link member connected to the staple pulley and the first reciprocating member connected to the first link member may be moved toward the distal end of the cartridge, and the second link member connected to the staple pulley, the second reciprocating member connected to the second link member, and the operation member in

contact with the second reciprocating member may be moved toward the distal end of the cartridge. [0273] In the present disclosure, in operation (d), the operation member may be moved toward the distal end of the cartridge.

[0274] In the present disclosure, the surgical instrument may further include a staple wire coupled to the staple pulley to rotate the staple pulley, wherein a bidirectional rotation of the staple pulley may be converted into a reciprocating linear motion of the reciprocating assembly by the staple wire.

[0275] In the present disclosure, as the operation member is moved toward the distal end of the cartridge, a wedge of the operation member may sequentially push and raise a plurality of staples in the cartridge to perform a stapling motion, and simultaneously, a blade formed on one side of the wedge of the operation member may be moved toward the distal end of the cartridge to perform a cutting motion.

[0276] In the present disclosure, operations (a) to (d) may be repeatedly performed.

MODE OF DISCLOSURE

[0277] While the present disclosure is susceptible to various modifications and alternative forms, specific embodiments thereof are shown by way of example in the drawings and will herein be described in detail. However, it should be understood that there is no intent to limit the present disclosure to the particular forms disclosed herein, rather, the present disclosure should be construed to cover various modifications, equivalents, and alternatives of embodiments of the present disclosure. In describing the present disclosure, detailed description of known related arts will be omitted when it is determined that the gist of the present disclosure may be unnecessarily obscured.

[0278] Although terms such as “first,” “second,” and the like may be used to describe various components, such components should not be limited to the above terms. The terms are only used to distinguish one component from another.

[0279] The terms used herein are for the purpose of describing particular embodiments only and are not intended to be limiting to the present disclosure. Singular forms are intended to include plural forms as well, unless the context clearly indicates otherwise. In the present application, it will be further understood that the terms “comprise,” “comprising,” “include,” and/or “including”, when used in this specification, specify the presence of stated features, integers, steps, operations, elements, components and/or groups thereof but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components and/or groups thereof.

[0280] Hereinafter, the embodiments of the present disclosure will be described below in detail with reference to the accompanying drawings, and when the embodiments of the present disclosure are described with reference to the drawings, the same or corresponding components are given the same reference numerals, and repetitive descriptions thereof will be omitted.

[0281] Further, in describing the various embodiments of the present disclosure, it is to be understood that each embodiment is not intended to be interpreted or implemented independently, and that the technical ideas described in each embodiment may be interpreted or implemented in combination with other embodiments described separately.

[0282] In a surgical instrument according to the present disclosure, when a manipulation part is rotated in one direction for at least any one of pitch, yaw, and actuation motions, an end tool is rotated in intuitively the same direction as a direction in which the manipulation part is moved.

[0283] FIG. 1A is a conceptual diagram of a pitch motion of a conventional surgical instrument, and FIG. 1B is a conceptual diagram of a yaw motion thereof.

[0284] Referring to FIG. 1A, in performing a pitch motion of a conventional surgical instrument, in a state in which an end tool **120a** is formed in front of a rotation center **121a** of the end tool, and a manipulation part **110a** is formed at the rear of a rotation center **111a** of the manipulation part, when the manipulation part **110a** is rotated in a clockwise direction, the end tool **120a** is also rotated in the clockwise direction, and when the manipulation part **120a** is rotated in a

counterclockwise direction, the end tool **120a** is also rotated in the counterclockwise direction. Referring to FIG. **1B**, in performing a yaw motion of the conventional surgical instrument, in a state in which the end tool **120a** is formed in front of the rotation center **121a** of the end tool, and the manipulation part **110a** is formed at the rear of the rotation center **111a** of the manipulation part, when the manipulation part **110a** is rotated in the clockwise direction, the end tool **120a** is also rotated in the clockwise direction, and when the manipulation part **120a** is rotated in the counterclockwise direction, the end tool **120a** is also rotated in the counterclockwise direction. In this case, in view of left and right directions of a user, when the user moves the manipulation part **110a** to the left, the end tool **120a** is moved to the right, and when the user moves the manipulation part **110a** to the right, the end tool **120a** is moved to the left. As a result, a manipulation direction of the user and an operation direction of the end tool are opposite to each other, which may cause the user to make a mistake, and user's manipulation may not be easy.

[0285] FIG. **1C** is a conceptual diagram of a pitch motion of another conventional surgical instrument, and FIG. **1D** is a conceptual diagram of a yaw motion thereof.

[0286] Referring to FIG. **1C**, in the conventional surgical instrument, which is partially formed in a mirror symmetrical shape, in performing a pitch motion, in a state in which an end tool **120b** is formed in front of a rotation center **121b** of the end tool, and a manipulation part **110b** is formed at the rear of a rotation center **111b** of the manipulation part, when the manipulation part **110b** is rotated in the clockwise direction, the end tool **120b** is rotated in the counterclockwise direction, and when the manipulation part **110b** is rotated in the counterclockwise direction, the end tool **120b** is rotated in the clockwise direction. In this case, in view of rotation directions of the manipulation part and the end tool, a rotation direction in which the user rotates the manipulation part **110b** and a rotation direction of the end tool **120b** according thereto are opposite to each other. As a result, the user may be confused with the manipulation direction, and as the operation of a joint is not intuitive, the user may make an error. Further, referring to FIG. **1D**, in performing a yaw motion, in a state in which the end tool **120b** is formed in front of the rotation center **121b** of the end tool, and the manipulation part **110b** is formed at the rear of the rotation center **111b** of the manipulation part, when the manipulation part **110b** is rotated in the clockwise direction, the end tool **120b** is rotated in the counterclockwise direction, and when the manipulation part **110b** is rotated in the counterclockwise direction, the end tool **120b** is rotated in the clockwise direction. In this case, in view of rotation directions of the manipulation part and the end tool, a rotation direction in which the user rotates the manipulation part **110b** and a rotation direction of the end tool **120b** according thereto are opposite to each other. As a result, the user may be confused with the manipulation direction, and as the operation of the joint is not intuitive, the user may make an error. In the user's pitch or yaw manipulation of the conventional surgical instrument, the user's manipulation direction and the end tool's operation direction do not match each other in view of one of the rotation direction and the left and right directions. This is because the configurations of the end tool and the manipulation part are different from each other in the joint configuration of the conventional surgical instrument. That is, this is because the manipulation part is formed at the rear of the rotation center of the manipulation part, while the end tool is formed in front of the rotation center of the end tool. In order to address the above problems, in a surgical instrument according to an embodiment of the present disclosure, which is illustrated in FIGS. **1E** and **1F**, an end tool **120c** is formed in front of a rotation center **121c** of the end tool and a manipulation part **110c** is also formed in front of a rotation center **111c** of the manipulation part, so that the operations of the manipulation part **110c** and the end tool **120c** are intuitively matched with each other. In other words, unlike existing examples such as those shown in FIGS. **1A**, **1B**, **1C**, and **1D**, in which the manipulation part is close to a user with respect to the joint thereof (that is, away from the end tool), the surgical instrument according to an embodiment of the present disclosure, which is illustrated in FIGS. **1E** and **1F**, is formed such that at least a portion of the manipulation part is closer (than a joint thereof) to the end tool with respect to the joint thereof at any one moment or

more in a manipulation process.

[0287] In other words, in the conventional surgical instrument as illustrated in FIGS. 1A, 1B, 1C, and 1D, the manipulation part is formed at the rear of the rotation center thereof, while the end tool is located in front of the rotation center thereof, and thus the end tool is moved at a front side thereof with a rear side fixed through a motion of the manipulation part that is moved at a rear side thereof with a front side thereof fixed, which is not an intuitively matching structure. Accordingly, a mismatch may occur between the manipulation of the manipulation part and the motion of the end tool in view of the left and right directions or in view of the rotation direction, which may cause confusion to the user, and the manipulation of the manipulation part may be difficult to perform intuitively and quickly and may cause mistakes. In contrast, in the surgical instrument according to an embodiment of the present disclosure, since both the end tool and the manipulation part are moved with respect to the rotation center formed at the rear side thereof, it may be said that the motions are intuitively matched with each other in terms of structure. In other words, moving portions of the manipulation part are moved with respect to the rotation center formed at the rear side thereof just as moving portions of the end tool are moved with respect to the rotation center formed at the rear side thereof, and thus it may be said that the motions are intuitively matched with each other in terms of structure. This allows the user to intuitively and quickly perform a control in a direction toward the end tool, and a possibility of making a mistake may be significantly reduced. Hereinafter, a detailed mechanism enabling the above-described function will be described below.

First Embodiment of Surgical Instrument

[0288] FIG. 2 is a perspective view illustrating a surgical instrument according to a first embodiment of the present disclosure, and FIG. 3 is a side view of the surgical instrument of FIG. 2. FIGS. 4 and 5 are perspective views illustrating an end tool of the surgical instrument of FIG. 2. FIG. 6 is a perspective view illustrating an end tool hub of the end tool of the surgical instrument of FIG. 2. FIGS. 7 and 8 are plan views illustrating the end tool of the surgical instrument of FIG. 2. FIG. 9 is a side view illustrating the end tool of the surgical instrument of FIG. 2. FIGS. 10 and 11 are exploded perspective views of the end tool of the surgical instrument of FIG. 2. FIG. 12 is a perspective view illustrating a first jaw pulley of the surgical instrument of FIG. 2. FIG. 13 is an exploded perspective view illustrating a staple pulley and a staple link of the surgical instrument of FIG. 2. FIG. 14 is a plan view illustrating a first jaw of the surgical instrument of FIG. 2, and FIG. 15 is a plan view illustrating a second jaw of the surgical instrument of FIG. 2. FIG. 16 is a plan view illustrating opening and closing motions of the first jaw of the surgical instrument of FIG. 2, FIG. 17 is a plan view illustrating opening and closing motions of the second jaw of the surgical instrument of FIG. 2, and FIG. 18 is a plan view illustrating opening and closing motions of the first jaw and the second jaw of the surgical instrument of FIG. 2. FIG. 19 is a perspective view illustrating opening and closing motions of the end tool of the surgical instrument of FIG. 2, and FIG. 20 is a plan view illustrating opening and closing motions of the end tool of the surgical instrument of FIG. 2.

[0289] First, referring to FIGS. 2 and 3, a surgical instrument **10** according to a first embodiment of the present disclosure includes an end tool **100**, a manipulation part **200**, a power transmission part **300**, and a connection part **400**.

[0290] Here, the connection part **400** is formed in the shape of a hollow shaft, and one or more wires and electric wires may be accommodated therein. The manipulation part **200** is coupled to one end portion of the connection part **400**, the end tool **100** is coupled to the other end portion thereof, and the connection part **400** may serve to connect the manipulation part **200** and the end tool **100**. Here, the connection part **400** of the surgical instrument **10** according to the first embodiment of the present disclosure includes a straight part **401** and a bent part **402**, wherein the straight part **401** is formed at a side coupled to the end tool **100**, and the bent part **402** is formed at a side to which the manipulation part **200** is coupled. As such, since the end portion of the

connection part **400** at the side of the manipulation part **200** is formed to be bent, a pitch manipulation part **201**, a yaw manipulation part **202**, and an actuation manipulation part **203** may be formed along an extension line of the end tool **100** or adjacent to the extension line. From another perspective, it may be said that the pitch manipulation part **201** and the yaw manipulation part **202** are at least partially accommodated in a concave portion formed by the bent part **402**. Due to the above-described shape of the bent part **402**, the shapes and motions of the manipulation part **200** and the end tool **100** may be further intuitively matched with each other.

[0291] Meanwhile, a plane on which the bent part **402** is formed may be substantially the same plane as a pitch plane, that is, an XZ plane of FIG. 2. As such, as the bent part **402** is formed on substantially the same plane as the XZ plane, interference with the manipulation part may be reduced. Of course, for intuitive motions of the end tool and the manipulation part, any form other than the XZ plane may be possible.

[0292] Meanwhile, a connector **410** may be formed on the bent part **402**. The connector **410** may be connected to an external power source (not shown), and the connector **410** may also be connected to the end tool **100** via an electric wire, and may transmit, to the end tool **100**, electric energy supplied from the external power source (not shown). In addition, the electric energy transmitted to the end tool **100** as described above may produce a driving force for rotating a staple pulley (see **161** of FIG. 5) to be described later in the clockwise or counterclockwise direction.

[0293] The manipulation part **200** is formed at the one end portion of the connection part **400** and provided as an interface to be directly controlled by a medical doctor, for example, a tongs shape, a stick shape, a lever shape, or the like, and when the medical doctor controls the manipulation part **200**, the end tool **100**, which is connected to the interface and inserted into the body of a surgical patient, performs a certain motion, thereby performing surgery. Here, the manipulation part **200** is illustrated in FIG. 2 as being formed in a handle shape that is rotatable while the finger is inserted therein, the concept of the present disclosure is not limited thereto, and various types of manipulation parts that are connected to the end tool **100** and manipulate the end tool **100** may be possible.

[0294] The end tool **100** is formed on the other end portion of the connection part **400**, and performs necessary motions for surgery by being inserted into a surgical site. In an example of the end tool **100**, as illustrated in FIG. 2, a pair of jaws **103** for performing a grip motion may be used. However, the concept of the present disclosure is not limited thereto, and various devices for performing surgery may be used as the end tool **100**. For example, a configuration of a cantilever cautery may also be used as the end tool. The end tool **100** is connected to the manipulation part **200** by the power transmission part **300**, and receives a driving force of the manipulation part **200** through the power transmission part **300** to perform a motion necessary for surgery, such as gripping, cutting, suturing, or the like.

[0295] Here, the end tool **100** of the surgical instrument **10** according to the first embodiment of the present disclosure is formed to be rotatable in at least one direction, for example, the end tool **100** may perform a pitch motion around a Y-axis of FIG. 2 and simultaneously perform a yaw motion and an actuation motion around a Z-axis of FIG. 2.

[0296] Here, each of the pitch, yaw, and actuation motions used in the present disclosure are defined as follows.

[0297] First, the pitch motion means a motion of the end tool **100** rotating in a vertical direction with respect to an extension direction of the connection part **400** (an X-axis direction of FIG. 2), that is, a motion rotating around the Y-axis of FIG. 2. In other words, the pitch motion means a motion of the end tool **100**, which is formed to extend from the connection part **400** in the extension direction of the connection part **400** (the X-axis direction of FIG. 2), rotating vertically around the Y-axis with respect to the connection part **400**.

[0298] Next, the yaw motion means a motion of the end tool **100** rotating in the left and right directions, that is, a motion rotating around the Z-axis of FIG. 2, with respect to the extension

direction of the connection part **400** (the X-axis direction of FIG. 2). In other words, the yaw motion means a motion of the end tool **100**, which extends from the connection part **400** in the extension direction of the connection part **400** (the X-axis direction of FIG. 2), rotating horizontally around the Z-axis with respect to the connection part **400**. That is, the yaw motion means a motion of the two jaws **103**, which are formed on the end tool **100**, rotating around the Z-axis in the same direction.

[0299] Meanwhile, the actuation motion may mean a motion of the end tool **100** rotating around the same shaft of rotation as that of the yaw motion, while the two jaws **103** rotating in the opposite directions so as to be closed or opened. That is, the actuation motion means rotating motions of the two jaws **103**, which are formed on the end tool **100**, in the opposite directions around the Z-axis.

[0300] The power transmission part **300** may connect the manipulation part **200** to the end tool **100**, transmit the driving force of the manipulation part **200** to the end tool **100**, and include a plurality of wires, pulleys, links, sections, gears, or the like.

[0301] The end tool **100**, the manipulation part **200**, and the power transmission part **300** of the surgical instrument **10** of FIG. 2 will be described in detail later.

(Intuitive Driving)

[0302] Hereinafter, intuitive driving of the surgical instrument **10** of the present disclosure will be described.

[0303] First, while holding a first handle **204** with the palm of the hand, the user may rotate a first handle **204** around the Y-axis (i.e., a rotation shaft **246** of FIG. 25) to perform a pitch motion, and rotate the first handle **204** around the Z-axis (i.e., a rotation shaft **243** of FIG. 43) to perform a yaw motion. In addition, the user may perform an actuation motion by manipulating the actuation manipulation part **203** while inserting the thumb and the index finger into a first actuation extension part **252** and/or a second actuation extension part **257** in the form of a hand ring formed at one end portion of the actuation manipulation part **203**.

[0304] Here, in the surgical instrument **10** according to the first embodiment of the present disclosure, when the manipulation part **200** is rotated in one direction with respect to the connection part **400**, the end tool **100** is rotated in a direction that is intuitively the same as a manipulation direction of the manipulation part **200**. In other words, when the first handle **204** of the manipulation part **200** is rotated in one direction, the end tool **100** is also rotated in a direction intuitively the same as the one direction, so that a pitch motion or a yaw motion is performed. Here, the phrase “intuitively the same direction” may be further explained as meaning that a direction of movement of the user's finger gripping the manipulation part **200** and a direction of movement of a distal end of the end tool **100** form substantially the same direction. Of course, “the same direction” as used herein may not be a perfectly matching direction on a three-dimensional coordinate, and may be understood to be equivalent to the extent that, for example, when the user's finger moves to the left, the distal end of the end tool **100** is moved to the left, and when the user's finger moves down, the end portion of the end tool **100** is moved down.

[0305] In addition, to this end, in the surgical instrument **10** according to the first embodiment of the present disclosure, the manipulation part **200** and the end tool **100** are formed in the same direction with respect to a plane perpendicular to the extension axis (X-axis) of the connection part **400**. That is, when viewed based on a YZ plane of FIG. 2, the manipulation part **200** is formed to extend in a positive (+) X-axis direction, and the end tool **100** is also formed to extend in the positive (+) X-axis direction. In other words, it may be said that a formation direction of the end tool **100** on one end portion of the connection part **400** is the same as a formation direction of the manipulation part **200** on the other end portion of the connection part **400** on the basis of the YZ plane. Further, in other words, it may be said that the manipulation part **200** may be formed in a direction away from the body of a user holding the manipulation part **200**, that is, in a direction in which the end tool **100** is formed. That is, in the parts such as the first handle **204**, a first actuation manipulation part **251**, a second actuation manipulation part **256**, and the like, which are moved by

the user's grip for actuation motion, yaw motion, and pitch motions, a corresponding portion that is moved for the motion is formed to extend in the positive (+) X-axis direction from the rotation center of a corresponding joint for the motion. In this manner, the manipulation part **200** may be configured in the same manner as the end tool **100** in which each moving portion is formed to extend in the positive (+) X-axis direction from the rotation center of a corresponding joint for the motion, and as described with reference to FIG. 1, the manipulation direction of the user may be identical to the operation direction of the end tool from the viewpoint of the rotation directions and the left and right directions. As a result, intuitively the same manipulation may be achieved.

[0306] In detail, in the case of the conventional surgical instrument, a direction in which a user manipulates the manipulation part is different from a direction in which the end tool is actually operated, that is, intuitively different from the direction in which the end tool is actually operated, and thus, a surgical operator may not easily intuitively manipulate the surgical instrument and may spend a long time to learn a skill of operating the end tool in desired directions, and in some cases, malfunctions may occur, which may cause damage to patients.

[0307] In order to address such problems, the surgical instrument **10** according to the first embodiment of the present disclosure is configured such that the manipulation direction of the manipulation part **200** and the operation direction of the end tool **100** are intuitively identical to each other. To this end, the manipulation part **200** is configured like the end tool **100**, that is, in the manipulation part **200**, portions that are actually moved for actuation, yaw, and pitch motions extend respectively from rotation centers of corresponding joints in the positive (+) X-axis direction.

[0308] Hereinafter, the end tool **100**, the manipulation part **200**, the power transmission part **300**, and the like of the surgical instrument **10** of FIG. 2 will be described in more detail.

(Power Transmission Part)

[0309] Hereinafter, the power transmission part **300** of the surgical instrument **10** of FIG. 2 will be described in more detail.

[0310] Referring to FIGS. 2 to 20, 43, and the like, the power transmission part **300** of the surgical instrument **10** according to an embodiment of the present disclosure may include a wire **301**, a wire **302**, a wire **303**, a wire **304**, a wire **305**, a wire **306**, a wire **307**, and a wire **308**.

[0311] Here, the wires **301** and **305** may be paired to serve as first jaw wires. The wires **302** and **306** may be paired to serve as second jaw wires. Here, the components encompassing the wires **301** and **305**, which are first jaw wires, and the wires **302** and **306**, which are second jaw wires, may be referred to as jaw wires. In addition, the wires **303** and **304** may be paired to serve as pitch wires. In addition, the wires **307** and **308**, may be paired to serve as staple wires.

[0312] Further, the power transmission part **300** of the surgical instrument **10** according to an embodiment of the present disclosure may include a coupling member (see **321** of FIG. 7) a coupling member **323**, a coupling member **324**, a coupling member **326**, a coupling member **327**, and a coupling member (see **329** of FIG. 62) that are coupled to respective end portions of the wires to respectively couple the wires to the pulleys. Here, each of the coupling members may have various shapes as necessary, such as a ball shape, a tube shape, and the like.

[0313] Here, in the end tool **100** side, the coupling member (see **321** of FIG. 7) may serve as a pitch wire-end tool coupling member, the coupling member **323** may serve as a first jaw wire-end tool coupling member, the coupling member **326** may serve as a second jaw wire-end tool coupling member, and the coupling member (see **329** of FIG. 62) may serve as a staple wire-end tool coupling member.

[0314] Further, in the manipulation part **200** side, the coupling member **324** may serve as a first jaw wire-manipulation part coupling member, and the coupling member **327** may serve as a second jaw wire-manipulation part coupling member. In addition, although not shown in the drawings, a pitch wire-manipulation part coupling member and a staple wire-manipulation part coupling member may be further formed in the manipulation part **200** side.

[0315] The coupling relationship between the wires, the coupling members, and the respectively pulleys will be described in detail as follows.

[0316] First, the wires **301** and **305**, which are first jaw wires, may be a single wire. The coupling member **323**, which is a first jaw wire-end tool coupling member, is inserted at an intermediate point of the first jaw wire, which is a single wire, and the coupling member **323** is crimped and fixed, and then, both strands of the first jaw wire centered on the coupling member **323** may be referred to as the wire **301** and the wire **305**, respectively.

[0317] Alternatively, the wires **301** and **305**, which are first jaw wires, may also be formed as separate wires, and connected by the coupling member **323**.

[0318] In addition, by coupling the coupling member **323** to a pulley **111**, the wires **301** and **305** may be fixedly coupled to the pulley **111**. This allows the pulley **111** to rotate as the wires **301** and **305** are pulled and released.

[0319] Meanwhile, the first jaw wire-manipulation part coupling member (see **324** of FIG. **43**) may be coupled to end portions of the wires **301** and **305**, which are opposite to one end portions to which the coupling member **323** is coupled.

[0320] In addition, by coupling the first jaw wire-manipulation part coupling member (see **324** of FIG. **43**) to a pulley **210** as described above, the wires **301** and **305** may be fixedly coupled to the pulley **210**. As a result, when the pulley **210** is rotated by a motor or a human force, the pulley **111** of the end tool **100** may be rotated as the wire **301** and the wire **305** are pulled and released.

[0321] In the same manner, each of the wires **302** and **306**, which are second jaw wires, is coupled to the coupling member (see **326** of FIG. **43**), which is a second jaw wire-end tool coupling member, and the second jaw wire-manipulation part coupling member (see **327** of FIG. **43**). In addition, the coupling member (see **326** of FIG. **43**) is coupled to a pulley **121**, and the second jaw wire-manipulation part coupling member (see **327** of FIG. **43**) is coupled to a pulley **220**. As a result, when the pulley **220** is rotated by a motor or a human force, the pulley **121** of the end tool **100** may be rotated as the wire **302** and the wire **306** are pulled and released.

[0322] In the same manner, the wires **303** and **304**, which are pitch wires, are respectively coupled to the coupling member (see **321** of FIG. **7**), which is a pitch wire-end tool coupling member, and the pitch wire-manipulation part coupling member (not shown). In addition, the coupling member (see **321** of FIG. **7**) is coupled to a pulley **131**, and the pitch wire-manipulation part coupling member (not shown) is coupled to a pulley **231**. As a result, when the pulley **231** is rotated by a motor or a human force, the pulley **131** of the end tool **100** may be rotated as the wire **303** and the wire **304** are pulled and released.

[0323] In the same manner, the wires **307** and **308**, which are staple wires, are respectively coupled to the coupling member (see **329** of FIG. **62**), which is a staple wire-end tool coupling member, and the staple wire-manipulation part coupling member (not shown). In addition, the coupling member (see **329** of FIG. **62**) is coupled to the staple pulley **161**, and the staple wire-manipulation part coupling member (not shown) is coupled to a pulley (see **269** of FIG. **47**). As a result, when the pulley **269** is rotated by a motor or a human force, the staple pulley **161** of the end tool **100** may be rotated as the wires **307** and **308** are pulled and released.

(End Tool)

[0324] Hereinafter, the end tool **100** of the surgical instrument **10** of FIG. **2** will be described in more detail.

[0325] FIGS. **4** and **5** are perspective views illustrating the end tool of the surgical instrument of FIG. **2**, FIG. **6** is a perspective view illustrating an end tool hub of the end tool of the surgical instrument of FIG. **2**, and FIGS. **7** and **8** are plan views illustrating the end tool of the surgical instrument of FIG. **2**.

[0326] Here, FIG. **4** illustrates a state in which an end tool hub **180** and a pitch hub **107** are coupled, and FIG. **5** illustrates a state in which the end tool hub **180** is removed. Meanwhile, FIG. **7** is a view mainly illustrating the wires, and FIG. **8** is a view mainly illustrating the pulleys.

[0327] Referring to FIGS. **4** to **8** and the like, the end tool **100** of the first embodiment of the present disclosure includes a pair of jaws for performing a grip motion, that is, a first jaw **101** and a second jaw **102**. Here, each of the first jaw **101** and the second jaw **102**, or a component encompassing the first jaw **101** and the second jaw **102** may be referred to as the jaw **103**.

[0328] Further, the end tool **100** may include the pulley **111**, a pulley **112**, a pulley **113**, a pulley **114**, a pulley **115**, and a pulley **116** that are related to a rotational motion of the first jaw **101**. In addition, the end tool **100** may include the pulley **121**, a pulley **122**, a pulley **123**, a pulley **124**, a pulley **125**, and a pulley **126** that are related to a rotational motion of the second jaw **102**.

[0329] Here, the pulleys facing each other are illustrated in the drawings as being formed parallel to each other, but the concept of the present disclosure is not limited thereto, and each of the pulleys may be variously formed with a position and a size suitable for the configuration of the end tool.

[0330] Further, the end tool **100** of the first embodiment of the present disclosure may include the end tool hub **180** and the pitch hub **107**.

[0331] A rotation shaft **141** and a rotation shaft **142**, which will be described later, may be inserted through the end tool hub **180**, and the end tool hub **180** may internally accommodate at least some of the pulley **111** and the pulley **121**, which are axially coupled to the rotation shaft **141**. In addition, the end tool hub **180** may internally accommodate at least some of the pulley **112** and the pulley **122** that are axially coupled to the rotation shaft **142**.

[0332] In detail, referring to FIG. **6**, the end tool hub **180** includes a first jaw pulley coupling part **181**, a second jaw pulley coupling part **182**, a guide part **183**, and a pitch pulley coupling part **185**.

[0333] In detail, the first jaw pulley coupling part **181** and the second jaw pulley coupling part **182** are formed to face each other, and the pulley **111**, the pulley **121**, and the staple pulley **161** are accommodated inside the first jaw pulley coupling part **181** and the second jaw pulley coupling part **182**. In addition, a through hole is formed in each of the first jaw pulley coupling part **181** and the second jaw pulley coupling part **182** such that the rotation shaft **141** passes through and axially couples the first jaw pulley coupling part **181**, the pulley **111**, the staple pulley **161**, the pulley **121**, and the second jaw pulley coupling part **182**.

[0334] The first jaw pulley coupling part **181** and the second jaw pulley coupling part **182** are connected by the guide part **183**. That is, the first jaw pulley coupling part **181** and the second jaw pulley coupling part **182** parallel to each other are coupled by the guide part **183** formed in a direction substantially perpendicular thereto, so that the first jaw pulley coupling part **181**, the second jaw pulley coupling part **182**, and the guide part **183** form a substantially “C-shape,” in which the pulley **111**, the pulley **121**, and the staple pulley **161** are accommodated.

[0335] Here, the pulley **111**, which is a first jaw pulley, is disposed adjacent to the first jaw pulley coupling part **181** of the end tool hub **180**, and the pulley **121**, which is a second jaw pulley, is disposed adjacent to the second jaw pulley coupling part **182** of the end tool hub **180**, so that a staple assembly accommodation part may be formed between the first jaw pulley coupling part **181** and the second jaw pulley coupling part **182**. In addition, at least some of a staple pulley assembly (see **160** of FIG. **13**) and a staple link assembly (see **170** of FIG. **13**), which will be described later, may be formed in the staple assembly accommodation part. In other words, it may be said that at least some of the staple pulley **161** and a link member **171** are disposed between the first jaw pulley coupling part **181** and the second jaw pulley coupling part **182**. As such, according to the present disclosure, by disposing at least some of the staple pulley assembly (see **160** of FIG. **13**) and the staple link assembly (see **170** of FIG. **13**) between the pulley **111**, which is a first jaw pulley, and the pulley **121**, which is a second jaw pulley, the end tool **100** is allowed to perform pitch and yaw motions, as well as stapling and cutting motions using the staple pulley **161**. This will be described in more detail later.

[0336] Meanwhile, the pulley **131** serving as an end tool pitch pulley may be formed at one end portion of the end tool hub **180**. As shown in FIG. **6**, the pulley **131** may be integrally formed with

the end tool hub **180** as one body. That is, a disk-shaped pulley is formed at one end portion of the end tool hub **180**, and a groove around which a wire may be wound may be formed on an outer circumferential surface of the pulley. Alternatively, the pulley **131** may be formed as a separate member from the end tool hub **180** to be coupled to the end tool hub **180**. The wires **303** and **304** described above are coupled to the pulley **131** serving as an end tool pitch pulley, and a pitch motion is performed as the pulley **131** is rotated around a rotation shaft **143**.

[0337] The rotation shaft **143** and a rotation shaft **144**, which will be described later, are inserted through the pitch hub **107**, and the pitch hub **107** may be axially coupled to the end tool hub **180** (and the pulley **131**) by the rotation shaft **143**. Thus, the end tool hub **180** and the pulley **131** may be formed to be rotatable around the rotation shaft **143** with respect to the pitch hub **107**.

[0338] Further, the pitch hub **107** may internally accommodate at least some of the pulley **113**, the pulley **114**, the pulley **123**, and the pulley **124** that are axially coupled to the rotation shaft **143**. In addition, the pitch hub **107** may internally accommodate at least some of the pulley **115**, the pulley **116**, the pulley **125**, and the pulley **126** that are axially coupled to the rotation shaft **144**.

[0339] Further, the end tool **100** of the first embodiment of the present disclosure may include the rotation shaft **141**, the rotation shaft **142**, the rotation shaft **143**, and the rotation shaft **144**. As described above, the rotation shaft **141** and the rotation shaft **142** may be inserted through the end tool hub **180**, and the rotation shaft **143** and the rotation shaft **144** may be inserted through the pitch hub **107**.

[0340] The rotation shaft **141**, the rotation shaft **142**, the rotation shaft **143**, and the rotation shaft **144** may be arranged sequentially from a distal end **104** of the end tool **100** toward a proximal end **105**. Accordingly, starting from the distal end **104**, the rotation shaft **141** may be referred to as a first pin, the rotation shaft **142** may be referred to as a second pin, the rotation shaft **143** may be referred to as a third pin, and the rotation shaft **144** may be referred to as a fourth pin.

[0341] Here, the rotation shaft **141** may function as an end tool jaw pulley rotation shaft, the rotation shaft **142** may function as an end tool jaw auxiliary pulley rotation shaft, the rotation shaft **143** may function as an end tool pitch rotating shaft, and the rotation shaft **144** may function as an end tool pitch auxiliary rotating shaft of the end tool **100**.

[0342] Each of the rotation shafts **141**, **142**, **143**, and **144** may be fitted into one or more pulleys, which will be described in detail below.

[0343] Meanwhile, a rotation shaft **145** may be further formed on one side of the rotation shaft **141**, specifically, one side of the rotation shaft **141** at the distal end **104** side. The rotation shaft **145** is inserted through the first jaw **101** and the second jaw **102** to function as a jaw rotation shaft. This will be described in detail below.

[0344] The pulley **111** functions as an end tool first jaw pulley, and the pulley **121** functions as an end tool second jaw pulley. The pulley **111** may also be referred to as a first jaw pulley, and the pulley **121** may also be referred to as a second jaw pulley, and these two components may also be referred to collectively as an end tool jaw pulley or simply a jaw pulley.

[0345] The pulley **111** and the pulley **121**, which are end tool jaw pulleys, are formed to face each other, and are formed to be rotatable independently of each other around the rotation shaft **141**, which is an end tool jaw pulley rotation shaft. In this case, the pulley **111** and the pulley **121** are formed to be spaced apart from each other by a certain extent, and the staple assembly accommodation part may be formed therebetween. In addition, at least some of the staple pulley assembly **160** and the staple link assembly **170**, which will be described later, may be disposed inside the staple assembly accommodation part.

[0346] Here, in the drawings, it is illustrated that the pulley **111** and the pulley **121** are formed to rotate around one rotation shaft **141**, but it is of course possible that each end tool jaw pulley may be formed to be rotatable around a separate shaft. Here, the first jaw **101** is fixedly coupled to the pulley **111** and rotates together with the pulley **111**, and the second jaw **102** is fixedly coupled to the pulley **121** and rotates together with the pulley **121**. Yaw and actuation motions of the end tool

100 are performed according to the rotation of the pulley **111** and the pulley **121**. That is, when the pulley **111** and the pulley **121** are rotated in the same direction around the rotation shaft **141**, the yaw motion is performed, and when the pulley **111** and the pulley **121** are rotated in opposite directions around the rotation shaft **141**, the actuation motion is performed.

[0347] Here, the first jaw **101** and the pulley **111** may be formed as separate members and coupled to each other, or the first jaw **101** and the pulley **111** may be integrally formed as one body.

Similarly, the second jaw **102** and the pulley **121** may be formed as separate members and coupled to each other, or the second jaw **102** and the pulley **121** may be integrally formed as one body.

[0348] The pulley **112** functions as an end tool first jaw auxiliary pulley, and the pulley **122** functions as an end tool second jaw auxiliary pulley, and these two components may be collectively referred to as an end tool jaw auxiliary pulley or simply an auxiliary pulley.

[0349] In detail, the pulley **112** and the pulley **122**, which are end tool jaw auxiliary pulleys, may be additionally provided on one sides of the pulley **111** and the pulley **121**, respectively. In other words, the pulley **112**, which is an auxiliary pulley, may be disposed between the pulley **111** and the pulley **113**/pulley **114**. In addition, the pulley **122**, which is an auxiliary pulley, may be disposed between the pulley **121** and the pulley **123** or the pulley **124**. The pulley **112** and the pulley **122** may be formed to be rotatable independently of each other around the rotation shaft **142**. Here, in the drawings, it is illustrated that the pulley **112** and the pulley **122** are formed to rotate around one rotation shaft **142**, but it is of course possible that each of the pulley **112** and the pulley **122** may be formed to be rotatable around a separate shaft. Such auxiliary pulleys will be described in more detail later.

[0350] The pulley **113** and the pulley **114** function as end tool first jaw pitch main pulleys, and the pulley **123** and the pulley **124** function as end tool second jaw pitch main pulleys, and these two components may be referred to collectively as an end tool jaw pitch main pulley.

[0351] The pulley **115** and the pulley **116** function as end tool first jaw pitch sub-pulleys, and the pulley **125** and the pulley **126** function as end tool second jaw pitch sub-pulleys, and these two components may be collectively referred to as an end tool jaw pitch sub-pulley.

[0352] Hereinafter, components related to the rotation of the pulley **111** will be described.

[0353] The pulley **113** and the pulley **114** function as end tool first jaw pitch main pulleys. That is, the pulley **113** and the pulley **114** function as main rotation pulleys for a pitch motion of the first jaw **101**. Here, the wire **301**, which is a first jaw wire, is wound around the pulley **113**, and the wire **305**, which is a first jaw wire, is wound around the pulley **114**.

[0354] The pulley **115** and the pulley **116** function as end tool first jaw pitch sub-pulleys. That is, the pulley **115** and the pulley **116** function as sub rotation pulleys for a pitch motion of the first jaw **101**. Here, the wire **301**, which is a first jaw wire, is wound around the pulley **115**, and the wire **305**, which is a first jaw wire, is wound around the pulley **116**.

[0355] Here, the pulley **113** and the pulley **114** are disposed on one side of the pulley **111** and the pulley **112** to face each other. Here, the pulley **113** and the pulley **114** are formed to be rotatable independently of each other around the rotation shaft **143** that is an end tool pitch rotating shaft. In addition, the pulley **115** and the pulley **116** are disposed on one sides of the pulley **113** and the pulley **114**, respectively, to face each other. Here, the pulley **115** and the pulley **116** are formed to be rotatable independently of each other around the rotation shaft **144** that is an end tool pitch auxiliary rotating shaft. Here, in the drawings, it is illustrated that the pulley **113**, the pulley **115**, the pulley **114**, and the pulley **116** are all formed to be rotatable around the Y-axis direction, but the concept of the present disclosure is not limited thereto, and the rotating axes of the respective pulleys may be formed in various directions according to configurations thereof.

[0356] The wire **301**, which is a first jaw wire, is sequentially wound to make contact with at least portions of the pulley **115**, the pulley **113**, and the pulley **111**. In addition, the wire **305** connected to the wire **301** by the coupling member **323** is sequentially wound to make contact with at least portions of the pulley **111**, the pulley **112**, the pulley **114**, and the pulley **116** in turn.

[0357] In other words, the wires **301** and **305**, which are first jaw wires, are sequentially wound to make contact with at least portions of the pulley **115**, the pulley **113**, the pulley **111**, the pulley **112**, the pulley **114**, and the pulley **116** and are formed to move along the above pulleys while rotating the above pulleys.

[0358] Accordingly, when the wire **301** is pulled in the direction of an arrow **301** of FIG. **13**, the coupling member **323** to which the wire **301** is coupled and the pulley **111** coupled to the coupling member **323** are rotated in an arrow L direction of FIG. **13**. In contrast, when the wire **305** is pulled in the direction of an arrow **305** of FIG. **6**, the coupling member **323** to which the wire **305** is coupled and the pulley **111** coupled to the coupling member **323** are rotated in an arrow R direction of FIG. **6**.

[0359] Hereinafter, the pulley **112** and the pulley **122** serving as auxiliary pulleys will be described in more detail.

[0360] The pulley **112** and the pulley **122** may serve to enlarge rotation angles of the first jaw **101** and the second jaw **102**, respectively, by coming into contact with the wire **305**, which is a jaw wire, and the wire **302**, which is a second jaw wire, to change the arrangement path of the wires **305** and **302** to a certain extent.

[0361] That is, when the auxiliary pulleys are not disposed, each of the first jaw and the second jaw may be rotated up to a right angle, but in an embodiment of the present disclosure, the pulley **112** and the pulley **122**, which are auxiliary pulleys, are additionally provided, so that the maximum rotation angle may be increased by θ as shown in FIG. **8**. This enables a motion of the two jaws of the end tool **100** being opened for an actuation motion while the two jaws are yaw-rotated by 90° in the L direction. This is because the second jaw **102** is rotated by the additional angle θ as shown in FIG. **8**. Similarly, an actuation motion is possible even when the two jaws are yaw-rotated in the L direction. In other words, a feature of increasing the range of a yaw rotation in which an actuation motion is possible may be obtained through the pulley **112** and the pulley **122**.

[0362] This will be described below in more detail.

[0363] When the auxiliary pulleys are not disposed, since the first jaw wire is fixedly coupled to the end tool first jaw pulley, and the second jaw wire is fixedly coupled to the end tool second jaw pulley, each of the end tool first jaw pulley and the end tool second jaw pulley may be rotated up to 90° . In this case, when the actuation motion is performed while the first jaw and the second jaw are located at a 90° line, the first jaw may be opened, but the second jaw may not be rotated beyond 90° . Accordingly, when the first jaw and the second jaw perform a yaw motion over a certain angle, there was a problem that the actuation motion is not smoothly performed.

[0364] In order to address such a problem, in the surgical instrument **10** of the present disclosure, the pulley **112** and the pulley **122**, which are auxiliary pulleys, are additionally disposed at one sides of the pulley **111** and the pulley **121**, respectively. As described above, as the arrangement path of the wire **305**, which is a first jaw wire, and the wire **302**, which is a second jaw wire, is changed to a certain extent by disposing the pulley **112** and the pulley **122**, a tangential direction of the wires **305** and **302** is changed, and accordingly, the coupling member **323** for coupling the wire **301** and the pulley **111** may be rotated up to a line N of FIG. **8**. That is, the coupling member **323**, which is a coupling part of the wire **301** and the pulley **111**, is rotatable until the coupling member **323** is located on a common internal tangent of the pulley **111** and the pulley **112**. Similarly, the coupling member **326**, which is a coupling part of the wire **302** and the pulley **121**, is rotatable until the coupling member **326** is located on a common internal tangent of the pulley **121** and the pulley **122**, so that the range of rotation in the L direction may be increased.

[0365] In other words, the wires **301** and **305**, which are two strands of the first jaw wire wound around the pulley **111** by the pulley **112**, are disposed at one side with respect to a plane perpendicular to the Y-axis and passing through the X-axis. Simultaneously, the wires **302** and **306**, which are two strands of the second jaw wire wound around the pulley **121** by the pulley **122**, are disposed at the other side with respect to a plane perpendicular to the Y-axis and passing through

the X-axis.

[0366] In other words, the pulley **113** and the pulley **114** are disposed at one side with respect to the plane perpendicular to the Y-axis and passing through the X-axis, and the pulley **123** and the pulley **124** are disposed at the other side with respect to the plane perpendicular to the Y-axis and passing through the X-axis.

[0367] In other words, the wire **305** is located on the internal tangent of the pulley **111** and the pulley **112**, and the rotation angle of the pulley **111** is increased by the pulley **112**. In addition, the wire **302** is located on the internal tangent of the pulley **121** and the pulley **122**, and the rotation angle of the pulley **121** is increased by the pulley **122**.

[0368] According to the present disclosure, as the rotation radii of the jaw **101** and the jaw **102** increase, an effect of increasing a yaw motion range in which a normal opening/closing actuation motion is performed may be obtained.

[0369] Next, components related to the rotation of the pulley **121** will be described.

[0370] The pulley **123** and the pulley **124** function as end tool second jaw pitch main pulleys. That is, the second jaw **102** functions as a main rotation pulley for a pitch motion of the second jaw **102**. Here, the wire **306**, which is a second jaw wire, is wound around the pulley **123**, and the wire **302**, which is a second jaw wire, is wound around the pulley **124**.

[0371] The pulley **125** and the pulley **126** function as end tool second jaw pitch sub-pulleys. That is, the pulley **125** and the pulley **126** function as sub rotation pulleys for a pitch motion of the second jaw **102**. Here, the wire **306**, which is a second jaw wire, is wound around the pulley **125**, and the wire **302**, which is a second jaw wire, is wound around the pulley **126**.

[0372] On one side of the pulley **121**, the pulley **123** and the pulley **124** are disposed to face each other. Here, the pulley **123** and the pulley **124** are formed to be rotatable independently of each other around the rotation shaft **143** which is an end tool pitch rotating shaft. In addition, the pulley **125** and the pulley **126** are disposed on one sides of the pulley **123** and the pulley **124**, respectively, to face each other. Here, the pulley **125** and the J15 pulley **123J25** are formed to be rotatable independently of each other around the rotation shaft **144**, which is an end tool pitch auxiliary rotating shaft. Here, in the drawings, it is illustrated that all of the pulley **123**, the pulley **125**, the pulley **124**, and the pulley **126** are formed to be rotatable around the Y-axis direction, but the concept of the present disclosure is not limited thereto, and the rotating axes of the respective pulleys may be formed in various directions according to configurations thereof.

[0373] The wire **306**, which is a second jaw wire, is sequentially wound to make contact with at least portions of the pulley **125**, the pulley **123**, and the pulley **121**. In addition, the wire **302** connected to the wire **306** by the coupling member **326** is sequentially wound to make contact with at least portions of the pulley **121**, the pulley **122**, the pulley **124**, and the pulley **126**.

[0374] In other words, the wires **306** and **302**, which are second jaw wires, are sequentially wound to make contact with at least portions of the pulley **125**, the pulley **123**, the pulley **121**, the pulley **122**, the pulley **124**, and the pulley **126**, and are formed to move along the above pulleys while rotating the above pulleys.

[0375] Accordingly, when the wire **306** is pulled in the direction of an arrow **306** of FIG. **13**, the coupling member **322** to which the wire **306** is coupled and the pulley **121** coupled to the coupling member **322** are rotated in the arrow R direction of FIG. **13**. In contrast, when the wire **302** is pulled in the direction of an arrow **302** of FIG. **13**, the coupling member **326** to which the wire **302** is coupled and the pulley **121** coupled to the coupling member **326** are rotated in the arrow L direction of FIG. **13**.

[0376] Hereinafter, a pitch motion of the present disclosure will be described in more detail.

[0377] Meanwhile, when the wire **301** is pulled toward the arrow **301** of FIG. **7**, and simultaneously, the wire **305** is pulled toward the arrow **305** of FIG. **7** (that is, when both strands of the first jaw wire are pulled), as shown in FIG. **43**, since the wires **301** and **305** are wound around lower portions of the pulley **113** and the pulley **114** rotatable around the rotation shaft **143**, which is

an end tool pitch rotating shaft, the pulley **111** to which the wires **301** and **305** are fixedly coupled and the end tool hub **180** to which the pulley **111** is coupled are rotated as a whole in the counterclockwise direction around the rotation shaft **143**, so that the end tool **100** is rotated downward to perform a pitch motion. At this time, since the second jaw **102** and the wires **302** and **306** fixedly coupled thereto are wound around upper portions of the pulley **123** and the pulley **124** rotatable around the rotation shaft **143**, the wires **302** and **306** are unwound in an opposite directions of the wires **302** and **306**, respectively.

[0378] In contrast, when the wire **302** is pulled toward the arrow **302** of FIG. 7, and simultaneously, the wire **306** is pulled toward the arrow **306** of FIG. 7, as shown in FIG. 43, since the wires **302** and **306** are wound around the upper portions of the pulley **123** and the pulley **124** rotatable around the rotation shaft **143**, which is an end tool pitch rotating shaft, the pulley **121** to which the wires **302** and **306** are fixedly coupled and the end tool hub **180** to which the pulley **121** is coupled are rotated as a whole in the clockwise direction around the rotation shaft **143**, so that the end tool **100** is rotated upward to perform a pitch motion. At this time, since the first jaw **101** and the wires **301** and **305** fixedly coupled thereto are wound around lower portions of the pulley **113** and the pulley **114** rotatable around the rotation shaft **143**, the wires **302** and **306** are moved in opposite directions of the wires **301** and **305**, respectively.

[0379] Meanwhile, the end tool **100** of the surgical instrument **10** of the present disclosure may further include the pulley **131**, which is an end tool pitch pulley, the manipulation part **200** may further include the pulley **231** and the pulley **232**, which are manipulation part pitch pulleys, and the power transmission part **300** may further include the wires **303** and **304**, which are pitch wires. In detail, the pulley **131** of the end tool **100** is rotatable around the rotation shaft **143**, which is an end tool pitch rotating shaft, and may be integrally formed with the end tool hub **180** (or fixedly coupled to the end tool hub **180**) as one body. In addition, the wires **303** and **304** may serve to connect the pulley **131** of the end tool **100** and the pulley **231** and the pulley **232** of the manipulation part **200**.

[0380] Thus, when the pulley **231** and the pulley **232** of the manipulation part **200** are rotated, the rotation of the pulley **231** and the pulley **232** is transmitted to the pulley **131** of the end tool **100** through the wires **303** and **304** so that the pulley **131** is rotated together therewith, and as a result, the end tool **100** performs a pitch motion while rotating.

[0381] That is, in the surgical instrument **10** according to the first embodiment of the present disclosure, by providing the pulley **131** of the end tool **100**, the pulley **231** and the pulley **232** of the manipulation part **200**, and the wires **303** and **304** of the power transmission part **300** to transmit power for a pitch motion, the driving force for the pitch motion of the manipulation part **200** may be more completely transmitted to the end tool **100**, thereby improving operation reliability.

[0382] Here, a diameter of each of the pulley **113**, the pulley **114**, the pulley **123**, and the pulley **124**, which are end tool jaw pitch main pulleys, and a diameter of the pulley **131**, which is an end tool pitch pulley, may be the same as each other or different from each other. At this time, a ratio of the diameter of the end tool jaw pitch main pulley to the diameter of the end tool pitch pulley may be the same as a ratio of a diameter of the manipulation part pitch pulley of the manipulation part **200** to a diameter of a manipulation part pitch main pulley to be described later. This will be described in detail below.

(Components Related to Staple Pulley)

[0383] Hereinafter, the staple pulley **161** of the end tool **100** of the surgical instrument **10** of FIG. 2 will be described in more detail.

[0384] FIG. 9 is a side view illustrating the end tool of the surgical instrument of FIG. 2, and FIGS. 10 and 11 are perspective views illustrating the first jaw of the surgical instrument of FIG. 2. FIG. 12 is a perspective view illustrating the first jaw pulley of the surgical instrument of FIG. 2, and FIG. 13 is an exploded perspective view illustrating the staple pulley and the staple link of the surgical instrument of FIG. 2.

[0385] Referring to FIGS. 4 to 13 and the like, the end tool 100 of the first embodiment of the present disclosure may include the staple pulley 161, a staple auxiliary pulley 162, a pulley 163, a pulley 164, a pulley 165, and a pulley 166 that are related to a linear motion/rotational motion of respective pulleys and links for stapling and cutting.

[0386] The staple pulley 161 is formed to face each of the pulley 111 and the pulley 121, which are end tool jaw pulleys, and the staple pulley 161 and the pulley 111 and the pulley 121 are formed to be rotatable independently of each other around the rotation shaft 141, which is an end tool jaw pulley rotation shaft. Here, the staple pulley 161 is illustrated as being disposed between the pulley 111 and the pulley 121, but the concept of the present disclosure is not limited thereto, and the staple pulley 161 may be disposed at various positions adjacent to the pulley 111 or the pulley 121. [0387] Here, in the present disclosure, the staple pulley 161, the pulley 111, and the pulley 121 are formed to rotate around substantially the same shaft. As the staple pulley 161, the pulley 111, and the pulley 121 are formed to rotate around the same shaft as described above, it is possible to perform a pitch motion/yaw motion/actuation motion as well as stapling and cutting motions. This will be described in more detail later. Here, although the staple pulley 161, the pulley 111, and the pulley 121 are illustrated in the drawing as being formed to rotate around one rotation shaft 141, it is of course possible that each jaw pulley may be formed to be rotatable around a separate shaft that is concentric therewith.

[0388] In other words, it may also be described as a structure in which the first jaw pulley, pulley 111, the staple pulley 161, and the pulley 121 that is a second jaw pulley are sequentially stacked along the rotation shaft 141. Alternatively, it may be also described as a structure in which the staple pulley 161 is disposed between the pulley 111 and the pulley 121 facing each other. Here, the first jaw pulley 111, the staple pulley 161, and the second jaw pulley 121 may be formed to be rotatable independently of each other.

[0389] The staple auxiliary pulley 162 may be further provided on one side of the staple pulley 161. In other words, the staple auxiliary pulley 162 may be disposed between the staple pulley 161 and the pulley 163/pulley 164. The staple auxiliary pulley 162 may be formed to be rotatable independently of the pulley 112 and the pulley 122 around the rotation shaft 142. Here, although the staple auxiliary pulley 162, the pulley 112, and the pulley 122 are illustrated in the drawing as being formed to rotate around one rotation shaft 142, it is of course possible that each of the staple auxiliary pulley 162, the pulley 112, and the pulley 122 may be formed to be rotatable around a separate shaft that is concentric therewith. Such a staple auxiliary pulley will be described in more detail below.

[0390] The pulley 163 and the pulley 164 may function as staple pitch main pulleys, and the pulley 165 and the pulley 166 may function as staple pitch sub-pulleys.

[0391] Hereinafter, components related to the rotation of the staple pulley 161 will be described.

[0392] The pulley 163 and the pulley 164 function as the staple pitch main pulleys. Here, the wire 307, which is a staple wire, is wound around the pulley 163, and the wire 308, which is the staple wire, is wound around the pulley 164.

[0393] The pulley 165 and the pulley 166 function as staple pitch sub-pulleys. Here, the wire 307, which is a staple wire, is wound around the pulley 165, and the wire 308, which is a staple wire, is wound around the pulley 166.

[0394] Here, the pulley 163 and the pulley 164 are disposed on one side of the staple pulley 161 and the staple auxiliary pulley 162 to face each other. Here, the pulley 163 and the pulley 164 are formed to be rotatable independently of each other around the rotation shaft 143 that is an end tool pitch rotating shaft. In addition, the pulley 165 and the pulley 166 are respectively disposed on one sides of the pulley 163 and the pulley 164 to face each other. Here, the pulley 165 and the pulley 166 are formed to be rotatable independently of each other around the rotation shaft 144 which is an end tool pitch auxiliary rotating shaft. Here, in the drawings, it is illustrated that all of the pulley 163, the pulley 165, the pulley 164, and the pulley 166 are formed to be rotatable around the Y-axis

direction, but the concept of the present disclosure is not limited thereto, and the rotating axes of the respective pulleys may be formed in various directions according to configurations thereof. [0395] As described above, the rotation shaft **141**, the rotation shaft **142**, the rotation shaft **143**, and the rotation shaft **144** may be arranged sequentially from the distal end **104** of the end tool **100** toward the proximal end **105**. Accordingly, the staple pulley **161**, the staple auxiliary pulley **162**, the pulley **163**/pulley **164**, the pulley **165**/pulley **166** may be arranged sequentially from the distal end **104** of the end tool **100** toward the proximal end **105**.

[0396] The wire **307**, which is a staple wire, is sequentially wound to make contact with at least portions of the pulley **165**, the pulley **163**, the staple auxiliary pulley **162**, and the staple pulley **161**. In addition, the wire **308** connected to the wire **307** by the coupling member (see **329** of FIG. **62**) is sequentially wound to make contact with at least portions of the staple pulley **161**, the staple auxiliary pulley **162**, the pulley **164**, and the pulley **166**.

[0397] In other words, the wires **307** and **308**, which are the staple wires, are sequentially wound to make contact with at least portions of the pulley **165**, the pulley **163**, the staple auxiliary pulley **162**, the staple pulley **161**, the staple auxiliary pulley **162**, the pulley **164**, and the pulley **166**, and the wires **307** and **308** are formed to move along the above pulleys while rotating the above pulleys.

[0398] Accordingly, when the wire **307** is pulled, the coupling member (see **329** of FIG. **62**) to which the wire **307** is coupled and the staple pulley **161** coupled to the coupling member **329** are rotated in one direction. In contrast, when the wire **308** is pulled, the coupling member (see **329** of FIG. **62**) to which the wire **308** is coupled and the staple pulley **161** coupled to the coupling member **329** are rotated in a direction opposite to the one direction.

[0399] Hereinafter, the staple auxiliary pulley **162** will be described in more detail.

[0400] The staple auxiliary pulley **162** may serve to increase the rotation angle of the staple pulley **161** by coming into contact with the wires **307** and **308**, which are staple wires, to change the arrangement path of the wires **307** and **308** to a certain extent.

[0401] That is, when the staple auxiliary pulley is not disposed, the staple pulley may be rotated only up to a right angle, but in an embodiment of the present disclosure, by additionally providing the staple auxiliary pulley **162**, which is an auxiliary pulley, the maximum rotation angle may be increased by θ in both directions. This allows the staple pulley **161** to rotate for the stapling and cutting motions while two jaws of the end tool **100** are yaw-rotated together by 90° , thus enabling a linear motion of an operation member **540** to be described later. In other words, a feature of increasing the range of yaw rotation in which stapling and cutting motions are possible may be obtained through the staple auxiliary pulley **162**.

[0402] This will be described below in more detail.

[0403] In the case of the surgical instrument **10** of the present disclosure, the staple auxiliary pulley **162** is further disposed on one side of the staple pulley **161**. By changing the arrangement path of the wires **307** and **308**, which are staple wires, to a certain extent by disposing the staple auxiliary pulley **162** as described above, the tangential direction of the wires **307** and **308** is changed, and thus the rotation angle of the coupling member (see **329** of FIG. **62**) that couple the wires **307** and **308** to the staple pulley **161** is increased. That is, the coupling member (see **329** of FIG. **62**), which is a coupling part of the wires **307** and **308** and the staple pulley **161**, is rotatable until the coupling member **329** is located on the common internal tangent of the staple pulley **161** and the staple auxiliary pulley **122**.

[0404] In other words, the wires **307** and **308** are located on the internal tangent of the staple pulley **161** and the staple auxiliary pulley **162**, and the rotation angle of the staple pulley **161** is increased by the staple auxiliary pulley **162**.

[0405] According to the present disclosure, as the rotation radius of the staple pulley **161** increases, a yaw motion range in which normal stapling and cutting motions are performed may be increased. (Staple Drive Assembly)

[0406] Hereinafter, a staple drive assembly **150** will be described in more detail.

[0407] Referring to FIG. **13** and the like, the staple drive assembly **150** may include the staple pulley assembly **160** and the staple link assembly **170**. Here, the staple drive assembly **150** is connected to a reciprocating assembly **550** of a cartridge **500** to be described later, and converts a rotational motion of the staple pulley **161** into a linear motion of the reciprocating assembly **550**. In other embodiments of the present disclosure, which will be described later, the staple drive assembly may be understood as a concept including the staple pulley assembly and the staple link assembly.

[0408] The staple pulley assembly **160** may include one or more staple pulleys **161**. The staple pulley assembly **160** may be formed between the pulley **111** and the pulley **121** to be adjacent to each of the pulley **111** and the pulley **121**. In the present embodiment, it is assumed that the staple pulley assembly **160** includes one staple pulley **161**.

[0409] A shaft pass-through part **161a** may be formed in the staple pulley **161**. The shaft pass-through part **161a** may be formed in the form of a hole, and the rotation shaft **141**, which is an end tool jaw pulley rotation shaft, may be inserted through the shaft pass-through part **161a**. In addition, a link coupling part **161b** may be formed on the staple pulley **161**. The staple link assembly **170** to be described later may be coupled to the link coupling part **161b**. This will be described in more detail later.

[0410] Meanwhile, the end tool **100** of the first embodiment of the present disclosure may further include the staple link assembly **170** connected to the staple pulley assembly **160**. The staple link assembly **170** may include one or more link members **171**. The staple link assembly **170** may serve to connect the staple pulley assembly **160** to the reciprocating assembly **550** of the cartridge **500** to be described later. In the present embodiment, it is assumed that the staple link assembly **170** includes one link member **171**, and the link member **171** includes a first link **172** and a second link **173**.

[0411] The first link **172** is formed in the form of an elongated bar, which may have through holes formed at both end portions. The link coupling part **161b** of the staple pulley **161** may be inserted through the through hole at one end portion of the first link **172**. The second link **173** may be inserted through the through hole at the other end portion of the first link **172**.

[0412] The second link **173** is formed in the form of an elongated bar, and may be coupled to the first link **172**. The second link **173** may include a first protrusion **173a**, a second protrusion **173b**, and a coupling part **173c**.

[0413] In detail, the first protrusion **173a** may be formed at one end portion of the second link **173**. The first protrusion **173a** is axially coupled to the first link **172** by being fitted into the through hole of the first link **172**, so that the second link **173** may be coupled to the first link **172**. In addition, the first protrusion **173a** may be fitted into a guide groove **101b** of the first jaw **101**, which will be described later.

[0414] Meanwhile, the second protrusion **173b** may be formed in one region of a central portion of the second link **173**. The second protrusion **173b** may be fitted into the guide groove **101b** of the first jaw **101**, which will be described later.

[0415] As described above, as the first protrusion **173a** and the second protrusion **173b** are moved along the guide groove **101b** in a state in which the first protrusion **173a** and the second protrusion **173b** of the second link **173** formed in a protruding shape are fitted into the groove-shaped guide groove **101b**, the staple link assembly **170** is moved with respect to the first jaw **101** (and the cartridge **500** therein). This will be described in more detail later.

[0416] Meanwhile, the coupling part **173c** may be formed at the other end portion of the second link **173**. The coupling part **173c** may be coupled to a coupling part **551a** of the reciprocating assembly **550** of the cartridge **500**, which will be described later.

[0417] In the state of FIG. **13**, when the staple pulley **161** is rotated in the clockwise direction, the link member **171** connected to the staple pulley **161** may be moved as a whole toward the distal

end (see **101f** of FIG. **14**) of the first jaw **101**. In contrast, when the staple pulley **161** is rotated in the counterclockwise direction, the link member **171** connected to the staple pulley **161** may be moved as a whole toward the proximal end (see **101g** of FIG. **14**) of the first jaw **101**.

[0418] Thus, a bidirectional rotational motion of the staple pulley assembly **160** causes a reciprocating linear motion of the reciprocating assembly **550** of the cartridge **500** through the staple link assembly **170**. This will be described in more detail later.

(First and Second Jaws and Actuation Motion)

[0419] Hereinafter, the coupling structure of the first jaw **101** and the second jaw **102** of the end tool **100** of the surgical instrument **10** of FIG. **2** will be described in more detail.

[0420] FIG. **14** is a plan view illustrating the first jaw of the surgical instrument of FIG. **2**, and FIG. **15** is a plan view illustrating the second jaw of the surgical instrument of FIG. **2**. FIG. **16** is a plan view illustrating opening and closing motions of the first jaw of the surgical instrument of FIG. **2**, FIG. **17** is a plan view illustrating opening and closing motions of the second jaw of the surgical instrument of FIG. **2**, and FIG. **18** is a plan view illustrating opening and closing motions of the first jaw and the second jaw of the surgical instrument of FIG. **2**. FIG. **19** is a perspective view illustrating opening and closing motions of the end tool of the surgical instrument of FIG. **2**, and FIG. **20** is a plan view illustrating opening and closing motions of the end tool of the surgical instrument of FIG. **2**.

[0421] Referring to FIGS. **9** to **20** and the like, the first jaw **101** includes a cartridge accommodation part **101a**, the guide groove **101b**, a movable-coupling-allowed hole **101c**, a jaw pulley coupling hole **101d**, and a shaft pass-through part **101e**.

[0422] The first jaw **101** is formed entirely in the shape of an elongated bar, the cartridge **500** is accommodated in the distal end **101f** side, and the pulley **111** is coupled to the proximal end **101g**, so that the first jaw **101** is formed to be rotatable around the rotation shaft **141**. In other words, the first jaw **101** may be formed entirely in the form of a hollow box with one surface (upper surface) thereof is removed, such that the cartridge accommodation part **101a** capable of accommodating the cartridge **500** may be formed inside the first jaw **101**. That is, the first jaw **101** may be formed in an approximately “U” shape in cross section.

[0423] The guide groove **101b** configured to guide the movement of the staple link assembly **170**, which will be described later, may be formed on one side of the cartridge accommodation part **101a** of the first jaw **101**, e.g., on the proximal end **101g** side. The guide groove **101b** may be formed in the shape of a groove formed along a moving path of the staple link assembly **170**. In addition, as the first protrusion **173a** and the second protrusion **173b** are moved along the guide groove **101b** in a state in which the first protrusion **173a** and the second protrusion **173b** of the second link **173** formed in a protruding shape are fitted into the groove-shaped guide groove **101b**, the staple link assembly **170** is moved with respect to the first jaw **101** (and the cartridge **500** therein). That is, the staple link assembly **170** may be moved along the guide groove **101b** of the first jaw **101**.

[0424] Meanwhile, the movable-coupling-allowed hole **101c**, the jaw pulley coupling hole **101d**, and the shaft pass-through part **101e** may be formed on the proximal end side of the first jaw **101**.

[0425] Here, the movable-coupling-allowed hole **101c** may be formed to have a predetermined curvature, and may be formed in an approximately elliptical shape. A shaft coupling part **111a** of the pulley **111**, which will be described later, may be fitted into the movable-coupling-allowed hole **101c**. Here, a short radius of the movable-coupling-allowed hole **101c** may be formed to be substantially the same as or slightly greater than a radius of the shaft coupling part **111a**.

Meanwhile, a long radius of the movable-coupling-allowed hole **101c** may be formed to be greater than the radius of the shaft coupling part **111a**. Thus, in a state in which the shaft coupling part **111a** of the pulley **111** is fitted into the movable-coupling-allowed hole **101c** of the first jaw **101**, the shaft coupling part **111a** is movable to a certain extent in the movable-coupling-allowed hole **101c**. This will be described in more detail below.

[0426] Meanwhile, the jaw pulley coupling hole **101d** is formed in the form of a cylindrical hole,

and a jaw coupling part **111b** of the pulley **111**, which will be described later, may be fitted into the jaw pulley coupling hole **101d**. Here, a radius of the jaw pulley coupling hole **101d** may be formed to be substantially the same as or slightly greater than a radius of the jaw coupling part **111b**. Thus, the jaw coupling part **111b** of the pulley **111** may be formed to be rotatably coupled to the jaw pulley coupling hole **101d** of the first jaw **101**. This will be described in more detail below.

[0427] The shaft pass-through part **101e** may be formed at the distal end **101f** side of the first jaw **101** relative to the movable-coupling-allowed hole **101c** and the jaw pulley coupling hole **101d**. The shaft pass-through part **101e** may be formed in the form of a hole, and the rotation shaft **145**, which is a jaw rotation shaft, may be inserted through the shaft pass-through part **101e**.

[0428] The second jaw **102** includes an anvil **102a**, a movable-coupling-allowed hole **102c**, a jaw pulley coupling hole **102d**, and a shaft pass-through part **102e**.

[0429] The second jaw **102** is formed entirely in the shape of an elongated bar, the anvil **102a** is formed on a distal end **102f** side, and the pulley **112** is coupled to a proximal end **102g**, so that the second jaw **102** is formed to be rotatable around the rotation shaft **141**.

[0430] In detail, the anvil **102a** is formed in the form of a flat plane, on one surface of which shapes corresponding to the shapes of staples **530** to be described later may be formed. The above-described anvil **102a** may serve as a support for supporting the staple **530** on the opposite side of the operation member **540** when the operation member **540** pushes and raises the staple **530** during a stapling motion, so that the staple **530** is bent.

[0431] Meanwhile, the movable-coupling-allowed hole **102c**, the jaw pulley coupling hole **102d**, and the shaft pass-through part **102e** may be formed on the proximal end side of the second jaw **102**.

[0432] Here, the movable-coupling-allowed hole **102c** may be formed to have a predetermined curvature, and may be formed in an approximately elliptical shape. A shaft coupling part **121a** of the pulley **121**, which will be described later, may be fitted into the movable-coupling-allowed hole **102c**. Here, a short radius of the movable-coupling-allowed hole **102c** may be formed to be substantially the same as or slightly greater than a radius of the shaft coupling part **121a**. Meanwhile, a long radius of the movable-coupling-allowed hole **102c** may be formed to be greater than the radius of the shaft coupling part **121a**. Thus, in a state in which the shaft coupling part **121a** of the pulley **121** is fitted into the movable-coupling-allowed hole **102c** of the second jaw **102**, the shaft coupling part **121a** is movable to a certain extent in the movable-coupling-allowed hole **102c**. This will be described in more detail below.

[0433] Meanwhile, the jaw pulley coupling hole **102d** is formed in the form of a cylindrical hole, and a jaw coupling part **121b** of the pulley **121**, which will be described later, may be fitted into the jaw pulley coupling hole **102d**. Here, a radius of the jaw pulley coupling hole **102d** may be formed to be substantially the same as or slightly greater than a radius of the jaw coupling part **121b**. Thus, the jaw coupling part **121b** of the pulley **121** may be formed to be rotatably coupled to the jaw pulley coupling hole **102d** of the second jaw **102**. This will be described in more detail below.

[0434] Meanwhile, the shaft pass-through part **102e** may be formed at the distal end **102g** side of the second jaw **102** relative to the movable-coupling-allowed hole **102c** and the jaw pulley coupling hole **102d**. The shaft pass-through part **102e** may be formed in the form of a hole, and the rotation shaft **145**, which is the jaw rotation shaft, may be inserted through the shaft pass-through part **102e**.

[0435] The pulley **111**, which is a first jaw pulley, may include the shaft coupling part **111a** and the jaw coupling part **111b**. The pulley **111** is formed entirely in the form of a rotatable disk, and the shaft coupling part **111a** and the jaw coupling part **111b** may be formed to protrude to a certain extent from one surface of the pulley **111**. As described above, the shaft coupling part **111a** of the pulley **111** may be fitted into the movable-coupling-allowed hole **101c** of the first jaw **101**, and the jaw coupling part **111b** of the pulley **111** may be fitted into the jaw pulley coupling hole **101d** of the first jaw **101**. The pulley **111** may be formed to be rotatable around the center of the rotation shaft

141, which is an end tool jaw pulley rotation shaft.

[0436] Meanwhile, the pulley **121**, which is a second jaw pulley, may include the shaft coupling part **121a** and the jaw coupling part **121b**. The pulley **121** is formed entirely in the form of a rotatable disk, and the shaft coupling part **121a** and the jaw coupling part **121b** may be formed to protrude to a certain extent from one surface of the pulley **121**. As described above, the shaft coupling part **112a** of the pulley **112** may be inserted into the movable-coupling-allowed hole **102c** of the second jaw **102**, and the jaw coupling part **112b** of the pulley **112** may be inserted into the jaw pulley coupling hole **102d** of the second jaw **102**. The pulley **121** may be formed to be rotatable around the center of the rotation shaft **141**, which is an end tool jaw pulley rotation shaft.

[0437] The coupling relationship between the components described above is as follows.

[0438] The rotation shaft **141**, which is an end tool jaw pulley rotation shaft, is sequentially inserted through the shaft coupling part **111a** of the pulley **111**, the movable-coupling-allowed hole **101c** of the first jaw **101**, the shaft pass-through part **161a** of the staple pulley **161**, the movable-coupling-allowed hole **102c** of the second jaw **102**, and the shaft coupling part **121a** of the pulley **121**.

[0439] The rotation shaft **145**, which is a jaw rotation shaft, is sequentially inserted through the shaft pass-through part **101e** of the first jaw **101** and the shaft pass-through part **102e** of the second jaw **102**.

[0440] The shaft coupling part **111a** of the pulley **111** is fitted into the movable-coupling-allowed hole **101c** of the first jaw **101**, and the jaw coupling part **111b** of the pulley **111** is fitted into the jaw pulley coupling hole **101d** of the first jaw **101**.

[0441] At this time, the jaw pulley coupling hole **101d** of the first jaw **101** and the jaw coupling part **111b** of the pulley **111** are axially coupled to each other so as to be rotatable, and the movable-coupling-allowed hole **101c** of the first jaw **101** and the shaft coupling part **111a** of the pulley **111** are movably coupled to each other.

[0442] The shaft coupling part **121a** of the pulley **121** is fitted into the movable-coupling-allowed hole **102c** of the second jaw **102**, and the jaw coupling part **121b** of the pulley **121** is fitted into the jaw pulley coupling hole **102d** of the second jaw **102**.

[0443] At this time, the jaw pulley coupling hole **102d** of the second jaw **102** and the jaw coupling part **121b** of the pulley **121** are axially coupled to each other to be rotatable, and the movable-coupling-allowed hole **102c** of the second jaw **102** and the shaft coupling part **121a** of the pulley **121** are movably coupled to each other.

[0444] Here, the pulley **111** and the pulley **121** are rotated around the rotation shaft **141**, which is an end tool jaw pulley rotation shaft. The first jaw **101** and the second jaw **102** are rotated around the rotation shaft **145**, which is a jaw rotation shaft. That is, the pulley **111** and the first jaw **101** have different axes of rotation. Similarly, the pulley **121** and the second jaw **102** have different axes of rotation.

[0445] That is, the rotation angle of the first jaw **101** is limited to a certain extent by the movable-coupling-allowed hole **101c**, but is essentially rotated around the rotation shaft **145**, which is a jaw rotation shaft. Similarly, the rotation angle of the second jaw **102** is limited to a certain extent by the movable-coupling-allowed hole **102c**, but is essentially rotated around the rotation shaft **145**, which is a jaw rotation shaft.

[0446] Amplification of grip force due to the coupling relationship between the above-described components will be described.

[0447] In the surgical instrument **10** according to an embodiment of the present disclosure, the coupling structure of the first jaw **101** and the second jaw **102** forms an X-shaped structure, and thus, when the first jaw **101** and the second jaw **102** are rotated in directions close to each other (i.e., when the first jaw **101** and the second jaw **102** are closed), the grip force is greater in a direction in which the first jaw **101** and the second jaw **102** are closed. This will be described below in more detail.

[0448] As described above, in motions of the first jaw **101** and the second jaw **102** being opened and closed, there are two axes that become the center of rotation of the jaws. That is, the first jaw **101** and the second jaw **102** perform opening and closing motions around two axes of the rotation shaft **141** and the rotation shaft **145**. In this case, the center of rotation of the first jaw **101** and the second jaw **102** is the rotation shaft **145**, and the center of rotation of the pulley **111** and the pulley **121** is the rotation shaft **141**. At this time, the rotation shaft **141** is a shaft whose position is relatively fixed, and the rotation shaft **145** is a shaft whose position is moved linearly. In other words, when the pulley **111** and the pulley **121** are rotated in a state in which the position of the rotation shaft **141** is fixed, the first jaw **101** and the second jaw **102** are opened/closed while the rotation shaft **145**, which is a rotation shaft of the first jaw **101** and the second jaw **102**, is moved backward and forward. This will be described below in more detail.

[0449] In FIG. **17**, r_1 is a distance from the jaw coupling part **121b** of the pulley **121** to the shaft coupling part **121a**, and a length thereof is constant. Thus, a distance from the rotation shaft **141** inserted into the shaft coupling part **121a** to the jaw coupling part **121b** is also constant as r_1 .

[0450] Meanwhile, r_2 of FIG. **17** is a distance from the jaw pulley coupling hole **102d** of the second jaw **102** to the shaft pass-through part **102e**, and a length thereof is constant. Thus, a distance from the jaw coupling part **121b** of the pulley **121** inserted into the jaw pulley coupling hole **102d** to the rotation shaft **145** inserted into the shaft pass-through part **102e** is also constant as r_2 .

[0451] That is, the lengths of r_1 and r_2 are kept constant. Accordingly, when the pulley **111** and the pulley **121** are rotated in the direction of an arrow **A1** of FIG. **16** and the direction of an arrow **A2** of FIG. **17**, respectively, around the rotation shaft **141** to perform a closing motion, the first jaw **101** and the second jaw **102** are rotated around the rotation shaft **145** as the angle between r_1 and r_2 changes while the lengths of r_1 and r_2 remain constant, and at this time, the rotation shaft **145** itself is also linearly moved (i.e., is moved forward/backward) by as much as the arrow **B1** of FIG. **16** and the arrow **B2** of FIG. **17**.

[0452] That is, when it is assumed that the position of the rotation shaft **141**, which is an end tool jaw pulley rotation shaft, is fixed, when the first jaw **101** and the second jaw **102** are closed, a force is applied in the direction in which the rotation shaft **145**, which is a jaw rotation shaft, is moved forward (i.e., toward the distal end), and thus the grip force in the direction in which the first jaw **101** and the second jaw **102** are closed becomes larger.

[0453] In other words, since the lengths of r_1 and r_2 remain constant when the second jaw **102** is rotated around the jaw rotation shaft **145**, when the pulley **121** is rotated around the rotation shaft **141**, the angle between r_1 and r_2 changes while the lengths of r_1 and r_2 remain constant. That is, θ_2 , which is the angle between r_1 and r_2 when the second jaw **102** is closed as shown in FIG. **17A**, is greater than θ_1 , which is the angle between r_1 and r_2 when the second jaw **102** is opened as shown in FIG. **17B**.

[0454] Accordingly, when the second jaw **102** is rotated from the open state to the close state, the angle between r_1 and r_2 changes, and a force in a direction in which the rotation shaft **145** is moved forward is applied.

[0455] In this case, since the rotation shaft **141** is a shaft whose position is relatively fixed, the jaw rotation shaft **145** is moved forward in the direction of the arrow **B1** of FIG. **16** and the direction of the arrow **B2** of FIG. **17**, and the grip force is further increased in a direction in which the second jaw **102** is closed.

[0456] In other words, when the pulley **111** and the pulley **121** are rotated around the rotation shaft **141**, which is a shaft whose relative position is fixed, the angle θ between r_1 and r_2 changes while the distance between r_1 and r_2 remains constant. In addition, when the angle θ changes as described above, the first jaw **101** and the second jaw **102** push or pull the rotation shaft **145**, and thus the rotation shaft **145** is moved forward or backward. In this case, when the first jaw **101** and the second jaw **102** are rotated in the direction of closing, the grip force is further increased as the

rotation shaft **145** is moved forward in the direction of the arrow **B1** of FIG. **16** and the direction of the arrow **B2** of FIG. **17**. In contrast, when the first jaw **101** and the second jaw **102** are rotated in the direction of opening, the rotation shaft **145** is moved backward in directions opposite to the arrow **B1** of FIG. **16** and the direction of the arrow **B2** of FIG. **17**.

[0457] With this configuration, the grip force becomes stronger when the first jaw **101** and the second jaw **102** are closed, thereby enabling a surgical operator to perform the actuation motion powerfully even with a small force.

(Cartridge)

[0458] Hereinafter, the cartridge **500** of the surgical instrument **10** of FIG. **2** will be described in more detail.

[0459] FIG. **21** is a perspective view illustrating the first jaw and the cartridge of the surgical instrument of FIG. **2**. FIG. **22** is an exploded perspective view illustrating the cartridge of FIG. **21**, FIG. **23** is a coupled perspective view illustrating the cartridge of FIG. **21**, FIG. **24** is a side view illustrating the cartridge of FIG. **21**, FIG. **25** is a perspective cross-sectional view illustrating the cartridge of FIG. **21**, and FIG. **26** is a side cross-sectional view illustrating the cartridge of FIG. **21**. FIGS. **27** and **28** are perspective views illustrating the operation member of the cartridge of FIG. **21**. FIG. **29** is a side cross-sectional view illustrating a stapling-related structure of the end tool of the surgical instrument of FIG. **2**, and FIGS. **30** and **31** are perspective cross-sectional views illustrating a stapling structure of the end tool of the surgical instrument of FIG. **2**. FIGS. **32** to **35** are perspective views illustrating a ratchet drive operation of the end tool of FIG. **30**, and FIGS. **36** and **37** are plan views illustrating a ratchet drive operation of the end tool of FIG. **30**. FIG. **38** is a perspective view illustrating an entire ratchet drive operation of the end tool of FIG. **30**. FIGS. **39** and **40** are perspective views illustrating an entire stapling motion of the end tool of FIG. **30**.

[0460] Referring to FIGS. **21** to **40** and the like, the cartridge **500** is formed to be mountable to and dismountable from the first jaw **101**, and includes a plurality of staples **530** and a blade **542** therein to perform suturing and cutting tissue. Here, the cartridge **500** may include a cover **510**, a housing **520**, the staples **530**, withdrawal members **535**, the operation member **540**, and the reciprocating assembly **550**.

[0461] The housing **520** forms an outer shape of the cartridge **500**, and may be formed entirely in the form of a hollow box with one surface (upper surface) thereof removed to accommodate the reciprocating assembly **550**, the operation member **540**, and the staple **530** therein. Here, the housing **520** may be formed in an approximately “U” shape in cross section.

[0462] The cover **510** is formed to cover an upper portion of the housing **520**. Staple holes **511** through which the plurality of staples **530** may be ejected to the outside may be formed in the cover **510**. As the staples **530**, which are accommodated inside the housing **520** before a stapling operation, are pushed and raised upward by the operation member **540** during a stapling motion, and pass through the staple holes **511** of the cover **510** to be withdrawn to the outside of the cartridge **500**, stapling is performed.

[0463] Meanwhile, a slit **512** may be formed in the cover **510** along a length direction of the cover **510**. The blade **542** of the operation member **540** may protrude out of the cartridge **500** through the slit **512**. As the blade **542** of the operation member **540** passes along the slit **512**, staple-completed tissue may be cut.

[0464] The plurality of staples **530** may be disposed inside the housing **520**. As the operation member **540**, which will be described later, is linearly moved in one direction, the plurality of staples **530** are sequentially pushed and raised from the inside of the housing **520** to the outside, thereby performing sealing, that is, stapling. Here, the staples **530** may be made of a material that may include titanium, stainless steel, or the like.

[0465] Meanwhile, the withdrawal member **535** may be further disposed between the housing **520** and the staple **530**. In other words, it may be said that the staple **530** is disposed above the withdrawal member **535**. In this case, the operation member **540** is linearly moved in one direction

to push and raise the withdrawal member **535**, and the withdrawal member **535** may push and raise the staple **530**.

[0466] As such, the operation member **540** may be described as pushing and raising the staples **530** in both the case in which the operation member **540** directly pushes and raises the staples **530** and the case in which the operation member **540** pushes and raises the withdrawal members **535** and the withdrawal members **535** pushes and raises the staples **530** (i.e., the operation member **540** indirectly pushes and raises the staples **530**).

[0467] The reciprocating assembly **550** may be disposed at an inner lower side of the housing **520**. The reciprocating assembly **550** may include one or more reciprocating members **551**. In the present embodiment, it is illustrated that one reciprocating member **551** is provided, but in embodiments to be described later, a plurality of reciprocating members **551** may be provided.

[0468] In the present embodiment, the reciprocating member **551** may be a rack. The reciprocating member **551** may include recesses **551b** and the coupling part **551a**. In detail, the reciprocating member **551** may be formed in the form of an elongated bar, and a plurality of recesses **551b** having a sawtooth shape may be formed on one surface thereof. The recess **551b** may be formed to be in contact with the operation member **540** to be described later, in particular, a ratchet member **543** of the operation member **540**. In other words, the reciprocating member **551** may include the plurality of recesses **551b** shaped to engage with ratchets **543a** of the ratchet member **543**.

[0469] Meanwhile, although not shown in the drawings, in addition to a rack shape, the reciprocating member **551** may be provided as various shapes of members, which are directly or indirectly connected to the staple pulley **161** and may perform a linear reciprocating motion according to a rotational motion of the staple pulley **161**. For example, the reciprocating member **551** may be in the form of a clutch in which recesses are not present.

[0470] Here, the reciprocating member **551** is not fixedly coupled to the other components of the cartridge **500**, and may be formed to be movable relatively to the other components of the cartridge **500**. That is, the reciprocating member **551** may perform a reciprocating linear motion with respect to the housing **520** and the cover **510** coupled to the housing **520**.

[0471] Meanwhile, in the reciprocating member **551**, the coupling part **551a** may be formed at the proximal end **501** side adjacent to the pulley **111**, and the coupling part **551a** may be fastened and coupled to the staple link assembly **170** of the end tool **100**. Thus, when the staple link assembly **170** performs a reciprocating linear motion in the extension direction (i.e., the Y-axis direction) of the connection part **400**, the reciprocating member **551** coupled thereto may also perform a reciprocating linear motion in the extension direction (i.e., the Y-axis direction) of the connection part **400**. This will be described in more detail later.

[0472] The operation member **540** may be disposed inside the housing **520**. The operation member **540** is formed to be in contact with the reciprocating member **551**, and may be formed to linearly move in one direction according to the reciprocating linear motion of the reciprocating member **551**. In other words, the operation member **540** interacts with the reciprocating member **551** to perform stapling and cutting motions while moving in the extension direction of the connection part **400**.

[0473] The operation member **540** may include a wedge **541**, the blade **542**, the ratchet member **543**, an elastic member **544**, and a body **545**.

[0474] The body **545** may be formed in the shape of an elongated square column, and forms a base of the operation member **540**.

[0475] The wedge **541** is formed on at least one side of the body **545**, and may be formed to have a predetermined inclined surface. That is, the wedge **541** may be formed to be inclined to a certain extent in the extension direction of the connection part **400**. In other words, the wedge **541** may be formed to have a greater height at a proximal end **501** side of the cartridge **500** than a distal end **502** side of the cartridge **500**. In the drawing, it is illustrated that two wedges **541** are formed on each side of the body **545**, but the concept of the present disclosure is not limited thereto, and the

wedge **541** may be formed in various numbers and shapes depending on the shape of the staple **530** or the withdrawal member **535** that is in contact with the wedge **541**.

[0476] The wedge **541** may be formed to be in contact with the withdrawal members **535** or the plurality of staples **530** in turn and may serve to sequentially push and raise the staples **530**. As shown in FIG. **40** to be described later and elsewhere herein, the operation member **540** may serve to withdraw the staples **530** to the outside of the cartridge **500** by sequentially pushing and raising the staples **530** while moving toward the distal end **502**.

[0477] The blade **542** may be formed on one side of the wedge **541**, more specifically, on one side of the wedge **541** at the proximal end **501** side. An edge **542a** formed to be sharp to cut tissue is formed in one region of the blade **542**. As at least a portion of the edge **542a** is withdrawn to the outside of the first jaw **101** and the cartridge **500**, tissue disposed between the first jaw **101** and the second jaw **102** may be cut. The edge **542a** of the blade **542** may be always withdrawn to the outside of the first jaw **101**. Alternatively, the edge **542a** of the blade **542** may normally be accommodated inside the first jaw **101** or inside the cartridge **500**, and may be withdrawn to the outside of the first jaw **101** only when the operation member **540** is moved in a length direction.

[0478] The ratchet member **543** is formed on one side of the wedge **541**, more specifically, below wedge **541**, and may be formed to face the reciprocating member **551** to be described later. The ratchet member **543** may be formed in the form of a bar and may include a plurality of ratchets **543a** on one surface. The operation member **540** is moved only in one direction (i.e., toward the distal end) with respect to the reciprocating member **551** by the ratchet member **543**. The ratchets **543a** of the ratchet member **543** may be formed to be in contact with the recess **551b** of the reciprocating member **551** described above.

[0479] The elastic member **544** is formed on one side of the body **545** or the wedge **541** and serves to apply a predetermined elastic force to the ratchet member **543**. In an example, one region of the elastic member **544** may be connected to the wedge **541** or the body **545**, and another region of the elastic member **544** may be connected to the ratchet member **543**, so that the elastic member **544** may connect the wedge **541** or the body **545** to the ratchet member **543**. Here, the elastic member **544** may apply an elastic force in a direction in which the ratchet member **543** comes into close contact with the reciprocating member **551**. To this end, the elastic member **544** may be formed in the form of a leaf spring, and may be provided in various forms capable of providing a predetermined elastic force to the ratchet member **543**, such as a coil spring, a dish spring, and the like.

[0480] Here, the ratchet **543a** of the ratchet member **543** may be formed such that a first surface **543a1** (specifically, at the distal end **502** side) is formed to have a gentle slope with a predetermined angle, and a second surface **543a2** (specifically, at the proximal end **501** side) is formed to be vertical or near vertical.

[0481] In addition, in order to be engaged with the ratchet **543a** of the ratchet member **543**, the recess **551b** of the reciprocating member **551** may also be formed such that a first surface **551b1** (specifically, at the distal end **502** side) is formed to have a gentle slope with a predetermined angle, and a second surface **551b2** (specifically, at the proximal end **501** side) is formed to be vertical or near vertical.

[0482] In a state in which the reciprocating member **551** and the ratchet member **543** are coupled to each other (or engaged or in close contact with each other), the inclined first surface **543a1** of the ratchet **543a** and the inclined first surface **551b1** of the recess **551b** may be formed to face each other (that is, in contact with each other). In addition, the vertically formed second surface **543a2** of the ratchet **543a** and the vertically formed second surface **551b2** of the recess **551b** may be disposed to face each other (i.e., in contact with each other).

[0483] With this configuration, in a state in which the ratchet **543a** and the recess **551b** are coupled to (or engaged with) each other, the ratchet **543a** and the recess **551b** may be allowed to move only in one direction, acting as a kind of ratchet.

[0484] In an example, when it is assumed that the reciprocating member 551 is in a fixed state, the operation member 540 is movable in a direction in which the second surface 543a2 and the second surface 551b2, which are vertically formed, are away from each other, but when the second surface 543a2 and the second surface 551b2 are in contact with each other, the operation member 540 is not movable in a direction in which the second surface 543a2 and the second surface 551b2 are closer to each other.

[0485] In other words, when the reciprocating member 551 is moved toward the distal end 502 in a state in which the reciprocating member 551 and the ratchet member 543 are coupled to each other (or engaged or in close contact with each other), the ratchet member 543 is moved together toward the distal end 502 by the reciprocating member 551. That is, the vertically formed second surface 551b2 of the reciprocating member 551 pushes the vertically formed second surface 543a2 of the operation member 540 such that the ratchet member 543 is moved together toward the distal end 502 by the reciprocating member 551.

[0486] In contrast, when the reciprocating member 551 is moved toward the proximal end 501 in a state in which the reciprocating member 551 and the ratchet member 543 are coupled to each other (or engaged or in close contact with each other), only the reciprocating member 551 is moved alone toward the proximal end 501 while the ratchet member 543 is fixed. That is, the inclined first surface 551b1 of the reciprocating member 551 is moved along the inclined first surface 543a1 of the operation member 540 in a state in which the operation member 540 is fixed, so that only the reciprocating member 551 is moved alone toward the proximal end 501.

[0487] Referring to FIGS. 34 to 37, when the reciprocating member 551 is moved toward (in the direction of an arrow K1 of FIGS. 35 and 37) of the proximal end 501 in the state of FIGS. 34 and 36, as the inclined first surface 551b1 of the reciprocating member 551 is moved along the inclined first surface 543a1 of the operation member 540, the ratchet member 543 is pushed as a whole in the direction of an arrow K2 of FIG. 35. In addition, at this time, the elastic member 544 is elastically deformed to a certain extent.

[0488] In this state, when the reciprocating member 551 is further moved toward the proximal end 501, and the inclined first surface 551b1 of the reciprocating member 551 is moved beyond an end of the inclined first surface 543a1 of the operation member 540, the recess 551b of the reciprocating member 551 meets the next ratchet 543a of the ratchet member 543. In this case, since the elastic member 544 applies an elastic force in a direction in which the ratchet member 543 comes into close contact with the reciprocating member 551, front surfaces of the reciprocating member 551 and the ratchet member 543 are brought into close contact with each other again.

[0489] As a result, the cartridge 500 is accommodated in the cartridge accommodation part 101a of the first jaw 101, and in this case, the reciprocating member 551 of the cartridge 500 is coupled to the staple link assembly 170 of the end tool 100. Accordingly, the rotational motion of the staple pulley 161 of the end tool 100 is converted into a linear motion of the reciprocating member 551 through the staple link assembly 170.

[0490] In this case, when the coupling part 551a of the reciprocating member 551 is connected to the staple pulley 161 through the staple link assembly 170, and the staple pulley 161 is rotated alternately in the clockwise/counterclockwise directions, the reciprocating member 551 may be repeatedly moved forward and backward. In addition, when the reciprocating member 551 is moved forward, the operation member 540 may be moved forward together with the reciprocating member 551, and when the reciprocating member 551 is moved backward, only the reciprocating member 551 may be moved backward and the operation member 540 may remain stationary in place. As the operation member 540 is moved forward while repeating this process, the staple 530 may be stapled by the wedge 541 while the blade 542 cuts stapled tissue.

[0491] This will be described in more detail as follows.

(Stapling and Cutting Motions)

[0492] Referring to FIG. 38, a method of driving the surgical instrument according to an

embodiment of the present disclosure is described as follows.

[0493] First, when the staple pulley **161** is rotated in one of the clockwise direction and the counterclockwise direction, the reciprocating assembly **550** of the staple link assembly **170** connected to the staple pulley **161** and the cartridge **500** connected to the staple link assembly **170** are moved toward the distal end **502** of the cartridge **500**.

[0494] In addition, when the reciprocating assembly **550** is moved toward the distal end **502** of the cartridge **500**, the operation member **540** in contact with the reciprocating assembly **550** is moved toward the distal end **502** of the cartridge **500** together with the reciprocating assembly **550**.

[0495] In addition, as the operation member **540** is moved toward the distal end **502** of the cartridge **500**, the blade **542** of the operation member **540** is moved toward the distal end **502** of the cartridge **500** while the operation member **540** ejects the staples **530** out of the cartridge **500**.

[0496] Meanwhile, when the staple pulley **161** is rotated in the other one of the clockwise direction and the counterclockwise direction, the staple link assembly **170** connected to the staple pulley **161** and the reciprocating assembly **550** of the cartridge **500** connected to the staple link assembly **170** are moved toward the proximal end **501** of the cartridge **500**, and in this case, the operation member **540** is stationary.

[0497] In addition, as the above operations are repeatedly performed, a stapling motion by the wedge **541** and a cutting motion by the blade **542** are simultaneously performed.

[0498] This will be described in more detail as follows.

[0499] First, in the state shown in FIG. **38A**, when the staple pulley **161** is rotated in the direction of an arrow **A1** (i.e., in the clockwise direction) as shown in FIG. **38B**, the staple link assembly **170** connected to the staple pulley **161** and the reciprocating member **551** coupled to the staple link assembly **170** are moved in the direction of an arrow **B1** (i.e., toward the distal end). In this state, since the reciprocating member **551** and the operation member **540** are in close contact with each other by the elastic member (see **544** of FIG. **37**), when the reciprocating member **551** is moved in the direction of the arrow **B1**, the operation member **540** is also moved in the direction of the arrow **B1** together with the reciprocating member **551**.

[0500] Next, as shown in FIG. **38C**, when the staple pulley **161** is further rotated in the direction of arrow **A2**, the staple link assembly **170** connected to the staple pulley **161**, the reciprocating member **551**, and the operation member **540** are further moved in the direction of the arrow **B2**.

[0501] In this state, when the staple pulley **161** stops rotating, as shown in FIG. **38D**, the staple link assembly **170**, the reciprocating member **551**, and the operation member **540** also stop moving.

[0502] In this state, as shown in FIG. **38E**, when the staple pulley **161** begins to rotate in the direction of an arrow **A3** (i.e., in the counterclockwise direction), the staple link assembly **170** connected to the staple pulley **161** and the reciprocating member **551** coupled to the staple link assembly **170** are moved in the direction of an arrow **B3** (i.e., toward the proximal end). In this state, due to the coupling structure of the ratchet member **543** and the reciprocating member **551**, even when the reciprocating member **551** is moved in the direction of the arrow **B3**, only the ratchet member **543** is repeatedly spaced apart from and in contact with the reciprocating member **551** to a certain extent as the elastic member **544** is repeatedly elastically deformed and restored in a state in which the overall position of the operation member **540** remains unchanged. That is, (referring to FIGS. **35** and **37**), when the reciprocating member **551** is moved in the direction of the arrow **B3**, the operation member **540** remains stationary in place when viewed in the X-axis direction.

[0503] As shown in FIG. **38F**, when the staple pulley **161** is further rotated in the direction of an arrow **A4**, only the staple link assembly **170** connected to the staple pulley **161** and the reciprocating member **551** are further moved in the direction of an arrow **B4**.

[0504] In this state, when the staple pulley **161** stops rotating, as shown in FIG. **38A**, the staple link assembly **170**, the reciprocating member **551**, and the operation member **540** also stop moving.

[0505] When the staple pulley **161** is alternately rotated in the clockwise/counterclockwise

directions while repeating the above process, the reciprocating member **551** is repeatedly moved forward and backward, and the operation member **540** repeats moving forward and stopping, and as a result, the operation member **540** is moved toward the distal end **502**. In addition, as the operation member **540** is moved toward the distal end **502**, a stapling motion by the wedge **541** and a cutting motion by the blade **542** are simultaneously performed.

[0506] Hereinafter, a stapling motion of the surgical instrument according to an embodiment of the present disclosure will be described.

[0507] FIG. **39** is a perspective view illustrating a stapling motion of the end tool of FIG. **30** for each section, and FIG. **40** is a perspective view illustrating an entire stapling motion of the end tool of FIG. **30**.

[0508] Referring to FIGS. **39** and **40**, in the state as shown in FIG. **39A**, as the operation member **540** is moved in the direction of an arrow **A1** of FIG. **39B**, the wedge **541** of the operation member **540** pushes and raises the withdrawal member **535**, and the withdrawal member **535** pushes and raises one side of a lower portion of the staple **530**. In addition, due thereto, the staple **530** is ejected to the outside of the first jaw **101** and the cartridge **500**.

[0509] In this state, when the operation member **540** is further moved in the direction of an arrow **A2** of FIG. **39C**, the ejected staple **530** is continuously pushed and raised by the operation member **540** while in contact with the anvil **102a** of the second jaw **102**, so that stapling is performed while both end portions of the staple **530** are bent.

[0510] As such motions are continuously performed, stapling is sequentially performed from the staple **530** on the proximal end **501** side to the staple **530** on the distal end **502** side among the plurality of staples **530**, as illustrated in FIG. **40**.

(Manipulation Part)

[0511] FIGS. **41** and **42** are perspective views illustrating the manipulation part of the surgical instrument of FIG. **2**. FIG. **43** is a view schematically illustrating only a configuration of pulleys and wires constituting joints of the surgical instrument illustrated in FIG. **2**.

[0512] Referring to FIGS. **2** to **42**, the manipulation part **200** of the surgical instrument **10** according to the first embodiment of the present disclosure includes a first handle **204** that a user can grip, the actuation manipulation part **203** configured to control an actuation motion of the end tool **100**, the yaw manipulation part **202** configured to control a yaw motion of the end tool **100**, and the pitch manipulation part **201** configured to control a pitch motion of the end tool **100**. Here, it is understood that only the components related to the pitch/yaw/actuation motions of the surgical instrument **10** are illustrated in FIGS. **41** and **42**.

[0513] In addition, the manipulation part **200** of the surgical instrument **10** may further include a staple manipulation part **260** configured to control the motion of the staple pulley assembly **160** of the end tool **100** to perform stapling and cutting motions.

[0514] The manipulation part **200** may include the pulley **210**, a pulley **211**, a pulley **212**, a pulley **213**, a pulley **214**, a pulley **215**, a pulley **216**, a pulley **217**, and a pulley **218** that are related to a rotational motion of the first jaw **101**. In addition, the manipulation part **200** may include the pulley **220**, a pulley **221**, a pulley **222**, a pulley **223**, a pulley **224**, a pulley **225**, a pulley **226**, a pulley **227**, and a pulley **228** that are related to a rotational motion of the second jaw **102**. In addition, the manipulation part **200** may include the pulley **231**, the pulley **232**, a pulley **233**, and a pulley **234** that are related to a pitch motion thereof. In addition, the manipulation part **200** may include a pulley **235**, which is a relay pulley disposed at some places along the bent part **402** of the connection part **400**.

[0515] Here, the pulleys facing each other are illustrated in the drawings as being formed parallel to each other, but the concept of the present disclosure is not limited thereto, and each of the pulleys may be variously formed with a position and a size suitable for the configuration of the manipulation part.

[0516] Further, the manipulation part **200** of the first embodiment of the present disclosure may

include a rotation shaft **241**, a rotation shaft **242**, the rotation shaft **243**, a rotation shaft **244**, a rotation shaft **245**, and the rotation shaft **246**. Here, the rotation shaft **241** may function as a manipulation part first jaw actuation rotation shaft, and the rotation shaft **242** may function as a manipulation part second jaw actuation rotation shaft. In addition, the rotation shaft **243** may function as a manipulation part yaw main rotation shaft, and the rotation shaft **244** may function as a manipulation part yaw sub-rotation shaft. In addition, the rotation shaft **245** may function as a manipulation part pitch sub-rotation shaft, and the rotation shaft **246** may function as a manipulation part pitch main rotation shaft.

[0517] The rotation shaft **241**/rotation shaft **242**, the rotation shaft **243**, the rotation shaft **244**, the rotation shaft **245**, and the rotation shaft **246** may be sequentially disposed from a distal end **205** of the manipulation part **200** toward a proximal end **206**.

[0518] Each of the rotation shafts **241**, **242**, **243**, **244**, **245**, and **246** may be fitted into one or more pulleys, which will be described in detail later.

[0519] The pulley **210** functions as a manipulation part first jaw actuation pulley, the pulley **220** functions as a manipulation part second jaw actuation pulley, and these components may also be collectively referred to as a manipulation part actuation pulley.

[0520] The pulley **211** and the pulley **212** function as manipulation part first jaw yaw main pulleys, the pulley **221** and the pulley **222** function as manipulation part second jaw yaw main pulleys, and these components may also be collectively referred to as a manipulation part yaw main pulley.

[0521] The pulley **213** and the pulley **214** function as manipulation part first jaw yaw sub-pulleys, the pulley **223** and the pulley **224** function as manipulation part second jaw yaw sub-pulleys, and these components may also be collectively referred to as a manipulation part yaw sub-pulley.

[0522] The pulley **215** and the pulley **216** function as manipulation part first jaw pitch sub-pulleys, the pulley **225** and the pulley **226** function as manipulation part second jaw pitch sub-pulleys, and these components may also be collectively referred to as a manipulation part pitch sub-pulley.

[0523] The pulley **217** and the pulley **218** function as manipulation part first jaw pitch main pulleys, and the pulley **227** and the pulley **228** function as manipulation part second jaw pitch main pulleys, and these components may also be collectively referred to as the manipulation part pitch main pulley.

[0524] The pulley **231** and the pulley **232** function as manipulation part pitch wire main pulleys, and the pulley **233** and the pulley **234** function as manipulation part pitch wire sub-pulleys.

[0525] The above components are categorized from the perspective of the manipulation part for each motion (pitch/yaw/actuation) as follows.

[0526] The pitch manipulation part **201** configured to control a pitch motion of the end tool **100** may include the pulley **215**, the pulley **216**, the pulley **217**, the pulley **218**, the pulley **225**, the pulley **226**, the pulley **227**, the pulley **228**, the pulley **231**, the pulley **232**, and the pulley **234**. In addition, the pitch manipulation part **201** may include the rotation shaft **245** and the rotation shaft **246**. In addition, the pitch manipulation part **201** may further include a pitch frame **208**.

[0527] The yaw manipulation part **202** configured to control a yaw motion of the end tool **100** may include the pulley **211**, the pulley **212**, the pulley **213**, the pulley **214**, the pulley **221**, the pulley **222**, the pulley **223**, and the pulley **224**. In addition, the yaw manipulation part **202** may include the rotation shaft **243** and the rotation shaft **244**. In addition, the yaw manipulation part **202** may further include a yaw frame **207**.

[0528] The actuation manipulation part **203** configured to control an actuation motion of the end tool **100** may include the pulley **210**, the pulley **220**, the rotation shaft **241**, and the rotation shaft **242**. In addition, the actuation manipulation part **203** may further include the first actuation manipulation part **251** and the second actuation manipulation part **256**.

[0529] Hereinafter, each component of the manipulation part **200** will be described in more detail.

[0530] The first handle **204** may be formed to be gripped by a user with the hand, and in particular, may be formed to be grasped by the user by wrapping the first handle **204** with his/her palm. In

addition, the actuation manipulation part **203** and the yaw manipulation part **202** are formed on the first handle **204**, and the pitch manipulation part **201** is formed on one side of the yaw manipulation part **202**. In addition, the other end portion of the pitch manipulation part **201** is connected to the bent part **402** of the connection part **400**.

[0531] The actuation manipulation part **203** includes the first actuation manipulation part **251** and the second actuation manipulation part **256**. The first actuation manipulation part **251** includes the rotation shaft **241**, the pulley **210**, the first actuation extension part **252**, and a first actuation gear **253**. The second actuation manipulation part **256** includes the rotation shaft **242**, the pulley **220**, the second actuation extension part **257**, and a second actuation gear **258**. Here, end portions of the first actuation extension part **252** and the second actuation extension part **257** are formed in the shape of a hand ring, which may act as a second handle.

[0532] Here, the rotation shaft **241** and the rotation shaft **242**, which are actuation rotation axes, may be formed to form a predetermined angle with an XY plane on which the connection part **400** is formed. For example, the rotation shaft **241** and the rotation shaft **242** may be formed in a direction parallel to the Z-axis, and in this state, when the pitch manipulation part **201** or the yaw manipulation part **202** is rotated, the coordinate system of the actuation manipulation part **203** may change relatively. Of course, the concept of the present disclosure is not limited thereto, and the rotation shaft **241** and the rotation shaft **242** may be formed in various directions so as to be suitable for a structure of the hand of the user gripping the actuation manipulation part **203** according to an ergonomic design.

[0533] Meanwhile, the pulley **210**, the first actuation extension part **252**, and the first actuation gear **253** are fixedly coupled to each other to be rotatable together around the rotation shaft **241**. Here, the pulley **210** may be configured to be a single pulley or two pulleys fixedly coupled to each other.

[0534] Similarly, the pulley **220**, the second actuation extension part **257**, and the second actuation gear **258** are fixedly coupled to each other to be rotatable together around the rotation shaft **242**. Here, the pulley **220** may be configured to be a single pulley or two pulleys fixedly coupled to each other.

[0535] Here, the first actuation gear **253** and the second actuation gear **258** are formed to be engaged with each other such that, when any one gear is rotated in one direction, the other gear is rotated together in a direction opposite to the one direction.

[0536] The yaw manipulation part **202** may include the rotation shaft **243**, the pulleys **211** and **212**, which are manipulation part first jaw yaw main pulleys, the pulleys **221** and **222**, which are manipulation part second jaw yaw main pulleys, and the yaw frame **207**. In addition, the yaw manipulation part **202** may further include the pulleys **213** and **214**, which are manipulation part first jaw yaw sub-pulleys formed on one side of the pulleys **211** and **212**, and the pulleys **223** and **224** that are manipulation part second jaw yaw sub-pulleys formed on one side of the pulleys **221** and **222**. Here, the pulleys **213** and **214** and the pulleys **223** and **224** may be coupled to the pitch frame **208** to be described later.

[0537] Here, it is illustrated in the drawings that the yaw manipulation part **202** includes the pulleys **211** and **212** and the pulleys **221** and **222**, wherein the pulleys **211** and **212** and the pulleys **221** and **222** are each provided with two pulleys formed to face each other and independently rotatable, but the concept of the present disclosure is not limited thereto. That is, one or more pulleys having the same diameter or different diameters may be provided according to the configuration of the yaw manipulation part **202**.

[0538] In detail, the rotation shaft **243**, which is a manipulation part yaw main rotation shaft, is formed on one side of the actuation manipulation part **203** on the first handle **204**. At this time, the first handle **204** is formed to be rotatable around the rotation shaft **243**.

[0539] Here, the rotation shaft **243** may be formed to form a predetermined angle with the XY plane on which the connection part **400** is formed. For example, the rotation shaft **243** may be formed in a direction parallel to the Z-axis, and in this state, when the pitch manipulation part **201**

is rotated, the coordinate system of the rotation shaft **243** may change relatively as described above. Of course, the concept of the present disclosure is not limited thereto, and the rotation shaft **243** may be formed in various directions so as to be suitable for a structure of the hand of the user gripping the manipulation part **200** according to an ergonomic design.

[0540] Meanwhile, the pulleys **211** and **212** and the pulleys **221** and **222** are coupled to the rotation shaft **243** so as to be rotatable around the rotation shaft **243**. In addition, the wire **301** or the wire **305**, which is a first jaw wire, is wound around the pulleys **211** and **212**, and the wire **302** or the wire **306**, which is a second jaw wire, may be wound around the pulleys **221** and **222**. In this case, the pulleys **211** and **212** and the pulleys **221** and **222** may each be configured as two pulleys formed to face each other and independently rotatable. Accordingly, a wire being wound and a wire being released may be wound around respective separate pulleys so that the wires may perform motions without interference with each other.

[0541] The yaw frame **207** rigidly connects the first handle **204**, the rotation shaft **241**, the rotation shaft **242**, and the rotation shaft **243**, so that the first handle **204**, the yaw manipulation part **202**, and the actuation manipulation part **203** are integrally yaw-rotated around the rotation shaft **243**.

[0542] The pitch manipulation part **201** may include the rotation shaft **246**, the pulley **217** and the pulley **218**, which are manipulation part first jaw pitch main pulleys, the pulleys **227** and **228**, which are manipulation part second jaw pitch main pulleys, and the pitch frame **208**. In addition, the pitch manipulation part **201** may further include the rotation shaft **245**, the pulleys **215** and **216**, which are manipulation part first jaw pitch sub-pulleys formed on one side of the pulley **217** and the pulley **218**, and the pulleys **225** and **226**, which are manipulation part second jaw pitch sub-pulleys formed on one side of the pulley **227** and the pulley **228**. The pitch manipulation part **201** may be connected to the bent part **402** of the connection part **400** through the rotation shaft **246**.

[0543] In detail, the pitch frame **208** is a base frame of the pitch manipulation part **201**, and the rotation shaft **243** is rotatably coupled to one end portion thereof. That is, the yaw frame **207** is formed to be rotatable around the rotation shaft **243** with respect to the pitch frame **208**.

[0544] As described above, since the yaw frame **207** connects the first handle **204**, the rotation shaft **243**, the rotation shaft **241**, and the rotation shaft **242**, and the yaw frame **207** is also axially coupled to the pitch frame **208**, when the pitch frame **208** is pitch-rotated around the rotation shaft **246**, the yaw frame **207** connected to the pitch frame **208**, the first handle **204**, the rotation shaft **241**, the rotation shaft **242**, and the rotation shaft **243** are pitch-rotated together. That is, when the pitch manipulation part **201** is rotated around the rotation shaft **246**, the actuation manipulation part **203** and the yaw manipulation part **202** are rotated together with the pitch manipulation part **201**. In other words, when a user pitch-rotates the first handle **204** around the rotation shaft **246**, the actuation manipulation part **203**, the yaw manipulation part **202**, and the pitch manipulation part **201** are moved together.

[0545] The pulleys **217** and **218** and the pulleys **227** and **228** are coupled to the rotation shaft **246** so as to be rotatable around the rotation shaft **246** of the pitch frame **208**.

[0546] Here, the pulley **217** and the pulley **218** may be formed to face each other so as to be independently rotatable. Accordingly, a wire being wound and a wire being released may be wound around respective separate pulleys so that the wires may perform motions without interference with each other. Similarly, the pulley **227** and the pulley **228** may also be formed to face each other so as to be independently rotatable. Accordingly, a wire being wound and a wire being released may be wound around respective separate pulleys so that the wires may perform motions without interference with each other.

[0547] Next, a motion of each of the wires **303** and **304**, which are pitch wires, is described as follows.

[0548] The pulley **131**, which is an end tool pitch pulley, is fixedly coupled to the end tool hub **180** in the end tool **100**, and the pulley **231** and the pulley **232**, which are manipulation part pitch pulleys, are fixedly coupled to the pitch frame **208** in the manipulation part **200**. In addition, these

pulleys are connected to each other by the wires **303** and **304**, which are pitch wires, so that a pitch motion of the end tool **100** may be performed more easily according to the pitch manipulation of the manipulation part **200**. Here, the wire **303** is fixedly coupled to the pitch frame **208** via the pulley **231** and the pulley **233**, and the wire **304** is fixedly coupled to the pitch frame **208** via the pulley **232** and the pulley **234**. That is, the pitch frame **208** and the pulleys **231** and **232** are rotated together around the rotation shaft **246** by the pitch rotation of the manipulation part **200**, and as a result, the wires **303** and **304** are also moved, and thus, a driving force of additional pitch rotation may be transmitted separately from the pitch motion of the end tool by the wire **301**, the wire **302**, the wire **305**, and the wire **306**, which are jaw wires.

[0549] A connection relationship of each of the first handle **204**, the pitch manipulation part **201**, the yaw manipulation part **202**, and the actuation manipulation part **203** is summarized as follows. The rotation shafts **241** and **242**, the rotation shaft **243**, the rotation shaft **244**, the rotation shaft **245**, and the rotation shaft **246** may be formed on the first handle **204**. In this case, since the rotation shafts **241** and **242** are directly formed on the first handle **204**, the first handle **204** and the actuation manipulation part **203** may be directly connected to each other. Meanwhile, since the rotation shaft **243** is directly formed on the first handle **204**, the first handle **204** and the yaw manipulation part **202** may be directly connected to each other. On the other hand, since the pitch manipulation part **201** is formed on one side of the yaw manipulation part **202** so as to be connected to the yaw manipulation part **202**, the pitch manipulation part **201** is not directly connected to the first handle **204**, and the pitch manipulation part **201** and the first handle **204** may be formed to be indirectly connected to each other via the yaw manipulation part **202**.

[0550] Continuing to refer to the drawings, in the surgical instrument **10** according to the first embodiment of the present disclosure, the pitch manipulation part **201** and the end tool **100** may be formed on the same or parallel axis (X-axis). That is, the rotation shaft **246** of the pitch manipulation part **201** is formed at one end portion of the bent part **402** of the connection part **400**, and the end tool **100** is formed at the other end portion of the connection part **400**.

[0551] In addition, one or more relay pulleys **235** configured to change or guide paths of the wires may be disposed at some places along the connection part **400**, particularly in the bent part **402**. As at least some of the wires are wound around the relay pulleys **235** to guide the paths of the wires, these wires may be disposed along a bent shape of the bent part **402**.

[0552] Here, in the drawings, it is illustrated that the connection part **400** is formed to be curved with a predetermined curvature by having the bent part **402**, but the concept of the present disclosure is not limited thereto, and the connection part **400** may be formed linearly or to be bent one or more times as necessary, and even in this case, it may be said that the pitch manipulation part **201** and the end tool **100** are formed on substantially the same axis or parallel axes. In addition, although FIG. 3 illustrates that each of the pitch manipulation part **201** and the end tool **100** is formed on an axis parallel to the X-axis, the concept of the present disclosure is not limited thereto, and the pitch manipulation part **201** and the end tool **100** may be formed on different axes.

[0553] The staple manipulation part **260** is connected to the staple pulley **161** of the end tool **100** by the wires **307** and **308**, which are staple wires, and serves to alternately rotate the staple pulley **161** in the clockwise or counterclockwise direction.

[0554] To this end, although not shown in the drawings, the staple manipulation part **260** may include a motor (not shown). That is, the motor (not shown) is driven while the user presses the staple manipulation part **260** formed in the form of a button to alternately rotate the manipulation part staple pulley (see **269** of FIG. 47) in the clockwise or counterclockwise direction. In addition, due thereto, the staple pulley **161** of the end tool **100** may be alternately rotated in the clockwise or counterclockwise direction.

(Actuation, Yaw, and Pitch Motions)

[0555] Actuation, yaw, and pitch motions in the present embodiment will be described as follows.

[0556] First, the actuation motion will be described below.

[0557] In a state in which a user inserts his/her index finger in the hand ring formed on the first actuation extension part **252** and his/her thumb in the hand ring formed on the second actuation extension part **257**, when the user rotates the actuation rotation parts **252** and **257** using one or both of his/her index finger and thumb, the pulley **210** and the first actuation gear **253** fixedly coupled to the first actuation extension part **252** are rotated around the rotation shaft **241**, and the pulley **220** and the second actuation gear **258** fixedly coupled to the second actuation extension part **257** are rotated around the rotation shaft **242**. At this time, the pulley **210** and the pulley **220** are rotated in opposite directions, and thus the wires **301** and **305** fixedly coupled to the pulley **210** at one end portion thereof and the wires **302** and **306** fixedly coupled to the pulley **220** at one end portion thereof are also moved in opposite directions. In addition, a rotating force is transmitted to the end tool **100** through the power transmission part **300**, and two jaws **103** of the end tool **100** perform an actuation motion.

[0558] Here, as described above, the actuation motion refers to a motion in which the two jaws **101** and **102** are splayed or closed while being rotated in opposite directions. That is, when the actuation rotation parts **252** and **257** of the actuation manipulation part **203** are rotated in directions close to each other, the first jaw **101** is rotated in the counterclockwise direction, and the second jaw **102** is rotated in the clockwise direction, thereby closing the end tool **100**. That is, when the actuation rotation parts **252** and **257** of the actuation manipulation part **203** are rotated in directions away from each other, the first jaw **121** is rotated in the counterclockwise direction, and the second jaw **122** is rotated in the clockwise direction, thereby opening the end tool **100**.

[0559] In the present embodiment, for the actuation manipulation described above, the first actuation extension part **252** and the second actuation extension part **257** are provided to configure the second handle and manipulated by gripping the second handle with two fingers. However, for the actuation manipulation in which the two jaws of the end tool **100** are opened or closed, the actuation manipulation part **203** may be configured in a manner different from the above-described manner, such as configuring the two actuation pulleys (the pulley **210** and the pulley **220**) to act in opposition to each other with an actuation rotation part.

[0560] Next, the yaw motion will be described below.

[0561] When a user rotates the first handle **204** around the rotation shaft **243** while holding the first handle **204**, the actuation manipulation part **203** and the yaw manipulation part **202** are yaw-rotated around the rotation shaft **243**. That is, when the pulley **210** of the first actuation manipulation part **251** to which the wires **301** and **305** are fixedly coupled is rotated around the rotation shaft **243**, the wires **301** and **305** wound around the pulleys **211** and **212** are moved. Similarly, when the pulley **220** of the second actuation manipulation part **256**, to which the wires **302** and **306** are fixedly coupled, is rotated around the rotation shaft **243**, the wires **302** and **306** wound around the pulleys **221** and **222** are moved. At this time, the wires **301** and **305** connected to the first jaw **101** and the wires **302** and **306** connected to the second jaw **102** are wound around the pulleys **211** and **212** and the pulleys **221** and **222**, so that the first jaw **101** and the second jaw **102** are rotated in the same direction during yaw rotation. In addition, a rotating force is transmitted to the end tool **100** through the power transmission part **300**, and thus a yaw motion in which two jaws **103** of the end tool **100** are rotated in the same direction is performed.

[0562] At this time, since the yaw frame **207** connects the first handle **204**, the rotation shaft **241**, the rotation shaft **242**, and the rotation shaft **243**, the first handle **204**, the yaw manipulation part **202**, and the actuation manipulation part **203** are rotated together around the rotation shaft **243**.

[0563] Next, the pitch motion will be described below.

[0564] When a user rotates the first handle **204** around the rotation shaft **246** while holding the first handle **204**, the actuation manipulation part **203**, the yaw manipulation part **202**, and the pitch manipulation part **201** are pitch-rotated around the rotation shaft **246**. That is, when the pulley **210** of the first actuation manipulation part **251** to which the wires **301** and **305** are fixedly coupled is rotated around the rotation shaft **246**, the wires **301** and **305** wound around the pulley **217** and the

pulley **218** are moved. Similarly, when the pulley **220** of the second actuation manipulation part **256**, to which the wires **302** and **306** are fixedly coupled, is rotated around the rotation shaft **246**, the wires **302** and **306** wound around the pulley **227** and the pulley **228** are moved. At this time, as described with reference to FIG. 5, in order to allow the first jaw **101** and the second jaw **102** to pitch-rotate, the wires **301** and **305**, which are first jaw wires, are moved in the same direction and respectively wound around the pulley **217** and the pulley **218**, which are manipulation part pitch main pulleys, and the wires **302** and **306**, which are second jaw wires, are moved in the same direction and respectively wound around the pulley **227** and the pulley **228**, which are manipulation part pitch main pulleys. In addition, a rotating force is transmitted to the end tool **100** through the power transmission part **300**, and two jaws **103** of the end tool **100** perform a pitch motion.

[0565] At this time, since the pitch frame **208** is connected to the yaw frame **207**, and the yaw frame **207** connects the first handle **204**, the rotation shaft **241**, the rotation shaft **242**, and the rotation shaft **243**, when the pitch frame **208** is rotated around the rotation shaft **246**, the yaw frame **207**, the first handle **204**, the rotation shaft **241**, the rotation shaft **242**, and the rotation shaft **243** connected to the pitch frame **208** are rotated together. That is, when the pitch manipulation part **201** is rotated around the rotation shaft **246**, the actuation manipulation part **203** and the yaw manipulation part **202** are rotated together with the pitch manipulation part **201**.

[0566] In summary, in the surgical instrument **10** according to an embodiment of the present disclosure, the pulleys are formed on respective joint points (an actuation joint, a yaw joint, and a pitch joint), the wires (the first jaw wire or the second jaw wire) are wound around the pulleys, the rotational manipulations (actuation rotation, yaw rotation, and pitch rotation) of the manipulation part cause the movement of each wire, which in turn induces the desired motion of the end tool **100**. Furthermore, the auxiliary pulley may be formed on one side of each of the pulleys, and the wire may not be wound several times around one pulley due to the auxiliary pulley.

[0567] FIG. **43** is a view schematically illustrating only a configuration of pulleys and wires constituting joints of the surgical instrument **10** according to an embodiment of the present disclosure illustrated in FIG. **2**. In FIG. **43**, the relay pulleys for changing paths of the wires and not related to the operation of joints are omitted.

[0568] Referring to FIG. **43**, the manipulation part **200** may include the pulley **210**, the pulley **211**, the pulley **212**, the pulley **213**, the pulley **214**, the pulley **215**, the pulley **216**, the pulley **217**, and the pulley **218** that are related to a rotational motion of the first jaw **101**.

[0569] In addition, the manipulation part **200** may include the pulley **220**, the pulley **221**, the pulley **222**, the pulley **223**, the pulley **224**, the pulley **225**, the pulley **226**, the pulley **227**, and the pulley **228** that are related to a rotational motion of the second jaw **122** (the arrangement and structure of each of the pulleys of the manipulation part **200** are the same in principle as the arrangement and structure of each of the pulleys of the end tool **100**, and thus specific designations of some reference numerals are omitted in the drawings).

[0570] The pulleys **211** and **212** and the pulleys **221** and **222** may be formed to be rotatable independently of each other around the same shaft, that is the rotation shaft **243**. In this case, the pulleys **211** and **212** and the pulleys **221** and **222** may each be formed as two pulleys formed to face each other and formed to be independently rotatable.

[0571] The pulleys **213** and **214** and the pulleys **223** and **224** may be formed to be rotatable independently of each other around the same shaft, that is the rotation shaft **244**. Here, the pulleys **213** and **214** may be formed as two pulleys formed to face each other and formed to be independently rotatable, and in this case, the two pulleys may be formed to have different diameters. Similarly, the pulleys **223** and **224** may be formed as two pulleys formed to face each other and formed to be independently rotatable, and in this case, the two pulleys may be formed to have different diameters.

[0572] The pulleys **215** and **216** and the pulleys **225** and **226** may be formed to be rotatable independently of each other around the same shaft, that is the rotation shaft **245**. In this case, the

pulleys **215** and **216** may be formed to have different diameters. In addition, the pulleys **225** and **226** may be formed to have different diameters.

[0573] The pulleys **217** and **218** and the pulleys **227** and **228** may be formed to be rotatable independently of each other around the same shaft, that is the rotation shaft **246**.

[0574] The wire **301** is wound around the pulley **210** after sequentially passing through the pulley **217**, the pulley **215**, the pulley **213**, and the pulley **211** of the manipulation part **200**, and then is coupled to the pulley **210** by the coupling member **324**. Meanwhile, the wire **305** sequentially passes through the pulley **218**, the pulley **216**, the pulley **214**, and the pulley **212** of the manipulation part **200** and is coupled to the pulley **210** by the coupling member **324**. Thus, when the pulley **210** is rotated, the wires **301** and **305** are wound around or released from the pulley **210**, and accordingly, the first jaw **101** is rotated.

[0575] The wire **306** is wound around the pulley **220** after sequentially passing through the pulley **227**, the pulley **225**, the pulley **223**, and the pulley **221** of the manipulation part **200**, and then is coupled to the pulley **220** by the coupling member **327**. Meanwhile, the wire **302** sequentially passes through the pulley **228**, the pulley **226**, the pulley **224**, and the pulley **222** of the manipulation part **200** and is coupled to the pulley **220** by the coupling member **327**. Thus, when the pulley **220** is rotated, the wire **302** and the wire **306** are wound around or released from the pulley **220**, and accordingly, the second jaw **102** is rotated.

(Conceptual Diagram of Pulleys and Wires)

[0576] FIGS. **45** and **46** are diagrams illustrating a configuration of pulleys and wires, which are related to an actuation motion and a yaw motion of the surgical instrument **10** according to an embodiment of the present disclosure illustrated in FIG. **2**, in detail for each of the first jaw and the second jaw. FIG. **45** is a diagram illustrating only pulleys and wires related to the second jaw, and FIG. **46** is a diagram illustrating only pulleys and wires related to the first jaw. In addition, FIG. **44** is a perspective view illustrating a yaw motion of the surgical instrument of FIG. **2**. Here, in FIG. **44**, components related to stapling and cutting motions are omitted.

[0577] First, a wire motion of the actuation motion will be described.

[0578] Referring to FIG. **46**, when the first actuation extension part **252** is rotated around the rotation shaft **241** in the direction of an arrow OPA **1**, the pulley **210** connected to the first actuation extension part **252** is rotated, and the wire **301** and the wire **305** wound around the pulley **210** are moved in directions W1a and W1b, respectively, and as a result, the first jaw **101** of the end tool **100** is rotated in the direction of an arrow EPA1.

[0579] Referring to FIG. **45**, when the second actuation extension part **257** is rotated around the rotation shaft **242** in the direction of an arrow OPA **2**, the pulley **220** connected to the second actuation extension part **257** is rotated, and thus both strands of the wires **302** and **306** wound around the pulley **220** are moved in directions W2a and W2b, respectively, and as a result, the second jaw **102** of the end tool **100** is rotated in the direction of an arrow EPA2. Accordingly, when a user manipulates the first actuation extension part **252** and the second actuation extension part **257** in directions close to each other, a motion of the first jaw **101** and the second jaw **102** of the end tool being close to each other is performed.

[0580] Next, a wire motion of the yaw motion will be described.

[0581] First, since the rotation shaft **243** is connected to the rotation shafts **241** and **242** by the yaw frame (see **207** of FIG. **30**), the rotation shaft **243** and the rotation shafts **241** and **242** are integrally rotated together.

[0582] Referring to FIG. **46**, when the first handle **204** is rotated around the rotation shaft **243** in the direction of an arrow OPY1, the pulley **210** and the pulleys **211** and **212** and the wires **301** and **305** wound therearound are rotated as a whole around the rotation shaft **243**, and as a result, the wires **301** and **305** wound around the pulleys **211** and **212** are moved in the directions W1a and W1b, respectively, which in turn causes the first jaw **101** of the end tool **100** to rotate in the direction of an arrow EPY1.

[0583] Referring to FIG. 45, when the first handle 204 is rotated around the rotation shaft 243 in the direction of an arrow OPY2, the pulley 220 and the pulleys 221 and 222 and the wires 302 and 306 wound therearound are rotated as a whole around the rotation shaft 243, and as a result, the wires 302 and 306 wound around the pulleys 221 and 222 are respectively moved in a direction opposite to a direction W1a and a direction opposite to a direction W1b, which in turn causes the first jaw 101 of the end tool 100 to rotate in the direction of an arrow EPY2.

[0584] FIGS. 47 to 49 are diagrams illustrating a configuration of pulleys and wires, which are related to stapling and cutting motions of the surgical instrument 10 according to an embodiment of the present disclosure illustrated in FIG. 2, in detail for each of the first jaw and the second jaw. Here, FIGS. 47 to 49 are drawings mainly illustrating pulleys and wires related to the second jaw. [0585] Here, FIGS. 47 and 48 illustrate the process of actuation motion of closing two jaws, and FIGS. 48 and 49 illustrate the process of motion of stapling and cutting of tissue interposed between two jaws.

[0586] First, a wire motion of the actuation motion will be described.

[0587] Referring to FIGS. 47 and 48, when the first actuation extension part 252 of the first actuation manipulation part 251 is rotated in the direction of an arrow OPA 1 around the rotation shaft 241, the pulley 210 connected to the first actuation extension part 252 is rotated, and each of the wire (see 301 of FIG. 43) and the wire (see 305 of FIG. 43) wound around the pulley 210 is moved, which in turn causes the first jaw 101 of the end tool 100 to rotate in the direction of an arrow EPA1.

[0588] At this time, the manipulation part staple pulley 269 of the staple manipulation part 260 is formed to be rotatable around the rotation shaft 241 together with the first actuation manipulation part 251. Thus, when the first actuation extension part 252 is rotated around the rotation shaft 241, the staple manipulation part 260 is also rotated around the rotation shaft 241 together with the first actuation manipulation part 251.

[0589] As a result, in the actuation motion, when the pulley 111 is rotated in the end tool 100, the staple pulley 161 is also rotated together with the pulley 111.

[0590] Next, a wire motion of the stapling and cutting motions will be described.

[0591] Referring to FIGS. 48 and 49, when the staple manipulation part 260 is rotated in the direction of an arrow OPC1 around a rotation shaft 247, which is a manipulation part cutting rotation shaft, the manipulation part staple pulley 269 and the wires 307 and 308 that are staple wires wound therearound are rotated around the rotation shaft 247, and as a result, each of the wires 307 and 308 wound around the manipulation part staple pulley 269 is moved, which in turn causes the staple pulley 161 of the end tool 100 to rotate in the direction of an arrow EPC1.

[0592] Meanwhile, when the staple manipulation part 260 is rotated, the manipulation part staple pulley 269 is rotated around the rotation shaft 247, and at this time, the rotation of the staple manipulation part 260 does not affect the first actuation manipulation part 251.

[0593] As a result, when the manipulation part staple pulley 269 is rotated, the staple pulley 161 of the end tool 100 is rotated alone independent of the first jaw 101. In addition, when the staple pulley 161 is alternately rotated in the clockwise/counterclockwise directions, the staple link assembly 170 connected to the staple pulley 161 and the reciprocating assembly 550 of the cartridge 500 connected to the staple link assembly 170 perform a reciprocating linear motion, and due to this, as the operation member 540 of the cartridge 500 is moved toward the distal end 502, the stapling and cutting motions are performed.

[0594] Here, it is illustrated in the drawings that the staple manipulation part 260 is formed in the form of a bar and a user manually rotates the staple manipulation part 260, but the concept of the present disclosure is not limited thereto. That is, as described above, the staple manipulation part 260 may include a motor (not shown), and while the user presses the staple manipulation part 260 formed in the form of a button, the motor (not shown) may be driven to alternately rotate the manipulation part staple pulley 269 in the clockwise or counterclockwise direction. In addition, due

thereto, the staple pulley **161** of the end tool **100** may be alternately rotated in the clockwise or counterclockwise direction.

[0595] FIGS. **51** to **53** are diagrams illustrating a configuration of pulleys and wires, which are related to a pitch motion of the surgical instrument **10** according to an embodiment of the present disclosure illustrated in FIG. **2**, in detail for each of the first jaw and the second jaw. FIG. **51** is a diagram illustrating only pulleys and wires related to the second jaw, and FIG. **52** is a diagram illustrating only pulleys and wires related to the first jaw. FIG. **53** is a diagram illustrating only pulleys and wires related to the staple pulley. As shown in FIG. **9** and elsewhere herein, there are two pulleys related to the pitch motion, and both strands of each wire are wound in the same path, which is illustrated with one line in FIGS. **51** and **53**. In addition, FIG. **50** is a perspective view illustrating a pitch motion of the surgical instrument of FIG. **2**. Here, in FIG. **50**, components related to stapling and cutting motions are omitted.

[0596] Referring to FIG. **51**, when the first handle **204** is rotated around the rotation shaft **246** in the direction of an arrow OPP1, the pulley **210**, the pulley **215**, the pulley **217**, and the like, and the wire **301** and the like wound therearound are rotated as a whole around the rotation shaft **246**. At this time, since the wires **301** and **305**, which are first jaw wires, are wound around upper portions of the pulley **217** and the pulley **218** as shown in FIG. **51**, the wires **301** and **305** are moved in the direction of an arrow W1. As a result, as described with reference to FIG. **5**, the first jaw **101** of the end tool **100** is rotated in the direction of an arrow EPP1.

[0597] Referring to FIG. **52**, when the first handle **204** is rotated around the rotation shaft **246** in the direction of an arrow OPP2, the pulley **220**, the pulley **225**, the pulley **227**, and the like, and the wire **302** and the like wound therearound are rotated as a whole around the rotation shaft **246**. At this time, since the wires **302** and **306**, which are second jaw wires, are wound around lower portions of the pulley **227** and the pulley **228** as shown in FIG. **52**, the wires **302** and **306** are moved in the direction of an arrow W2. As a result, as described with reference to FIG. **5**, the second jaw **102** of the end tool **100** is rotated in the direction of an arrow EPP2.

[0598] Referring to FIG. **53**, when the first handle **204** is rotated around the rotation shaft **246** in the direction of an arrow OPC1, the manipulation part staple pulley **269**, a pulley **265**, a pulley **267**, and the like, and the wires **307** and **308** and the like wound therearound are rotated as a whole around the rotation shaft **246**. At this time, since the wires **307** and **308**, which are staple wires, are wound around lower portions of the pulley **267** and a pulley **268**, the wires **307** and **308** are moved in the direction of an arrow W3. As a result, as described with reference to FIG. **5**, the staple pulley **161** of the end tool **100** is rotated in the direction of an arrow EPC1.

[0599] As a result, in the pitch motion, when the pulley **111** is rotated around the rotation shaft **143** in the end tool **100**, the staple pulley **161** is also rotated around the rotation shaft **143** together with the pulley **111**.

[0600] Thus, the actuation, yaw, and pitch manipulations are manipulatable independent of each other.

[0601] As described with reference to FIG. **1**, the actuation manipulation part **203**, the yaw manipulation part **202**, and the pitch manipulation part **201** are configured such that the respective rotation shafts are located at the rear thereof to be identical to the joint configuration of the end tool, so that a user may intuitively perform matching manipulations.

[0602] In particular, in the surgical instrument **10** according to an embodiment of the present disclosure, the pulleys are formed on respective joint points (an actuation joint, a yaw joint, and a pitch joint), the wires (the first jaw wire or the second jaw wire) are formed to be wound around the pulleys, the rotational manipulations (actuation rotation, yaw rotation, and pitch rotation) of the manipulation part cause the movement of each wire, which in turn induces the desired motion of the end tool **100**. Furthermore, the auxiliary pulleys may be formed on one side of the respective pulleys, and these auxiliary pulleys may prevent the wire from being wound on one pulley multiple times, so that the wires wound on the pulley do not come into contact with each other, and paths of

the wire being wound around the pulley and the wire being released from the pulley are safely formed, so that safety and efficiency in the transmission of driving force of a wire may be improved.

[0603] Meanwhile, as described above, the yaw manipulation part **202** and the actuation manipulation part **203** are directly formed on the first handle **204**. Thus, when the first handle **204** is rotated around the rotation shaft **246**, the yaw manipulation part **202** and the actuation manipulation part **203** are also rotated together with the first handle **204**. Accordingly, the coordinate systems of the yaw manipulation part **202** and the actuation manipulation part **203** are not fixed, but are continuously changed relative to the rotation of the first handle **204**. That is, in FIG. 2 or the like, the yaw manipulation part **202** and the actuation manipulation part **203** are illustrated as being parallel to the z-axis. However, when the first handle **204** is rotated, the yaw manipulation part **202** and the actuation manipulation part **203** are not parallel to the Z-axis any longer. That is, the coordinate systems of the yaw manipulation part **202** and the actuation manipulation part **203** are changed according to the rotation of the first handle **204**. However, in the present specification, for convenience of description, unless described otherwise, the coordinate systems of the yaw manipulation part **202** and the actuation manipulation part **203** are described on the basis of a state in which the first handle **204** is located perpendicular to the connection part **400** as illustrated in FIG. 2.

(Correlation Between Stapling and Cutting Motions and Other Motions)

[0604] Hereinafter, a correlation between stapling and cutting motions and other motions (pitch, yaw, and actuation motions) will be described.

[0605] First, when the end tool **100** performs a pitch motion, the staple pulley **161** also performs a pitch motion. That is, when the pulley **111** and the pulley **121** perform pitch motions of being rotated in the same direction around the rotation shaft **143**, the staple pulley **161** should also be rotated in the same direction as the pulley **111** and the pulley **121**. If the staple pulley **161** is not rotated together with the pulley **111** and the pulley **121** when the pulley **111** and the pulley **121** are rotated around the rotation shaft **143**, there is a risk that the cartridge **500** connected to the staple pulley **161** is moved relative to the first jaw **101** and is disconnected from the first jaw **101**. Further, rotation of the staple pulley **161** that is not synchronized with that of the pulley **111** may cause the reciprocating member **551** to unintentionally move forward, which in turn may cause an unintended stapling motion.

[0606] Next, when the end tool **100** performs a yaw motion, the staple pulley **161** also performs a yaw motion. That is, when the pulley **111** and the pulley **121** perform yaw motions of being rotated in the same direction around the rotation shaft **141**, the staple pulley **161** should also be rotated in the same direction as the pulley **111** and the pulley **121**. If the staple pulley **161** is not rotated together with the pulley **111** and the pulley **121** when the pulley **111** and the pulley **121** are rotated around the rotation shaft **141**, there is a risk that the cartridge **500** connected to the staple pulley **161** is moved relative to the first jaw **101** and is disconnected from the first jaw **101**. Further, rotation of the staple pulley **161** that is not synchronized with that of the pulley **111** may cause the reciprocating member **551** to unintentionally move forward, which in turn may cause an unintended stapling motion.

[0607] Next, when the end tool **100** performs an actuation motion, the staple pulley **161** is rotated together with the pulley **111**. That is, when the pulley **111** and the pulley **121** perform actuation motions of being rotated the opposite directions around the rotation shaft **141**, the staple pulley **161** should be rotated in the same direction as the pulley **111**. If the staple pulley **161** is not rotated together with the pulley **111** when the pulley **111** is rotated around the rotation shaft **143**, there is a risk that the cartridge **500** connected to the staple pulley **161** is moved relative to the first jaw **101** and is disconnected from the first jaw **101**. Further, rotation of the staple pulley **161** that is not synchronized with that of the pulley **111** may cause the reciprocating member **551** to unintentionally move forward, which in turn may cause an unintended stapling motion.

[0608] Meanwhile, when the end tool **100** performs stapling and cutting motions, the pulley **111** and the pulley **121** are not rotated. That is, when the staple pulley **161** is rotated around the rotation shaft **141** and the reciprocating member **551** of the link member **171** and the cartridge **500** connected thereto performs a linear reciprocating motion, the pulley **111** and the pulley **121** should not be rotated. Otherwise, the first jaw **101** or the second jaw **102** is rotated during the stapling and cutting motion, and the stapling and cutting motions will not be performed normally.

[0609] As a result, when the pulley **111**, which is a first jaw pulley, is rotated, the staple pulley **161** accommodated in the first jaw **101** should be also rotated together with the pulley **111**. On the other hand, when the staple pulley **161** is rotated for the stapling and cutting motions, the pulley **111** and the pulley **121** should be formed to maintain positions thereof without rotating. As such, the correlation between the stapling and cutting motions and other motions (the yaw and actuation motions) has been discussed above.

[0610] In other words, the pulley **111** and the pulley **121** may be said to be independent of the rotation of the staple pulley **161**. That is, even when the staple pulley **161** is rotated by the staple wire, the pulley **111** and the pulley **121** may not be rotated. In contrast, the staple pulley **161** may be said to be dependent of the rotation of the pulley **111**. That is, when the pulley **111** is rotated by the jaw wire, the staple pulley **161** may be formed to be rotated together with the pulley **111**.

[0611] FIGS. **54** and **56** are views illustrating a state in which the jaws are yaw-rotated by -90° , and FIGS. **55** and **57** are views illustrating a process of performing an actuation motion in a state in which the jaws are yaw-rotated by -90° . Here, FIGS. **54** and **55** are views in which the pulley **111** is illustrated, and FIGS. **56** and **57** are views in which the pulley **111** is omitted.

[0612] FIGS. **58** and **60** are views illustrating a state in which the jaws are yaw-rotated by $+90^\circ$, and FIGS. **59** and **61** are views illustrating a process of performing an actuation motion in a state in which the jaws are yaw-rotated by $+90^\circ$. Here, FIGS. **58** and **59** are views in which the pulley **111** is illustrated, and FIGS. **60** and **61** are views in which the pulley **111** is omitted.

[0613] As shown in FIGS. **54** to **61**, the end tool of the surgical instrument according to the first embodiment of the present disclosure is formed to normally perform an actuation motion even when the jaws are yaw-rotated by $+90^\circ$ or -90° .

[0614] FIG. **62** is a plan view illustrating stapling and cutting motions of the end tool of the surgical instrument of FIG. **2**, and is a view illustrating a process of performing the stapling and cutting motions in a state in which the jaws are yaw-rotated by $+90^\circ$. As shown in FIG. **62**, the end tool of the surgical instrument according to the first embodiment of the present disclosure is formed to normally perform the stapling and cutting motions even when the jaws are yaw-rotated by $+90^\circ$.

[0615] In detail, in a state in which the pulley **111**, the pulley **121**, and the staple pulley **161** are rotated by $+90^\circ$ around the rotation shaft **141**, when the staple pulley **161** is alternately rotated in the clockwise/counterclockwise directions, the link member **171** and the reciprocating member **551** connected thereto are repeatedly moved forward and backward. In addition, when the reciprocating member **551** is moved forward, the operation member **540** may be moved forward together with the reciprocating member **551**, and when the reciprocating member **551** is moved backward, only the reciprocating member **551** is moved backward and the operation member **540** remains stationary in place. By repeating this process, the stapling and cutting motions are performed while the operation member **540** is moved toward the distal end **502**.

[0616] FIG. **63** is a plan view illustrating stapling and cutting motions of the end tool of the surgical instrument of FIG. **2**, and is a view illustrating a process of performing the stapling and cutting motions in a state in which the jaws are yaw-rotated by -90° . As shown in FIG. **63**, the end tool of the surgical instrument according to the first embodiment of the present disclosure is formed to normally perform the stapling and cutting motions even when the jaws are yaw-rotated by -90° .

[0617] In detail, in a state in which the pulley **111**, the pulley **121**, and the staple pulley **161** are rotated by -90° around the rotation shaft **141**, when the staple pulley **161** is alternately rotated in the clockwise/counterclockwise directions, the link member **171** and the reciprocating member **551**

connected thereto are repeatedly moved forward and backward. In addition, when the reciprocating member **551** is moved forward, the operation member **540** may be moved forward together with the reciprocating member **551**, and when the reciprocating member **551** is moved backward, only the reciprocating member **551** is moved backward and the operation member **540** remains stationary. By repeating this process, the stapling and cutting motions are performed while the operation member **540** is moved toward the distal end **502**.

[0618] FIG. **64** is a view illustrating a state in which the jaws are pitch-rotated by -90° , and FIG. **65** is a view illustrating a process of performing an actuation motion in a state in which the jaws are pitch-rotated by -90° . FIG. **66** is a view illustrating a state in which the jaws are pitch-rotated by $+90^\circ$, and FIG. **67** is a view illustrating a process of performing an actuation motion in a state in which the jaws are pitch-rotated by $+90^\circ$.

[0619] Referring to FIGS. **64** to **67**, it can be seen that, in performing a pitch motion, the motions of the manipulation part **200** and the end tool **100** are intuitively matched. That is, when the manipulation part **200** is rotated in a positive (+) direction with respect to the pitch rotation shaft (Y-axis), the end tool **100** is also rotated in the positive (+) direction with respect to the pitch rotation shaft (Y-axis). In addition, when the manipulation part **200** is rotated in a negative (−) direction with respect to the pitch rotation shaft (Y-axis), the end tool **100** is also rotated in the negative (−) direction with respect to the pitch rotation shaft (Y-axis). Here, the rotation angle of the manipulation part **200** and the rotation angle of the end tool **100** may be variously set according to the ratio of the pulleys.

[0620] FIG. **68** is a view illustrating a state in which the jaws are yaw-rotated by $+90^\circ$, and FIG. **69** is a view illustrating a process of performing an actuation motion in a state in which the jaws are yaw-rotated by $+90^\circ$. FIG. **70** is a view illustrating a state in which the jaws are yaw-rotated by -90° , and FIG. **71** is a view illustrating a process of performing an actuation motion in a state in which the jaws are yaw-rotated by -90° .

[0621] Referring to FIGS. **68** to **71**, it can be seen that, in performing a yaw motion, the motions of the manipulation part **200** and the end tool **100** are intuitively matched. That is, when the manipulation part **200** is rotated in a positive (+) direction with respect to the yaw rotation shaft (Z-axis), the end tool **100** is also rotated in the positive (+) direction with respect to the yaw rotation shaft (Z-axis). In addition, when the manipulation part **200** is rotated in a negative (−) direction with respect to the yaw rotation shaft (Z-axis), the end tool **100** is also rotated in the negative (−) direction with respect to the yaw rotation shaft (Z-axis). Here, the rotation angle of the manipulation part **200** and the rotation angle of the end tool **100** may be variously set according to the ratio of the pulleys.

[0622] FIG. **72** is a view illustrating a state in which the jaws are pitch-rotated by -90° and at the same time yaw-rotated by $+90^\circ$, and FIG. **73** is a view illustrating a process of performing an actuation motion in the state in which the jaws are pitch-rotated by -90° and at the same time yaw-rotated by $+90^\circ$, FIG. **74** is a view illustrating a state in which the jaws are pitch-rotated by $+90^\circ$ and at the same time yaw-rotated by -90° , and FIG. **75** is a view illustrating a process of performing an actuation motion in the state in which the jaws are pitch-rotated by $+90^\circ$ and at the same time yaw-rotated by -90° ,

[0623] Referring to FIGS. **72** to **75**, it can be seen that the motions of the manipulation part **200** and the end tool **100** are intuitively matched, even when performing the pitch and yaw motions simultaneously.

Second Embodiment—Engrave

[0624] Hereinafter, an end tool **700** of a surgical instrument according to a second embodiment of the present disclosure will be described. Here, the end tool **700** of the surgical instrument according to the second embodiment of the present disclosure is different from the end tool (see **100** of FIG. **2** or the like) of the surgical instrument according to the first embodiment of the present disclosure described above in that a configuration of an end tool hub **780** serving as an auxiliary pulley is

different. The configuration changed from the first embodiment as described above will be described in detail later.

[0625] FIG. **87** is a perspective view illustrating the end tool of the surgical instrument according to the second embodiment of the present disclosure, and FIGS. **88** and **89** are magnified views of the end tool of the surgical instrument of FIG. **87**. FIGS. **90** and **91** are magnified views illustrating the end tool of the surgical instrument of FIG. **87** from a different angle. FIGS. **92** and **93** are views illustrating a state in which jaws of the end tool of the surgical instrument of FIG. **87** are yaw-rotated by 90° in a counterclockwise direction. FIG. **94** is a magnified perspective view illustrating the end tool hub of the surgical instrument of FIG. **87**.

[0626] Here, FIGS. **89** and **91** illustrate a state in which wires are removed, and FIG. **93** illustrates a state in which jaw pulley coupling parts and jaw pulleys of the end tool hub are removed.

[0627] Referring to FIGS. **87** to **94**, the end tool **700** of the second embodiment of the present disclosure includes a pair of jaws, that is, a first jaw **701** and a second jaw **702** for performing a grip motion, and here, each of the first jaw **701** and the second jaw **702** or a component encompassing the first jaw **701** and the second jaw **702** may be referred to as a jaw **703**.

[0628] Meanwhile, the end tool **700** includes a plurality of pulleys including a pulley **711**, a pulley **713**, and a pulley **714** that are related to a rotational motion of the first jaw **701**. The pulleys related to the rotational motion of the first jaw **701** described in the present embodiment are substantially the same as the pulley **113**, the pulley **114**, the pulley **115**, and the pulley **116** described with reference to FIG. **8** or the like of the first embodiment, and thus, detailed descriptions thereof will be omitted herein.

[0629] Meanwhile, the end tool **700** includes a plurality of pulleys including a pulley **721** related to a rotational motion of the second jaw **702**. The pulleys related to the rotational motion of the second jaw **702** described in the present embodiment are substantially the same as the pulley **123**, the pulley **124**, the pulley **125**, and the pulley **126** described with reference to FIG. **8** or the like of the first embodiment, and thus, detailed descriptions thereof will be omitted herein.

[0630] In addition, the end tool **700** of the second embodiment of the present disclosure may include a rotation shaft **741**, a rotation shaft **743**, and a rotation shaft **744**. Here, the rotation shaft **741** may be inserted through the end tool hub **780**, and the rotation shaft **743** and the rotation shaft **744** may be inserted through a pitch hub **707**. The rotation shaft **741**, the rotation shaft **743**, and the rotation shaft **744** may be arranged sequentially from a distal end **704** of the end tool **700** toward a proximal end **705**.

[0631] In addition, the end tool **700** of the second embodiment of the present disclosure may include the end tool hub **780** and the pitch hub **707**.

[0632] The rotation shaft **741**, which will be described later, is inserted through the end tool hub **780**, and the pulley **711** and the pulley **721**, which are axially coupled to the rotation shaft **741**, and at least some of the first jaw **701** and the second jaw **702** coupled to the pulley **711** and the pulley **721** may be accommodated inside the end tool hub **780**. Here, in an embodiment of the present disclosure, a guide part **783** serving as an auxiliary pulley is formed in the end tool hub **780**. That is, the guide part **783** configured to guide paths of the wires **305** and **302** may be formed in the end tool hub **780**. The guide part **783** of the end tool hub **780** may serve as the auxiliary pulleys (see **112**, **122**, and **162** of FIG. **9**) of the first embodiment and change the paths of the wires, and the guide part **783** of the end tool hub **780** serving as the auxiliary pulley will be described in more detail later.

[0633] Meanwhile, a pulley **731** serving as an end tool pitch pulley may be formed at one end portion of the end tool hub **780**. The pulley **731** may be formed as a separate member from the end tool hub **780** to be coupled to the end tool hub **780**. Alternatively, the pulley **731** may be integrally formed with the end tool hub **780** as one body. In addition, the wire (see **303** of FIG. **5**) and the wire (see **304** of FIG. **5**) are coupled to the pulley **731** serving as an end tool pitch pulley, and the pulley **731** is rotated around the rotation shaft **743** to perform a pitch motion.

[0634] The rotation shaft **743** and the rotation shaft **744** are inserted through the pitch hub **707**, and the pitch hub **707** may be axially coupled to the end tool hub **780** and the pulley **731** by the rotation shaft **743**. Accordingly, the end tool hub **780** and the pulley **731** may be formed to be pitch-rotatable around the rotation shaft **743** with respect to the pitch hub **707**.

[0635] Meanwhile, the end tool **700** of the second embodiment of the present disclosure may further include components such as a staple drive assembly (see **150** of FIG. **13**) including a staple pulley assembly (see **160** of FIG. **13**) and a staple link assembly (see **170** of FIG. **13**) to perform stapling and cutting motions.

[0636] The staple pulley assembly (see **160** of FIG. **13**) may be formed between the pulley **711** and the pulley **721** to be adjacent to the pulley **711** and the pulley **721**. In the present embodiment, it is assumed that the staple pulley assembly (see **160** of FIG. **13**) includes one staple pulley **761**.

[0637] The staple link assembly (see **170** of FIG. **13**) may include one or more link members **771**. The staple link assembly (see **170** of FIG. **13**) may serve to connect the staple pulley assembly **760** to a reciprocating assembly (see **550** of FIG. **22**) of a cartridge (see **500** of FIG. **22**). In the present embodiment, it is assumed that the staple link assembly (see **170** of FIG. **13**) includes one link member **771**, and the link member **771** includes a first link **772** and a second link **773**.

[0638] Meanwhile, in the second embodiment of the present disclosure, by disposing the staple pulley assembly (see **160** of FIG. **13**) and the staple link assembly (see **170** of FIG. **13**) between the pulley **711**, which is a first jaw pulley, and the pulley **721**, which is a second jaw pulley, the end tool **700** is allowed to perform pitch and yaw motions, as well as stapling and cutting motions using the cartridge (see **500** of FIG. **22**). In the present embodiment, components for performing stapling and cutting motions are substantially the same as those described in the first embodiment, and thus detailed descriptions thereof will be omitted herein.

[0639] The surgical instrument according to the second embodiment of the present disclosure may include a wire **301**, a wire **302**, a wire **303**, a wire **304**, a wire **305**, a wire **306**, a wire **307**, and a wire **308** as in the first embodiment of the present disclosure described with reference to FIG. **7** or the like.

[0640] In addition, the surgical instrument according to the second embodiment of the present disclosure may include a coupling member **321**, a coupling member **323**, a coupling member **324**, a coupling member **326**, a coupling member **327**, and a coupling member **329**, which are coupled to respective end portions of the wires to couple the wires and the pulleys, as in the first embodiment of the present disclosure described with reference to FIG. **7** or the like.

[0641] Hereinafter, the end tool hub **780** of the second embodiment of the present disclosure will be described in more detail, and in particular, the guide part **783** of the end tool hub **780** serving as an auxiliary pulley will be mainly described.

[0642] The end tool hub **780** includes a first jaw pulley coupling part **781**, a second jaw pulley coupling part **782**, a guide part **783**, a guide groove **784**, and a pitch pulley coupling part **785**.

[0643] In detail, the first jaw pulley coupling part **781** and the second jaw pulley coupling part **782** are formed to face each other, and the pulley **711**, the pulley **721**, and the staple pulley **761** are accommodated inside the first jaw pulley coupling part **781** and the second jaw pulley coupling part **782**. In addition, a through hole is formed in each of the jaw pulley coupling parts **781** and **782**, and thus the rotation shaft **741** passes through the jaw pulley coupling parts **781** and **782**, the pulley **711**, the pulley **721**, and the staple pulley **761** and axially couples them.

[0644] The first jaw pulley coupling part **781** and the second jaw pulley coupling part **782** are connected by the guide part **783**. That is, the first jaw pulley coupling part **781** and the second jaw pulley coupling part **782** parallel to each other are coupled by the guide part **783** formed in a direction substantially perpendicular thereto, so that the first jaw pulley coupling part **781**, the second jaw pulley coupling part **782**, and the guide part **783** form a substantially “C-shape,” in which the pulley **711**, the pulley **721**, and the staple pulley **761** are accommodated.

[0645] In other words, it may be said that the first jaw pulley coupling part **781** and the second jaw

pulley coupling part **782** are formed to extend in an X-axis direction from both end portions of the guide part **783** that is formed long in a Z-axis direction.

[0646] Here, the guide part **783** may be formed in a cylindrical shape with a cross section that is approximately semi-circular. In addition, the semi-circular portion may protrude toward the pulley **711**, the pulley **721**, and the staple pulley **761**. In other words, it may be said that the guide part **783** is formed to protrude toward a space formed by the first jaw pulley coupling part **781**, the second jaw pulley coupling part **782**, and the guide part **783**. In other words, it may be said that a region of the guide part **783** adjacent to the jaw pulley coupling parts **781** and **782** is formed such that a cross section thereof is curved with a predetermined curvature.

[0647] Alternatively, in other words, it may be also said that the guide part **783** functions as a kind of pulley member around which the wire **305**, the wire **302**, the wire **307**, and the wire **308** are wound to guide the paths of the wire **305**, the wire **302**, the wire **307**, and the wire **308**. However, here, the guide part **783** is not a member that is rotated around a certain axis like the original meaning pulley does, and it may be said that the guide part **783** is formed to be fixed as a portion of the end tool hub **780** and performs some similar functions of a pulley by winding a wire therearound.

[0648] Here, the guide part **783** is illustrated in the drawing as being formed in a cylindrical shape with a cross section that is approximately semi-circular. That is, at least a portion of the cross section of the guide part **783** on the XY plane is illustrated as having a certain arc shape. However, the concept of the present disclosure is not limited thereto, and the cross section may have a predetermined curvature like an oval or a parabola, or a corner of a polygonal column is rounded to a certain extent, so that the cross section may have various shapes and sizes suitable for guiding the paths of the wire **305**, the wire **302**, the wire **307**, and the wire **308**.

[0649] Here, in order to more effectively guide the paths of the wire **305**, the wire **302**, the wire **307**, and the wire **308**, the guide groove **784** may be further formed at a portion of the guide part **783** in contact with the wire **305**, the wire **302**, the wire **307**, and the wire **308**. The guide groove **784** may be formed in the form of a groove recessed to a certain extent from a protruding surface of the guide part **783**.

[0650] Here, although the guide groove **784** is illustrated in the drawing as being formed in the entire arc surface of the guide part **783**, the concept of the present disclosure is not limited thereto, and the guide groove **784** may be formed only in a portion of the arc surface of the guide part **783** as necessary.

[0651] By further forming the guide groove **784** in the guide part **783** as described above, the durability of the wire may be improved.

[0652] The pitch pulley coupling part **785** may be further formed in a direction opposite to a direction of formation of the jaw pulley coupling parts **781** and **782** in the guide part **783**. The pitch pulley coupling part **785** may be formed in a direction parallel to the pulley **731** that is a pitch pulley, i.e., on the XZ plane. A through hole through which the rotation shaft **743** may be inserted may be formed in the pitch pulley coupling part **785**, and the rotation shaft **743** may be inserted through the pitch pulley coupling part **785** and the pulley **731** to couple the pitch pulley coupling part **785** and the pulley **731**. Here, the pitch pulley coupling part **785** is formed to be biased to a certain extent from the center to one side when viewed from the XY plane, thereby being balanced as a whole when coupled with the pulley **731**.

[0653] Hereinafter, the role and function of the guide part **783** will be described in more detail.

[0654] The guide part **783** may be in contact with the wire **305** and the wire **302** and may change the arrangement path of the wires **305** and **302** to a certain extent to serve to increase a rotation radius of each of the first jaw **701** and the second jaw **702**.

[0655] In addition, the guide part **783** may serve to increase the radius of rotation of the staple pulley **761** by coming into contact with the wires **307** and **308**, which are blade wires, to change the arrangement path of the wires **307** and **308** to a certain extent.

[0656] That is, when the auxiliary pulley is not disposed, each of the first jaw pulley **711**, the second jaw pulley **721**, and the staple pulley **761** may be rotated up to a right angle, but in the second embodiment of the present disclosure, by additionally providing the guide part **783** in the end tool hub **780**, the maximum rotation angle of each pulley may be increased.

[0657] This enables a motion that two jaws of the end tool **700** have to open for an actuation motion in a state in which the two jaws of the end tool **700** are yaw-rotated by 90°. In other words, a feature of increasing the range of a yaw rotation in which an actuation motion is possible may be obtained through the configuration of the guide part **783** of the end tool hub **780**. In other words, a feature of increasing the range of a yaw rotation in which an actuation motion is possible may be obtained through the configuration of the guide part **783** of the end tool hub **780**.

[0658] In addition, the guide part **783** of the end tool hub **780**, which serves as an auxiliary pulley, allows a staple pulley **761** to further rotate for a cutting motion in a state in which the two jaws are yaw-rotated by 90°. In other words, a feature of increasing the range of a yaw rotation in which a cutting motion is possible may be obtained through the configuration of the guide part **783** of the end tool hub **780**.

[0659] Furthermore, by forming the guide part **783** in the end tool hub **780**, which already exists, without adding a separate structure such as an auxiliary pulley, the range of rotation may be increased without adding a component and a manufacturing process.

[0660] As described above, since there is no need to additionally dispose a separate structure for increasing the rotation angle, the number of components is decreased and the manufacturing process is simplified, and also a length of the end tool is shortened by as much as the size of the auxiliary pulley, so that the length of the end tool is shortened during a pitch motion. Accordingly, a surgical motion may be more easily performed in a narrow space.

[0661] This will be described below in more detail.

[0662] In the end tool **700** of the surgical instrument according to the second embodiment of the present disclosure, the arrangement path of the wires may be changed without a separate structure by forming the guide part **783** capable of changing the path of the wire on an inner side wall of the end tool hub **780**. As described above, as the arrangement paths of the wires **305**, **302**, **307**, and **308** are changed to a certain extent by forming the guide part **783** on the end tool hub **780**, tangential directions of the wires **305**, **302**, **307**, and **308** are changed, and accordingly, the rotation angles of the coupling members **323**, **326**, and **329**, which respectively couple the wires and the pulleys, are increased.

[0663] That is, the coupling member **326** that couples the wire **302** and the pulley **721** is rotatable until being located on a common internal tangent of the pulley **721** and the guide part **783**.

Similarly, the coupling member (see **323** of FIG. **8**) that couples the wire **305** and the pulley **711** is rotatable until being located on a common internal tangent of the pulley **711** and the guide part **783**, so that a rotation angle of the coupling member (see **323** of FIG. **8**) may be increased. Similarly, the coupling member **329** that couples the wires **307** and **308** and the pulley **761** is rotatable until being located on a common internal tangent of the pulley **761** and the guide part **783**, so that a rotation angle of the coupling member **329** may be increased.

[0664] In other words, due to the guide part **783**, the wire **301** and the wire **305** wound around the pulley **711** are disposed on one side with respect to a plane perpendicular to the Y-axis and passing through the X-axis. Simultaneously, due to the guide part **783**, the wires **302** and **306** wound around the pulley **721** are disposed on the other side with respect to the plane perpendicular to the Y-axis and passing through the X-axis.

[0665] In other words, the pulley **713** and the pulley **714** are disposed at one side with respect to the plane perpendicular to the Y-axis and passing through the X-axis, and a pulley **723** and a pulley **724** are disposed at the other side with respect to the plane perpendicular to the Y-axis and passing through the X-axis.

[0666] In other words, the wire **305** is located on the internal tangent of the pulley **711** and the

guide part **783**, and the rotation angle of the pulley **711** is increased by the guide part **783**. In addition, the wire **302** is located on the internal tangent of the pulley **721** and the guide part **783**, and the rotation angle of the pulley **721** is increased by the guide part **783**.

[0667] In the present modified example in which an auxiliary pulley is not formed and the guide part **783** capable of changing the path of the wire is formed on the inner side wall of the end tool hub **780**, the length of the end tool of the surgical instrument may be shortened as compared to the surgical instrument of the first embodiment in which a separate auxiliary pulley is formed. Since the length of the end tool is shortened as described above, a surgical operator may easily manipulate a surgical instrument, and a side effect of surgery may be reduced when the surgery is performed in a narrow surgical space in the human body.

[0668] According to the present disclosure described above, as the rotation radii of the pulley **711**, which is a first jaw pulley, the pulley **721**, which is a second jaw pulley, and the staple pulley **761** increase, a yaw motion range in which a normal opening/closing actuation motion and a normal cutting motion may be performed may be increased.

Third Embodiment—Direct-Connection Type

[0669] Hereinafter, an end tool **800** of a surgical instrument according to a third embodiment of the present disclosure will be described. Here, the end tool **800** of the surgical instrument according to the third embodiment of the present disclosure is different from the end tool (see **100** of FIG. 2 or the like) of the surgical instrument according to the first embodiment of the present disclosure described above in that the configuration and coupling relationship of a first jaw **801** and a second jaw **802** are different. Hereinafter, the configuration that is different from that of the first embodiment will be described in detail.

[0670] FIGS. **95** and **96** are perspective views illustrating the end tool of the surgical instrument according to the third embodiment of the present disclosure. Here, FIG. **96** illustrates a state in which an end tool hub is removed. FIG. **97** is a perspective view illustrating a state in which the end tool of the surgical instrument of FIG. **96** is opened, and FIG. **98** is a perspective view illustrating a state in which the end tool of the surgical instrument of FIG. **96** is closed. FIG. **99** is a side view illustrating the end tool of the surgical instrument of FIG. **96**. FIGS. **100** and **101** are exploded perspective views of the end tool of the surgical instrument of FIG. **96**. FIG. **102** is a plan view illustrating a state in which a first jaw of the end tool of the surgical instrument of FIG. **96** is fully opened. FIGS. **103** and **104** are plan views illustrating opening and closing motions of the end tool of the surgical instrument of FIG. **96**.

[0671] Referring to FIGS. **95** to **104**, the end tool **800** of the third embodiment of the present disclosure includes a pair of jaws **803** for performing a grip motion, that is, the first jaw **801** and the second jaw **802**. Here, each of the first jaw **801** and the second jaw **802**, or a component encompassing the first jaw **801** and the second jaw **802** may be referred to as the jaw **803**.

[0672] Meanwhile, the end tool **800** includes a plurality of pulleys including a pulley **811** and a pulley **812** that are related to a rotational motion of the first jaw **801**. The pulleys related to the rotational motion of the first jaw **801** described in the present embodiment are substantially the same as the pulley **111**, the pulley **112**, the pulley **113**, the pulley **114**, the pulley **115**, and the pulley **116** described with reference to FIG. **8** or the like of the first embodiment, and thus, detailed descriptions thereof will be omitted herein.

[0673] Meanwhile, the end tool **800** includes a plurality of pulleys including a pulley **821** and a pulley **822** that are related to a rotational motion of the second jaw **802**. The pulleys related to the rotational motion of the second jaw **802** described in the present embodiment are substantially the same as the pulley **121**, the pulley **122**, the pulley **123**, the pulley **124**, the pulley **125**, and the pulley **126** described with reference to FIG. **8** or the like of the first embodiment, and thus, detailed descriptions thereof will be omitted herein.

[0674] In addition, the end tool **800** of the third embodiment of the present disclosure may include a rotation shaft **841**, a rotation shaft **842**, a rotation shaft **843**, and a rotation shaft **844**. Here, the

rotation shaft **841** and the rotation shaft **842** may be inserted through an end tool hub **880**, and the rotation shaft **843** and the rotation shaft **844** may be inserted through a pitch hub **807**. The rotation shaft **841**, the rotation shaft **842**, the rotation shaft **843**, and the rotation shaft **844** may be arranged sequentially from a distal end **804** of the end tool **800** toward a proximal end **805**.

[0675] In addition, the end tool **800** of the third embodiment of the present disclosure may include the end tool hub **880** and the pitch hub **807**.

[0676] The rotation shaft **841** and the rotation shaft **842**, which will be described later, may be inserted through the end tool hub **880**, and the pulley **811** and the pulley **821** axially coupled to the rotation shaft **841** and at least some of the first jaw **801** and the second jaw **802** coupled to the pulley **811** and the pulley **821** may be accommodated inside the end tool hub **880**.

[0677] The rotation shaft **843** and the rotation shaft **844** may be inserted through the pitch hub **807**, and the pitch hub **807** may be axially coupled to the end tool hub **880** by the rotation shaft **843**.

[0678] Thus, the end tool hub **880** may be formed to be pitch-rotatable around the rotation shaft **843** with respect to the pitch hub **807**.

[0679] Meanwhile, the end tool **800** of the third embodiment of the present disclosure may further include a component such as a staple pulley assembly (see **160** of FIG. **13**) to perform stapling and cutting motions.

[0680] The staple pulley assembly (see **160** of FIG. **13**) may be formed between the pulley **811** and the pulley **821** to be adjacent to the pulley **811** and the pulley **821**. In the present embodiment, it is assumed that the staple pulley assembly (see **160** of FIG. **13**) includes one staple pulley **861**.

[0681] Meanwhile, in the third embodiment of the present disclosure, by disposing the staple pulley assembly (see **160** of FIG. **13**) between the pulley **811**, which is a first jaw pulley, and the pulley **821**, which is a second jaw pulley, the end tool **800** is allowed to perform pitch and yaw motions as well as stapling and cutting motions using a cartridge (see **500** of FIG. **22**). In the present embodiment, components for performing the stapling and cutting motions are substantially the same as those described in the first embodiment, and thus detailed descriptions thereof will be omitted herein.

[0682] Hereinafter, the first jaw **801** and the second jaw **802** of the end tool **800** of the surgical instrument according to the third embodiment of the present disclosure will be described in more detail.

[0683] The end tool **800** of the surgical instrument according to the third embodiment of the present disclosure includes one rotation shaft **841** that simultaneously performs the roles of a jaw pulley rotation shaft and a jaw rotation shaft, instead including the jaw pulley rotation shaft and the jaw rotation shaft separately. In other words, in the present embodiment, a direct connection structure is adopted for connecting the first jaw **801** and the second jaw **802** instead of the X-shaped structure of the first embodiment.

[0684] That is, in the first embodiment of the present disclosure, the rotation shaft (see **141** of FIG. **17**), which is a jaw pulley rotation shaft, and the rotation shaft (see **145** of FIG. **17**), which is a jaw rotation shaft, are separately provided, so that a stronger grip force may be provided when the jaws are closed. In contrast, in the third embodiment of the present disclosure, the rotation shaft **841** serves as both the jaw rotation shaft and the jaw pulley rotation shaft, allowing each jaw and each jaw pulley to rotate together as one body.

[0685] In detail, the first jaw **801** includes a cartridge accommodation part (see **101a** of FIG. **14**) and a guide groove **801b**. The first jaw **801** is formed entirely in the shape of an elongated bar, the cartridge **500** is accommodated in a distal end **801f** side, and the pulley **811** is coupled to a proximal end **801g**, so that the first jaw **801** is formed to be rotatable around the rotation shaft **841**.

[0686] Here, the first jaw **801** and the pulley **811** are illustrated in the drawings as being integrally formed, but a configuration in which the first jaw **801** and pulley **811** are formed as separate members and coupled together is also possible.

[0687] The second jaw **802** includes an anvil **802a**. The second jaw **802** is formed entirely in the

shape of an elongated bar, the anvil **802a** is formed in a distal end **802f** side, and the pulley **812** is coupled to a proximal end **802g**, so that the second jaw **802** is formed to be rotatable around the rotation shaft **841**.

[0688] Here, the second jaw **802** and the pulley **821** are illustrated in the drawings as being integrally formed, but a configuration in which the second jaw **802** and pulley **821** are formed as separate members and coupled together is also possible.

[0689] The rotation shaft **841**, which is an end tool jaw pulley rotation shaft, is sequentially inserted through the pulley **811** coupled to (or integrally formed with) the first jaw **801**, the staple pulley **861**, and the pulley **821** coupled to (or integrally formed with) the second jaw **802**.

[0690] Thus, the pulley **811** coupled to (or integrally formed with) the first jaw **801**, the staple pulley **861**, and the pulley **821** coupled to (or integrally formed with) the second jaw **802** are all rotated around the rotation shaft **841**.

[0691] In addition, in the third embodiment of the present disclosure, the auxiliary pulley **812** is provided. The auxiliary pulley **812** may be in contact with the wire **305** and the wire **302** and may change the arrangement path of the wires **305** and **302** to a certain extent to serve to increase a rotation radius of each of the first jaw **801** and the second jaw **802**.

[0692] As a result, unlike the first embodiment, in the present embodiment, since there is no separate jaw rotation shaft for restricting the first jaw **801** and the second jaw **802**, a rotation angle between the first jaw **801** and the second jaw **802** is increased to approximately 120°. In addition, by coupling the first jaw **801** and the second jaw **802** by a single shaft without a separate connection structure for connecting the X-shaped structure between the first jaw **801** and the second jaw **802**, the number of parts may be reduced and the manufacturing may be easily performed.

Fourth Embodiment—Dual Rack

[0693] Hereinafter, an end tool **900** of a surgical instrument according to a fourth embodiment of the present disclosure will be described. Here, the end tool **900** of the surgical instrument according to the fourth embodiment of the present disclosure is different from the end tool (see **100** of FIG. 2 or the like) of the surgical instrument according to the first embodiment of the present disclosure described above in that a reciprocating assembly **950** of a cartridge **910** and a staple link assembly **970** are different. Hereinafter, the configuration that is different from that of the first embodiment will be described in detail.

[0694] FIGS. **105** and **106** are perspective views illustrating the end tool of the surgical instrument according to the fourth embodiment of the present disclosure. Here, FIG. **106** illustrates a state in which an end tool hub is removed. FIGS. **107** and **108** are perspective views of the end tool of the surgical instrument of FIG. **105** viewed from a different angle. Here, FIG. **108** illustrates a state in which the end tool hub is removed. FIG. **109** is a side view illustrating the end tool of the surgical instrument of FIG. **105**. FIGS. **110** and **111** are exploded perspective views of the end tool of the surgical instrument of FIG. **105**. FIG. **112** is an exploded perspective view of the staple link assembly of the surgical instrument of FIG. **105**. FIGS. **113** and **114** are a side view and a plan view illustrating respective operating states of the staple link assembly of FIG. **105**. FIG. **115** is a perspective view illustrating an internal structure of the end tool of the surgical instrument of FIG. **105**. FIGS. **116** and **117** are perspective views illustrating respective operating states of the end tool of FIG. **115**. FIGS. **118** and **119** are perspective views illustrating respective operating states of the end tool of FIG. **115**. Here, FIGS. **118** and **119** mainly illustrate operations of the reciprocating assembly and the staple link assembly. FIGS. **120** and **121** are perspective views illustrating respective operating states of the end tool of FIG. **115**. Here, FIGS. **120** and **121** mainly illustrate operations of the reciprocating assembly and an operation member.

[0695] Referring to FIGS. **105** to **121**, the end tool **900** of the fourth embodiment of the present disclosure includes a pair of jaws **903** for performing a grip motion, that is, a first jaw **901** and a second jaw **902**. Here, each of the first jaw **901** and the second jaw **902**, or a component

encompassing the first jaw **901** and the second jaw **902** may be referred to as the jaw **903**.

[0696] Meanwhile, the end tool **900** includes a plurality of pulleys including a pulley **911** and a pulley **912** that are related to a rotational motion of the first jaw **901**. The pulleys related to the rotational motion of the first jaw **901** described in the present embodiment are substantially the same as the pulley **111**, the pulley **112**, the pulley **113**, the pulley **114**, the pulley **115**, and the pulley **116** described with reference to FIG. **8** or the like of the first embodiment, and thus, detailed descriptions thereof will be omitted herein.

[0697] Meanwhile, the end tool **900** includes a plurality of pulleys including a pulley **921** and a pulley **922** that are related to a rotational motion of the second jaw **902**. The pulleys related to the rotational motion of the second jaw **902** described in the present embodiment are substantially the same as the pulley **121**, the pulley **122**, the pulley **123**, the pulley **124**, the pulley **125**, and the pulley **126** described with reference to FIG. **8** or the like of the first embodiment, and thus, detailed descriptions thereof will be omitted herein.

[0698] In addition, the end tool **900** of the fourth embodiment of the present disclosure may include a rotation shaft **941**, a rotation shaft **942**, a rotation shaft **933**, and a rotation shaft **944**. Here, the rotation shaft **941** and the rotation shaft **942** may be inserted through an end tool hub **980**, and the rotation shaft **933** and the rotation shaft **944** may be inserted through a pitch hub **907**. The rotation shaft **941**, the rotation shaft **942**, the rotation shaft **933**, and the rotation shaft **944** may be arranged sequentially from a distal end **904** of the end tool **900** toward a proximal end **905**.

[0699] In addition, the end tool **900** of the fourth embodiment of the present disclosure may include the end tool hub **980** and the pitch hub **907**.

[0700] The rotation shaft **941** and the rotation shaft **942** may be inserted through the end tool hub **980**, and the pulley **911** and the pulley **921** axially coupled to the rotation shaft **941** and at least some of the first jaw **901** and the second jaw **902** coupled to the pulley **911** and the pulley **921** may be accommodated inside the end tool hub **980**.

[0701] The rotation shaft **933** and the rotation shaft **944** may be inserted through the pitch hub **907**, and the pitch hub **907** may be axially coupled to the end tool hub **980** by the rotation shaft **933**. Thus, the end tool hub **980** may be formed to be pitch-rotatable around the rotation shaft **933** with respect to the pitch hub **907**.

[0702] Meanwhile, the end tool **900** of the fourth embodiment of the present disclosure may further include components, such as a staple drive assembly (see **150** of FIG. **13**) including a staple pulley assembly **960** and the staple link assembly **970**, to perform stapling and cutting motions.

[0703] The staple pulley assembly **960** may be formed between the pulley **911** and the pulley **921** to be adjacent to the pulley **911** and the pulley **921**. In the present embodiment, it is assumed that the staple pulley assembly **960** includes one staple pulley **961**.

[0704] In the fourth embodiment of the present disclosure, by disposing the staple pulley assembly **960** between the pulley **911**, which is a first jaw pulley, and the pulley **921**, which is a second jaw pulley, the end tool **900** is allowed to perform pitch and yaw motions as well as stapling and cutting motions using the cartridge **910**.

[0705] Hereinafter, the staple pulley assembly **960**, the staple link assembly **970**, and the reciprocating assembly **950** of the cartridge **910** of the end tool **900** of the surgical instrument according to the fourth embodiment of the present disclosure will be described in more detail.

[0706] In the end tool **900** of the surgical instrument according to the fourth embodiment of the present disclosure, the staple link assembly **970** includes a first link member **971** and a second link member **976**, and the reciprocating assembly **950** of the cartridge **910** includes a first reciprocating member **951** and a second reciprocating member **952**, so that a kind of dual rack structure is formed in the end tool **900**.

[0707] Referring to FIGS. **105** to **121** and the like, the staple pulley assembly **960** may include one or more staple pulleys **961**.

[0708] A shaft pass-through part **961a** may be formed in the staple pulley **961**. The shaft pass-

through part **961a** may be formed in the form of a hole, and the rotation shaft **941**, which is an end tool jaw pulley rotation shaft, may be inserted through the shaft pass-through part **961a**.

[0709] In addition, a first link coupling part **961b** and a second link coupling part **961c** may be formed in the staple pulley **961**. The first link member **971** of the staple link assembly **970** may be coupled to the first link coupling part **961b**, and the second link member **976** of the staple link assembly **970** may be coupled to the second link coupling part **961c**. Here, the first link coupling part **961b** and the second link coupling part **961c** may be disposed on opposite sides with respect to a central axis of the staple pulley **961**.

[0710] Meanwhile, the end tool **900** of the fourth embodiment of the present disclosure may further include the staple link assembly **970** connected to the staple pulley assembly **960**. Here, the staple link assembly **970** may serve to connect the staple pulley assembly **960** to the reciprocating assembly **950** of the cartridge **910** to be described later. In the end tool **900** of the fourth embodiment of the present disclosure, the staple link assembly **970** includes two pairs of link members in the first link member **971** and the second link member **976**.

[0711] The first link member **971** may include a first link **972** and a second link **973**.

[0712] The first link **972** is formed in the form of an elongated bar, which may have through holes formed at both end portions. The first link coupling part **961b** of the staple pulley **961** may be inserted through the through hole at one end portion of the first link **972**. The second link **973** may be inserted through the through hole at the other end portion of the first link **972**.

[0713] The second link **973** is formed in the form of an elongated bar, and may be coupled to the first link **972**. The second link **973** may include a first protrusion **973a**, a second protrusion **973b**, and a coupling part **973c**.

[0714] In detail, the first protrusion **973a** may be formed at one end portion of the second link **973**. The first protrusion **973a** is axially coupled to the first link **972** by being fitted into the through hole of the first link **972**, so that the second link **973** may be coupled to the first link **972**. In addition, the first protrusion **973a** may be fitted into a first guide groove **901b** of the first jaw **901**, which will be described later.

[0715] Meanwhile, the second protrusion **973b** may be formed in one region of a central portion of the second link **973**. The second protrusion **973b** may be fitted into the first guide groove **901b** of the first jaw **901**, which will be described later.

[0716] As described above, as the first protrusion **973a** and the second protrusion **973b** are moved along the first guide groove **901b** in a state in which the first protrusion **973a** and the second protrusion **973b** of the second link **973** formed in a protruding shape are fitted into the groove-shaped first guide groove **901b**, the first link member **971** is moved with respect to the first jaw **901** (and the cartridge **910** therein). This will be described in more detail later.

[0717] Meanwhile, the coupling part **973c** may be formed at the other end portion of the second link **973**. The coupling part **973c** may be coupled to a coupling part **951a** of the first reciprocating member **951** of the cartridge **910**, which will be described later.

[0718] In the state of FIG. **113**, when the staple pulley **961** is rotated in the direction of an arrow **A1** of FIG. **114** (i.e., in the clockwise direction), the first link member **971** connected to the staple pulley **961** may be moved in the direction of an arrow **B1** of FIG. **114**, in other words, toward a distal end **901f** of the first jaw **901**. In contrast, when the staple pulley **961** is rotated in the counterclockwise direction, the first link member **971** connected to the staple pulley **961** may be moved in the direction of an arrow **C1** of FIG. **114**, in other words, toward a proximal end **901g** of the first jaw **901**.

[0719] Accordingly, a bidirectional rotational motion of the staple pulley assembly **960** may cause a reciprocating linear motion of the first reciprocating member **951** of the cartridge **910** through the first link member **971** of the staple link assembly **970**. This will be described in more detail later.

[0720] The second link member **976** may include a third link **977** and a fourth link **978**.

[0721] The third link **977** is formed in the form of an elongated bar, which may have through holes

formed at both end portions. The second link coupling part **961c** of the staple pulley **961** may be inserted through the through hole at one end portion of the third link **977**. The fourth link **978** may be inserted through the through hole at the other end portion of the third link **977**.

[0722] The fourth link **978** is formed in the form of a bar bent one or more times, and may be coupled to the third link **977**. The fourth link **978** may include a first protrusion **978a**, a second protrusion **978b**, and a coupling part **978c**.

[0723] In detail, the first protrusion **978a** may be formed at one end portion of the fourth link **978**. The first protrusion **978a** is axially coupled to the third link **977** by being fitted into the through hole of the third link **977**, so that the fourth link **978** may be coupled to the third link **977**. In addition, the first protrusion **978a** may be fitted into a second guide groove **901c** of the first jaw **901**, which will be described later.

[0724] Meanwhile, the second protrusion **978b** may be formed in one region of a central portion of the fourth link **978**. The second protrusion **978b** may be fitted into the second guide groove **901c** of the first jaw **901**, which will be described later.

[0725] As described above, as the first protrusion **978a** and the second protrusion **978b** are moved along the second guide groove **901c** in a state in which the first protrusion **978a** and the second protrusion **978b** of the fourth link **978** formed in a protruding shape are fitted into the groove-shaped second guide groove **901c**, the second link member **976** is moved with respect to the first jaw **901** (and the cartridge **910** therein). This will be described in more detail later.

[0726] Meanwhile, the coupling part **978c** may be formed at the other end portion of the fourth link **978**. The coupling part **978c** may be coupled to a coupling part **952a** of the second reciprocating member **952** of the cartridge **910**, which will be described later.

[0727] Here, the fourth link **978** may include two horizontal regions and one vertical region connecting the two horizontal regions.

[0728] In detail, since the first link coupling part **961b** and the second link coupling part **961c** are disposed on opposite sides with respect to the central axis of the staple pulley **961**, a height difference (with respect to a z-axis direction) exists between the first link **972** coupled to the first link coupling part **961b** and the third link **977** coupled to the second link coupling part **961c**.

[0729] In contrast, it is structurally advantageous for the first reciprocating member **951** and the second reciprocating member **952** of the reciprocating assembly **950**, which will be described later, to be located at the same height (with respect to the Z-axis direction) as each other.

[0730] Accordingly, in order to connect the third link **977** and the second reciprocating member **952**, the fourth link **978** may be formed in the form of a bar that is bent one or more times.

[0731] Here, the first protrusion **978a** and the second protrusion **978b** are formed in a first horizontal region, and the coupling part **978c** may be formed in a second horizontal region.

[0732] In the state of FIG. **113**, when the staple pulley **961** is rotated in the direction of the arrow **A1** of FIG. **114** (i.e., in the clockwise direction), the second link member **976** connected to the staple pulley **961** may be moved in the direction of the arrow **C1** of FIG. **114**, in other words, toward the proximal end **901g** of the first jaw **901**. In contrast, when the staple pulley **961** is rotated in the counterclockwise direction, the second link member **976** connected to the staple pulley **961** may be moved in the direction of the arrow **B1** of FIG. **114**, in other words, toward the distal end **901f** of the first jaw **901**.

[0733] Accordingly, a bidirectional rotational motion of the staple pulley assembly **960** may cause a reciprocating linear motion of the second reciprocating member **952** of the cartridge **910** through the second link member **976** of the staple link assembly **970**. This will be described in more detail later.

[0734] As described above, the first link coupling part **961b** and the second link coupling part **961c** may be disposed on opposite sides with respect to the central axis of the staple pulley **961**. In addition, the first link member **971** is coupled to the first link coupling part **961b**, and the second link member **976** is coupled to the second link coupling part **961c**.

[0735] Accordingly, when the staple pulley **961** is rotated in one direction (e.g., the clockwise direction), the first link member **971** is moved forward and the second link member **976** is moved backward. Meanwhile, when the staple pulley **961** is rotated in the other direction (e.g., the counterclockwise direction), the first link member **971** is moved backward and the second link member **976** is moved forward.

[0736] With this configuration, when the staple pulley **961** is alternately rotated in the clockwise and counterclockwise directions, the first link member **971** and the second link member **976** perform a linear reciprocating motion, and in this case, the first link member **971** and the second link member **976** are moved in opposite directions. That is, when the first link member **971** is moved forward (i.e., toward the distal end), the second link member **976** is moved backward (i.e., toward the proximal end). Conversely, when the first link member **971** is moved backward (i.e., toward the proximal end), the second link member **976** is moved forward (i.e., toward the distal end).

(Cartridge)

[0737] Hereinafter, the cartridge **910** of the end tool **900** of the surgical instrument according to the fourth embodiment of the present disclosure will be described in more detail.

[0738] In the cartridge **910** of the end tool **900** of the surgical instrument according to the fourth embodiment of the present disclosure, the reciprocating assembly **950** includes the first reciprocating member **951** and the second reciprocating member **952**.

[0739] In detail, referring to FIGS. **115** to **117** and the like, the cartridge **910** is formed to be mountable to and dismountable from the first jaw **101** and includes a plurality of staples (see **530** of FIG. **22**) and an operation member **940** therein to perform suturing and cutting tissue.

[0740] Here, the cartridge **910** may include a cover (see **510** of FIG. **22**), a housing **920**, a staple (see **530** of FIG. **22**), the operation member **940**, and the reciprocating assembly **950**.

[0741] The housing **920** forms an outer shape of the cartridge **910**, and may be formed entirely in the form of a hollow box with one surface (upper surface) removed to accommodate the reciprocating assembly **950**, the operation member **940**, and the staple (see **530** of FIG. **22**) therein. Here, the housing **920** may be formed in an approximately “U” shape in cross section.

[0742] A plurality of staples (see **530** of FIG. **22**) may be disposed inside the housing **920**. As the operation member **940**, which will be described later, is linearly moved in one direction, the plurality of staples (see **530** of FIG. **22**) are sequentially pushed and raised from the inside of the housing **920** to the outside, thereby performing sealing, that is, stapling.

[0743] The reciprocating assembly **950** may be disposed at an inner lower side of the housing **920**. In the present embodiment, the reciprocating assembly **950** includes the first reciprocating member **951** and the second reciprocating member **952**.

[0744] In the present embodiment, the first reciprocating member **951** and the second reciprocating member **952** may form a rack. The first reciprocating member **951** may include a recess **951b** and the coupling part **951a**.

[0745] In detail, the first reciprocating member **951** may be formed in the form of an elongated bar, and a plurality of recesses **951b** having a sawtooth shape may be formed on one surface thereof. The recess **951b** may be formed to be in contact with an operation member **940** to be described later, in particular, a ratchet member **943** of the operation member **940**. In other words, the first reciprocating member **951** may include the plurality of recesses **951b** shaped to engage with first ratchets **943a** of the ratchet member **943**.

[0746] Similarly, the second reciprocating member **952** may include a recess **952b** and the coupling part **952a**. That is, the second reciprocating member **952** may include a plurality of recesses **952b** shaped to engage with second ratchets **943b** of the ratchet member **943**.

[0747] Here, the first reciprocating member **951** and the second reciprocating member **952** are not fixedly coupled to other components of the cartridge **910**, and may be formed to be movable relative to other components of the cartridge **910**. That is, the first reciprocating member **951** and

the second reciprocating member **952** may perform a reciprocating linear motion with respect to the cover **510** coupled to the housing **920** and the housing **920**.

[0748] Meanwhile, in the first reciprocating member **951**, the coupling part **951a** may be formed at the proximal end side adjacent to the pulley **911** and the coupling part **951a** may be fastened and coupled to the first link member **971** of the staple link assembly **970** of the end tool **900**. Thus, when the first link member **971** performs a reciprocating linear motion in the extension direction (i.e., the Y-axis direction) of the connection part **400**, the first reciprocating member **951** coupled thereto may also perform a reciprocating linear motion in the extension direction (i.e., the Y-axis direction) of the connection part **400**.

[0749] Similarly, in the second reciprocating member **952**, the coupling part **952a** may be formed at the proximal end side adjacent to the pulley **911** and the coupling part **952a** may be fastened and coupled to the second link member **976** of the staple link assembly **970** of the end tool **900**. Thus, when the second link member **976** performs a reciprocating linear motion in the extension direction (i.e., the Y-axis direction) of the connection part **400**, the second reciprocating member **952** coupled thereto may also perform a reciprocating linear motion in the extension direction (i.e., the Y-axis direction) of the connection part **400**.

[0750] The operation member **940** may be disposed inside the housing **920**. The operation member **940** is formed to be in contact with the first reciprocating member **951** and the second reciprocating member **952**, and may be formed to linearly move in one direction according to the reciprocating linear motion of the first reciprocating member **951** and the second reciprocating member **952**. In other words, the operation member **940** interacts with the first reciprocating member **951** and the second reciprocating member **952** to perform stapling and cutting while moving in the extension direction of the connection part **400**.

[0751] The operation member **940** may include a wedge (see **541** of FIG. 22), a blade (see **542** of FIG. 22), the ratchet member **943**, an elastic member (see **544** of FIG. 22), and the like. Here, in the drawings, components other than the ratchet member **943**, that is, the wedge, the blade, and the like are omitted from the operation member **940**, but it is of course possible that such components may be included in the operation member **940**.

[0752] The ratchet member **943** may include the first ratchet **943a** and the second ratchet **943b**. Here, the first ratchet **943a** may be formed to be engageable with the first reciprocating member **951**, and the second ratchet **943b** may be formed to be engageable with the second reciprocating member **952**. That is, a motion, in which the first ratchet **943a** is engaged with (or in close contact with) the first reciprocating member **951** and the operation member **940** is moved forward by the first reciprocating member **951**, and a motion, in which the second ratchet **943b** is engaged with (or in close contact with) the second reciprocating member **952** and the operation member **940** is moved forward by the second reciprocating member **952**, are alternately performed.

[0753] Here, due to the structural shape of the ratchet member **943**, even when a separate elastic member is not provided, the ratchet member **943** may alternately come into close contact with the first reciprocating member **951** and the second reciprocating member **952**. This will be described in more detail below.

(Operations of Reciprocating Assembly and Operation Member)

[0754] Hereinafter, operations of the reciprocating assembly **950** and the operation member **940** will be described in more detail.

[0755] FIGS. **120** and **121** are views illustrating respective operating states of the reciprocating assembly **950** and the operation member **940**. Here, FIG. **120A** is a view illustrating a state in which the first ratchet **943a** of the ratchet member **943** and the first reciprocating member **951** come into close contact with each other, and FIG. **120E** is a view illustrating a state in which the second ratchet **943b** of the ratchet member **943** and the second reciprocating member **952** come into close contact with each other, and FIG. **120I** is a view illustrating a state in which the first ratchet **943a** of the ratchet member **943** and the first reciprocating member **951** come into close

contact with each other.

[0756] In addition, FIGS. **120A** to **120E** illustrate operating states in which the staple pulley **961** is rotated in the counterclockwise direction (in a direction opposite to the direction of the arrow **A1** of FIG. **114**) and thus the second reciprocating member **952** is moved forward, and FIGS. **120E** to **120I** illustrate operating states in which the staple pulley **961** is rotated in the clockwise direction (in the direction of the arrow **A1** of FIG. **114**) and thus the first reciprocating member **951** is moved forward.

[0757] First, operations of FIGS. **120A** to **120E** will be described.

[0758] FIG. **120A** illustrates a state in which the first ratchet **943a** of the ratchet member **943** and the first reciprocating member **951** come into close contact with each other.

[0759] In this state, when the staple pulley **961** is rotated in the counterclockwise direction (in the opposite direction of the arrow **A1** of FIG. **114**), the first reciprocating member **951** is moved in the direction of an arrow **b1** (i.e., is moved backward), and the second reciprocating member **952** is moved in the direction of an arrow **b2** (i.e., is moved forward) as shown in FIG. **120B**. Then, the recess **951b** of the first reciprocating member **951** pushes the first ratchet **943a** of the ratchet member **943** toward the second reciprocating member **952**, which causes the entire ratchet member **943** to be rotationally moved in the direction of an arrow **b3**. In addition, when the ratchet member **943** is rotationally moved in the direction of the arrow **b3**, the first ratchet **943a** of the ratchet member **943** starts to be spaced apart from the recess **951b** of the first reciprocating member **951**, and the second ratchet **943b** of the ratchet member **943** starts to come into contact with the recess **952b** of the second reciprocating member **952**.

[0760] In this state, when the staple pulley **961** is further rotated in the counterclockwise direction (in the opposite direction of the arrow **A1** of FIG. **114**), the first reciprocating member **951** is further moved in the direction of an arrow **c1** (i.e., is moved backward), and the second reciprocating member **952** is further moved in the direction of an arrow **c2** (i.e., is moved forward) as shown in FIG. **120C**. Then, the recess **951b** of the first reciprocating member **951** further pushes the first ratchet **943a** of the ratchet member **943** toward the second reciprocating member **952**, so that the entire ratchet member **943** is further rotationally moved in the direction of an arrow **c3**. In addition, when the ratchet member **943** is rotationally moved in the direction of the arrow **c3**, the first ratchet **943a** of the ratchet member **943** is further spaced apart from the recess **951b** of the first reciprocating member **951**, and the second ratchet **943b** of the ratchet member **943** is further brought into contact with the recess **952b** of the second reciprocating member **952**.

[0761] In this state, when the staple pulley **961** is further rotated in the counterclockwise direction (in the opposite direction of the arrow **A1** of FIG. **114**), the first reciprocating member **951** is further moved in the direction of an arrow **d1** (i.e., is moved backward), and the second reciprocating member **952** is further moved in the direction of an arrow **d2** (i.e., is moved forward) as shown in FIG. **120D**. Then, the recess **951b** of the first reciprocating member **951** further pushes the first ratchet **943a** of the ratchet member **943** toward the second reciprocating member **952**, which causes the entire ratchet member **943** to be further rotationally moved in the direction of an arrow **d3**. In addition, when the ratchet member **943** is rotationally moved in the direction of the arrow **d3**, the first ratchet **943a** of the ratchet member **943** is completely spaced apart from the recess **951b** of the first reciprocating member **951**, and the second ratchet **943b** of the ratchet member **943** is brought into close contact with the recess **952b** of the second reciprocating member **952**.

[0762] In this state, when the staple pulley **961** is further rotated in the counterclockwise direction (in the opposite direction of the arrow **A1** of FIG. **114**), the first reciprocating member **951** is further moved in the direction of an arrow **e1** (i.e., is moved backward), and the second reciprocating member **952** is further moved in the direction of an arrow **e2** (i.e., is moved forward) as shown in FIG. **120E**. Then, since the second ratchet **943b** of the ratchet member **943** is in a state of being in close contact with the recess **952b** of the second reciprocating member **952**, the ratchet

member **943** is linearly moved in the direction of an arrow **e4** by the second reciprocating member **952**.

[0763] Next, operations of FIGS. **120E** to **120I** will be described.

[0764] FIG. **120E** illustrates a state in which the second ratchet **943b** of the ratchet member **943** and the second reciprocating member **952** come into close contact with each other.

[0765] In this state, when the staple pulley **961** is rotated in the clockwise direction (in the direction of the arrow **A1** of FIG. **114**), as shown in FIG. **120F**, the first reciprocating member **951** is moved in the direction of an arrow **f1** (i.e., is moved forward), and the second reciprocating member **952** is moved in the direction of an arrow **f2** (i.e., is moved backward). Then, the recess **952b** of the second reciprocating member **952** pushes the second ratchet **943b** of the ratchet member **943** toward the first reciprocating member **951**, which causes the entire ratchet member **943** to be rotationally moved in the direction of an arrow **f3**. In addition, when the ratchet member **943** is rotationally moved in the direction of the arrow **f3**, the second ratchet **943b** of the ratchet member **943** starts to be spaced apart from the recess **952b** of the second reciprocating member **952**, and the first ratchet **943a** of the ratchet member **943** starts to come into contact with the recess **951b** of the first reciprocating member **951**.

[0766] In this state, when the staple pulley **961** is further rotated in the clockwise direction (in the direction of the arrow **A1** of FIG. **114**), as shown in FIG. **120G**, the first reciprocating member **951** is further moved in the direction of an arrow **g1** (i.e., is moved forward), and the second reciprocating member **952** is further moved in the direction of an arrow **g2** (i.e., is moved backward). Then, the recess **952b** of the second reciprocating member **952** further pushes the second ratchet **943b** of the ratchet member **943** toward the first reciprocating member **951**, which causes the entire ratchet member **943** to be further rotationally moved in the direction of an arrow **g3**. In addition, when the ratchet member **943** is rotationally moved in the direction of the arrow **g3**, the second ratchet **943b** of the ratchet member **943** is further spaced apart from the recess **952b** of the second reciprocating member **952**, and the first ratchet **943a** of the ratchet member **943** is further brought into contact with the recess **951b** of the first reciprocating member **951**.

[0767] In this state, when the staple pulley **961** is further rotated in the clockwise direction (in the direction of the arrow **A1** of FIG. **114**), as shown in FIG. **120H**, the first reciprocating member **951** is further moved in the direction of an arrow **h1** (i.e., is moved forward), and the second reciprocating member **952** is further moved in the direction of an arrow **h2** (i.e., is moved backward). Then, the recess **952b** of the second reciprocating member **952** further pushes the second ratchet **943b** of the ratchet member **943** toward the first reciprocating member **951**, which causes the entire ratchet member **943** to be further rotationally moved in the direction of an arrow **h3**. In addition, when the ratchet member **943** is rotationally moved in the direction of the arrow **h3**, the second ratchet **943b** of the ratchet member **943** is completely spaced apart from the recess **952b** of the second reciprocating member **952**, and the first ratchet **943a** of the ratchet member **943** is brought into close contact with the recess **951b** of the first reciprocating member **951**.

[0768] In this state, when the staple pulley **961** is further rotated in the clockwise direction (in the direction of the arrow **A1** of FIG. **114**), as shown in FIG. **120I**, the first reciprocating member **951** is further moved in the direction of an arrow **i1** (i.e., is moved forward), and the second reciprocating member **952** is further moved in the direction of an arrow **i2** (i.e., is moved backward). Then, since the first ratchet **943a** of the ratchet member **943** is in close contact with the recess **951b** of the first reciprocating member **951**, the ratchet member **943** is linearly moved in the direction of an arrow **i4** by the first reciprocating member **951**.

[0769] FIG. **121** is a view illustrating by extracting some operations of FIG. **120**, and is a view in which the forward moving motion of the operation member **940** is mainly described.

[0770] FIG. **121A** is a view illustrating the state in which the first ratchet **943a** of the ratchet member **943** and the first reciprocating member **951** come into close contact with each other, and FIG. **121E** is a view illustrating the state in which the second ratchet **943b** of the ratchet member

943 and the second reciprocating member **952** come into close contact with each other, and FIG. **1211** is a view illustrating the state in which the first ratchet **943a** of the ratchet member **943** and the first reciprocating member **951** come into close contact with each other.

[0771] In the state of FIG. **121A**, when the staple pulley **961** is rotated in the counterclockwise direction (in the opposite direction of the arrow **A1** of FIG. **114**), the first reciprocating member **951** is moved in the direction of an arrow **e1** (i.e., is moved backward), and the second reciprocating member **952** is moved in the direction of an arrow **e2** (i.e., is moved forward) as shown in FIG. **121E**.

[0772] Then, the recess **951b** of the first reciprocating member **951** pushes the first ratchet **943a** of the ratchet member **943** toward the second reciprocating member **952**, which causes the entire ratchet member **943** to be rotationally moved in the direction of an arrow **e3**, so that the first ratchet **943a** of the ratchet member **943** is spaced apart from the recess **951b** of the first reciprocating member **951**, and the second ratchet **943b** of the ratchet member **943** comes into close contact with the recess **952b** of the second reciprocating member **952**.

[0773] In this state, when the staple pulley **961** is further rotated in the counterclockwise direction (in the opposite direction of the arrow **A1** of FIG. **114**), since the second ratchet **943b** of the ratchet member **943** is in a state of being in close contact with the recess **952b** of the second reciprocating member **952**, the ratchet member **943** is linearly moved in the direction of an arrow **e4** by the second reciprocating member **952**.

[0774] Meanwhile, in the state of FIG. **121E**, when the staple pulley **961** is rotated in the clockwise direction (in the direction of the arrow **A1** of FIG. **114**), as shown in FIG. **1211**, the first reciprocating member **951** is moved in the direction of an arrow **i1** (i.e., is moved forward), and the second reciprocating member **952** is moved in the direction of an arrow **i2** (i.e., is moved backward).

[0775] Then, the recess **952b** of the second reciprocating member **952** pushes the second ratchet **943b** of the ratchet member **943** toward the first reciprocating member **951**, which causes the entire ratchet member **943** to be rotationally moved in the direction of an arrow **i3**, so that the second ratchet **943b** of the ratchet member **943** is spaced apart from the recess **952b** of the second reciprocating member **952**, and the first ratchet **943a** of the ratchet member **943** is brought into close contact with the recess **951b** of the first reciprocating member **951**.

[0776] In this state, when the staple pulley **961** is further rotated in the clockwise direction (in the direction of the arrow **A1** of FIG. **114**), since the first ratchet **943a** of the ratchet member **943** is in a state of being in close contact with the recess **951b** of the first reciprocating member **951**, the ratchet member **943** is linearly moved in the direction of an arrow **i4** by the first reciprocating member **951**.

[0777] Hereinafter, the overall operation of the present embodiment will be described.

[0778] FIGS. **118** and **119** are perspective views illustrating respective operating states of the end tool of FIG. **115**. Here, FIG. **118** is a view illustrating when the staple pulley is rotated in the clockwise direction, and FIG. **119** is a view illustrating when the staple pulley is rotated in the counterclockwise direction.

[0779] FIG. **118A** illustrates a state in which the first ratchet **943a** of the ratchet member **943** and the first reciprocating member **951** come into close contact with each other. In this state, when the staple pulley **961** is continuously rotated in a direction **b1** of FIG. **118B** and a direction **c1** of FIG. **118C**, the first reciprocating member **951** is moved in a direction **b2** of FIG. **118B** and a direction **c2** of FIG. **118C** by the first link member **971** connected to the staple pulley **961**. At the same time, the second reciprocating member **952** is moved in a direction **b3** of FIG. **118B** and a direction **c3** of FIG. **118C** by the second link member **976** connected to the staple pulley **961**. In addition, since the first ratchet **943a** of the ratchet member **943** is in close contact with the recess **951b** of the first reciprocating member **951**, the ratchet member **943** is linearly moved in a direction **b4** of FIG. **118B** and a direction **c4** of FIG. **118C** by the first reciprocating member **951**.

[0780] FIG. 119A illustrates a state in which the second ratchet **943b** of the ratchet member **943** and the second reciprocating member **952** come into close contact with each other. In this state, when the staple pulley **961** is continuously rotated in a direction **b1** of FIG. 119B and a direction **c1** of FIG. 118C, the first reciprocating member **951** is moved in a direction **b2** of FIG. 119B and a direction **c2** of FIG. 119C by the first link member **971** connected to the staple pulley **961**. At the same time, the second reciprocating member **952** is moved in a direction **b3** of FIG. 119B and a direction **c3** of FIG. 119C by the second link member **976** connected to the staple pulley **961**. In addition, since the second ratchet **943b** of the ratchet member **943** is in close contact with the recess **952b** of the second reciprocating member **952**, the ratchet member **943** is linearly moved in a direction **b4** of FIG. 119B and a direction **c4** of FIG. 119C by the second reciprocating member **952**.

[0781] In conclusion, when the staple pulley **961** is rotated in one direction, the first reciprocating member **951** is moved forward and the second reciprocating member **952** is moved backward, and in this case, the backward-moving second reciprocating member **952** pushes the operation member **940** toward the first reciprocating member **951** to bring the operation member **940** into close contact with the first reciprocating member **951**, and the forward-moving first reciprocating member **951** moves the operation member **940**, which is in close contact therewith, forward.

[0782] Meanwhile, when the staple pulley **961** is rotated in the other direction, the second reciprocating member **952** is moved forward and the first reciprocating member **951** is moved backward, and in this case, the backward-moving first reciprocating member **951** pushes the operation member **940** toward the second reciprocating member **952** to bring the operation member **940** into close contact with the second reciprocating member **952**, and the forward-moving second reciprocating member **952** moves the operation member **940**, which is in close contact therewith, forward.

[0783] As a result, when the staple pulley **961** is rotated in one direction, the first reciprocating member **951** moves the operation member **940** forward, and when the staple pulley **961** is rotated in the opposite direction, the second reciprocating member **952** moves the operation member **940** forward, and thus a forward-moving speed of the operation member **940** is approximately doubled as compared to the first embodiment, thereby reducing a stapling and cutting time.

Fifth Embodiment-Pin/Slot Type

[0784] Hereinafter, an end tool **1000** of a surgical instrument according to a fifth embodiment of the present disclosure will be described. Here, the end tool **1000** of the surgical instrument according to the fifth embodiment of the present disclosure is different from the end tool (see **100** of FIG. 2 or the like) of the surgical instrument according to the first embodiment of the present disclosure described above in that a configuration of a staple pulley assembly **1060** and a staple link assembly **1070** is different. Hereinafter, the configuration that is different from that of the first embodiment will be described in detail.

[0785] FIG. 122 is a perspective view illustrating the end tool of the surgical instrument according to the fifth embodiment of the present disclosure, and FIG. 123 is a magnified view of the end tool of the surgical instrument of FIG. 122. FIG. 124 is a side view illustrating the end tool of the surgical instrument of FIG. 122. FIGS. 125 and 126 are exploded perspective views of the end tool of the surgical instrument of FIG. 122. FIG. 127 is an exploded perspective view of a staple pulley assembly of the surgical instrument of FIG. 122. FIG. 128 is a plan view illustrating operating states of a staple pulley in the end tool of FIG. 122. FIG. 129 is a perspective view illustrating opening and closing motions of the end tool of the surgical instrument of FIG. 122, and FIG. 130 is a plan view of FIG. 129. FIGS. 131 and 132 are plan views illustrating opening and closing motions of the end tool of the surgical instrument of FIG. 122. FIGS. 133 and 134 are views illustrating a process in which the end tool of the surgical instrument of FIG. 122 is switched from a deactivated state to an activated state. FIGS. 135 to 139 are views illustrating a process of performing stapling and cutting motions as the end tool of the surgical instrument of FIG. 122 is

switched from the deactivated state to the activated state.

[0786] Here, FIG. **123** illustrates a state in which an end tool hub is removed. FIGS. **128** and **129** illustrate a state in which the end tool hub and a jaw pulley are removed. FIGS. **131** to **134** mainly describe operations of the staple pulley assembly, the staple link assembly, and a reciprocating assembly. FIGS. **135** to **139** mainly describe a coupling relationship between the reciprocating member and an operation member.

[0787] Referring to FIGS. **122** to **139**, the end tool **1000** of the fifth embodiment of the present disclosure includes a pair of jaws **1003** for performing a grip motion, that is, a first jaw **1001** and a second jaw **1002**. Here, each of the first jaw **1001** and the second jaw **1002**, or a component encompassing the first jaw **1001** and the second jaw **1002** may be referred to as the jaw.

[0788] Meanwhile, the end tool **1000** includes a plurality of pulleys including a pulley **1011** and a pulley **1012** that are related to a rotational motion of the first jaw **1001**. The pulleys related to the rotational motion of the first jaw **1001** described in the present embodiment are substantially the same as the pulley **111**, the pulley **112**, the pulley **113**, the pulley **114**, the pulley **115**, and the pulley **116** described with reference to FIG. **8** or the like of the first embodiment, and thus, detailed descriptions thereof will be omitted herein.

[0789] Meanwhile, the end tool **1000** includes a plurality of pulleys including a pulley **1021** and a pulley **1022** that are related to a rotational motion of the second jaw **1002**. The pulleys related to the rotational motion of the second jaw **1002** described in the present embodiment are substantially the same as the pulley **121**, the pulley **122**, the pulley **123**, the pulley **124**, the pulley **125**, and the pulley **126** described with reference to FIG. **8** or the like of the first embodiment, and thus, detailed descriptions thereof will be omitted herein.

[0790] In addition, the end tool **1000** of the fifth embodiment of the present disclosure may include a rotation shaft **1041**, a rotation shaft **1042**, a rotation shaft **1043**, and a rotation shaft **1044**. Here, the rotation shaft **1041** and the rotation shaft **1042** may be inserted through an end tool hub **1080**, and the rotation shaft **1043** and the rotation shaft **1044** may be inserted through a pitch hub **1007**. The rotation shaft **1041**, the rotation shaft **1042**, the rotation shaft **1043**, and the rotation shaft **1044** may be arranged sequentially from a distal end **1004** of the end tool **1000** toward a proximal end **1005**.

[0791] In addition, the end tool **1000** of the fifth embodiment of the present disclosure may include the end tool hub **1080** and the pitch hub **1007**.

[0792] The rotation shaft **1041** and the rotation shaft **1042** may be inserted through the end tool hub **1080**, and the pulley **1011** and the pulley **1021** axially coupled to the rotation shaft **1041** and at least some of the first jaw **1001** and the second jaw **1002** coupled to the pulley **1011** and the pulley **1021** may be accommodated inside the end tool hub **1080**.

[0793] The rotation shaft **1043** and the rotation shaft **1044** may be inserted through the pitch hub **1007**, and the pitch hub **1007** may be axially coupled to the end tool hub **1080** by the rotation shaft **1043**. Accordingly, the end tool hub **1080** may be formed to be pitch-rotatable around the rotation shaft **1043** with respect to the pitch hub **1007**.

[0794] Meanwhile, the end tool **1000** of the fifth embodiment of the present disclosure may further include components, such as a staple drive assembly (see **150** of FIG. **13**) including the staple pulley assembly **1060** and the staple link assembly **1070**, to perform stapling and cutting motions.

[0795] The staple pulley assembly **1060** may be formed between the pulley **1011** and the pulley **1021** to be adjacent to the pulley **1011** and the pulley **1021**. In the present embodiment, it is assumed that the staple pulley assembly **1060** includes one staple pulley **1061**.

[0796] In the fifth embodiment of the present disclosure, by disposing the staple pulley assembly **1060** between the pulley **1011**, which is a first jaw pulley, and the pulley **1021**, which is a second jaw pulley, the end tool **1000** is allowed to perform pitch and yaw motions, as well as stapling and cutting motions using a cartridge **1010**.

[0797] Hereinafter, the staple pulley assembly **1060**, the staple link assembly **1070**, and a

reciprocating assembly **1050** of the end tool **1000** of the surgical instrument according to the fifth embodiment of the present disclosure will be described in more detail.

[0798] In the end tool **1000** of the surgical instrument according to the fifth embodiment of the present disclosure, the staple pulley assembly **1060** and the staple link assembly **1070** form a pin-slot structure. In addition, with such a structure, a force for moving the reciprocating assembly **550** forward may be amplified. Furthermore, a deactivated state, in which the staple link assembly **1070** and the reciprocating assembly **1050** are separated from each other, may be switched to an activated state in which the staple link assembly **1070** and the reciprocating assembly **1050** are coupled to each other.

[0799] Referring to FIGS. **122** to **139** and the like, the staple pulley assembly **1060** may include one or more staple pulleys **1061**.

[0800] A shaft pass-through part **1061a** may be formed in the staple pulley **1061**. The shaft pass-through part **1061a** may be formed in the form of a hole, and the rotation shaft **1041**, which is an end tool jaw pulley rotation shaft, may be inserted through the shaft pass-through part **1061a**.

[0801] Further, a protruding member **1061b** may be formed on the staple pulley **1061**. A link member **1071** of the staple link assembly **1070** may be coupled to the protruding member **1061b**. Here, the protruding member **1061b** is formed in the form of a pin, and may be fitted into a slot **1072d** of the link member **1071**, which will be described later.

[0802] Meanwhile, the end tool **1000** of the fifth embodiment of the present disclosure may further include the staple link assembly **1070** connected to the staple pulley assembly **1060**, and the staple link assembly **1070** may include the link member **1071**. Here, the staple link assembly **1070** may serve to connect the staple pulley assembly **1060** to the reciprocating assembly **1050** of the cartridge **1010** to be described later.

[0803] In the present embodiment, the staple link assembly **1070** includes only one link. That is, by coupling the staple pulley assembly **1060** and the staple link assembly **1070** by a pin-slot structure, it is possible to convert a rotational motion of the staple pulley assembly **1060** into a linear motion of the staple link assembly **1070** even when the staple link assembly **1070** includes only one link.

[0804] In detail, the link member **1071** may be formed as a single link, which is a first link **1072**. In other words, the first link **1072** may be formed as a single member.

[0805] The first link **1072** is formed in the form of an elongated bar, and may include a first protrusion **1072a**, a second protrusion **1072b**, a coupling part **1072c**, and the slot **1072d**.

[0806] The first protrusion **1072a** and the second protrusion **1072b** may be formed in one region of a central portion of the first link **1072**. The first protrusion **1072a** and the second protrusion **1072b** may be fitted into a guide groove **1001b** of the first jaw **1001**.

[0807] As described above, as the first protrusion **1072a** and the second protrusion **1072b** are moved along the guide groove **1001b** in a state in which the first protrusion **1072a** and the second protrusion **1072b** of the first link **1072** formed in a protruding shape are fitted into the groove-shaped guide groove **1001b**, the link member **1071** is moved with respect to the first jaw **1001** (and the cartridge **500** therein). This will be described in more detail later.

[0808] Meanwhile, the coupling part **1072c** may be formed at one end portion of the first link **1072**. The coupling part **1072c** may be coupled to a coupling part **551a** of a reciprocating member **551** of the cartridge **500**.

[0809] Meanwhile, the slot **1072d** may be formed at one end portion of the first link **1072**. Here, the slot **1072d** may be formed in the form of an elongated hole, into which the protruding member **1061b** in the form of a pin may be fitted. In detail, the slot **1072d** may be formed to have a predetermined curvature, and may be formed in an approximately elliptical shape. Here, a short radius of the slot **1072d** may be formed to be substantially the same as or slightly greater than a radius of the protruding member **1061b**. Meanwhile, a long radius the slot **1072d** may be formed to be greater than a radius of the protruding member **1061b**. Accordingly, in a state in which the protruding member **1061b** of the staple pulley **1061** is fitted into the slot **1072d** of the first link

1072, the protruding member **1061b** is moved to a certain extent in the slot **1072d**.
[0810] Here, the slot **1072d** may be formed obliquely without being concentric with the staple pulley **1061**. Accordingly, when the staple pulley **1061** is rotated, the protruding member **1061b**, while in contact with the slot **1072d**, may push the slot **1072d** to move the first link **1072**. That is, when the staple pulley **1061** is rotated, the protruding member **1061b** is moved in the slot **1072d** while coming into contact with the slot **1072d**, which causes the first link **1072** to be linearly moved along the guide groove **1001b** of the first jaw **1001**.

(Displacement of Staple Link Assembly According to Rotation of Staple Pulley)

[0811] Hereinafter, the displacement of the staple link assembly **1070** according to the rotation of the staple pulley **1061** will be described.

[0812] Referring to FIG. **129**, in the fifth embodiment of the present disclosure, the staple pulley **1061** and the staple link assembly **1070** are coupled in the form of a pin-slot. That is, the protruding member **1061b** formed in the form of a pin on the staple pulley **1061** is coupled to the slot **1072d** formed in the staple link assembly **1070**. Accordingly, when the staple pulley **1061** is rotated in the direction of an arrow R, the displacement of the protruding member **1061b** of the staple pulley **1061** in the X-axis direction becomes D1. In addition, the displacement of the staple link assembly **1070** in the X-axis direction becomes D2.

[0813] Meanwhile, referring to FIG. **38** or the like, in the first embodiment of the present disclosure, the staple pulley **161** and the staple link assembly **170** are axially coupled to each other and thus the displacement of the protruding member **1061b** of the staple pulley **1061** in the X-axis direction becomes much greater than that of the fourth embodiment.

[0814] In other words, as compared with the case in which the staple pulley and the staple link assembly are axially coupled as in the first embodiment of the present disclosure or the like, when the staple pulley and the staple link assembly are pin-slot coupled with each other as in the present embodiment, the displacement of the staple link assembly in the X-axis direction is reduced even when the staple pulley is rotated by the same distance.

[0815] Meanwhile, since work is the product of force and displacement, assuming that the work for rotating the staple pulley is the same, the displacement and the force are inversely proportional to each other. Accordingly, when the displacement is reduced, the force is increased in inverse proportion to the displacement.

[0816] As a result, in the fifth embodiment of the present disclosure, the staple pulley **1061** and the staple link assembly **1070** are coupled in the form of a pin-slot, and the displacement of the staple link assembly **1070** in the X-axis direction caused by the rotation of the staple pulley **1061** is relatively reduced compared to the other embodiments, and thus the force received by the staple link assembly **1070** in the X-axis direction is relatively increased compared to the other embodiments.

[0817] According to the fifth embodiment of the present disclosure described above, a force for moving the staple link assembly **1070** and the reciprocating assembly **550** connected thereto forward is amplified, and thus a stapling motion may be performed more robustly.

(Activated State and Deactivated State)

[0818] Hereinafter, the activated state and the deactivated state according to the present embodiment will be described.

[0819] In the first embodiment of the present disclosure or the like described above, the staple link assembly and the reciprocating assembly are always in the coupled state, and thus the reciprocating assembly is always linearly moved when the staple pulley is rotated. In addition, the staple pulley is dependent on the jaw pulley, and thus, when the jaw pulley is rotated, i.e., during a yaw or actuation motion, the staple pulley is also rotated. As a result, the reciprocating assembly is also moved forward/backward during the yaw or actuation motion, and in this process, the operation member may be moved. Accordingly, when the yaw or actuation motion is repeated, an undesired stapling motion may be performed.

[0820] In order to address such a problem, in the fifth embodiment of the present disclosure, there are a deactivated state in which the reciprocating assembly **550** and the operation member **540** are not coupled to and separated from each other, and an activated state in which the reciprocating assembly **550** and the operation member **540** are coupled to each other. In addition, the deactivated state may be maintained in a jaw open state, and the deactivated state may be switched to the activated state in a jaw close state. In addition, after switching to the activated state, as the reciprocating assembly is repeatedly moved forward and backward according to an alternating rotational motion of the staple pulley, the operation member is moved forward.

[0821] This distinction between the activated and deactivated states may be made possible by the pin-slot structure described above. Hereinafter, this will be described in more detail.

[0822] Referring to FIGS. **130** to **132** and the like, in a state in which the jaw **1003** is closed as shown in FIG. **131**, when the first jaw **1001** is rotated in the direction of an arrow R of FIG. **132** and the second jaw **1002** is rotated in the opposite direction of the arrow R, the jaw **1003** is in an opened state. Here, it is assumed that the position of a jaw rotation shaft **1045** in the X-axis direction is fixed, while the other components move.

[0823] When the jaw **1003** is in the opened state, the first jaw pulley **1011** and the staple pulley **1061** are rotated in the same direction.

[0824] When the first jaw pulley **1011** is rotated in the direction of the arrow R of FIG. **132**, the pulley **1011**, the staple pulley **1061**, the staple link assembly **1070**, and the reciprocating assembly **550** are moved together toward a distal end **1001f** (i.e., are moved forward) in a state in which the jaw rotation shaft **1045** is fixed. Accordingly, A1 of FIG. **131**, which is a distance between the rotation shaft **1041** and the jaw rotation shaft **1045** in a state in which the jaw **1003** is closed, is greater than A2 of FIG. **132**, which is a distance between the rotation shaft **1041** and the jaw rotation shaft **1045** in a state in which the jaw **1003** is opened (i.e., $A1 > A2$).

[0825] At the same time, when the staple pulley **1061** is rotated in the direction of the arrow R in FIG. **132**, the staple link assembly **1070** connected to the staple pulley **1061** and the reciprocating assembly **550** connected to the staple link assembly **1070** are moved as a whole (i.e., are moved backward) toward a proximal end **1001g**. Accordingly, B1 of FIG. **131**, which is a distance between the rotation shaft **1041** and the staple link assembly **1070** (the second protrusion **1072b** thereof) in a state in which the jaw **1003** is closed, is greater than B2 of FIG. **132**, which is a distance between the rotation shaft **1041** and the staple link assembly **1070** (the second protrusion **1072b** thereof) in a state in which the jaw **1003** is opened (i.e., $B1 > B2$).

[0826] However, here, due to the pin-slot structure as described above, the displacement of the staple link assembly **1070** in the x-axis direction when the staple pulley **1061** is rotated is relatively small (compared to other embodiments).

[0827] That is, in the fifth embodiment of the present disclosure, a relationship of $(A1 - A2) > (B1 - B2)$ is established.

[0828] In other words, when the jaw **1003** is opened, the movement amount $(A1 - A2)$ of the pulley **1011**, the staple pulley **1061**, the staple link assembly **1070**, and the reciprocating assembly **550** toward the distal end is greater than the movement amount $(B1 - B2)$ of the staple link assembly **1070** and the reciprocating assembly **550** connected thereto toward the proximal end.

[0829] As a result, when the two movement components of the reciprocating assembly **550** described above are synthesized, the reciprocating assembly **550** is moved toward the distal end **1001f** when the jaw **1003** is opened. That is, the reciprocating assembly **550** is moved forward.

[0830] That is, in the first embodiment, the reciprocating member is moved backward when the moving-backward amount of the reciprocating assembly due to the rotation of the staple pulley is larger than the moving-forward amount of the reciprocating assembly due to the rotation of the jaw pulley, whereas in the present embodiment, the reciprocating member is moved forward when the moving-forward amount of the reciprocating assembly due to the rotation of the jaw pulley is larger than the moving-backward amount of the reciprocating assembly due to the rotation of the staple

pulley.

[0831] In addition, when the reciprocating assembly **550** is moved forward as described above, the reciprocating assembly **550** and the operation member **540** are further spaced apart from each other, and thus, when the jaw is opened, the deactivated state in which the reciprocating assembly **550** and the operation member **540** are not coupled is maintained.

[0832] In contrast, in a state in which the jaw **1003** is opened as shown in FIG. **132**, the first jaw **1001** is rotated in the opposite direction of the arrow R of FIG. **132**, and when the second jaw **1002** is rotated in the direction of the arrow R, the jaw **1003** is in a closed state as shown in FIG. **131**.

[0833] When the jaw **1003** is closed as described above, the reciprocating assembly **550** is moved toward the proximal end **1001g**. That is, the reciprocating assembly **550** is moved backward.

[0834] In the closed state as shown in FIG. **131**, a recess **550b** of the reciprocating assembly **550** and a ratchet **543a** of the ratchet member **543** of the operation member **540** are in contact with each other but are not coupled to each other.

[0835] Accordingly, the deactivated state in which the reciprocating assembly **550** and the operation member **540** are not coupled to and separated from each other is maintained between the state of FIG. **131**, in which the jaw **1003** is fully opened, and the state of FIG. **132**, in which the jaw **1003** is closed. That is, in this state, even when the staple pulley **1061** is rotated, the operation member **540** is not moved, and thus stapling and cutting motions are not performed.

[0836] Next, referring to FIGS. **133** and **134** and the like, in a state in which the jaw **1003** is closed as shown in FIG. **133**, the staple pulley **1061** is rotated in the direction of an arrow R of FIG. **134** for stapling and cutting motions.

[0837] When the staple pulley **1061** is rotated in the direction of the arrow R, the staple link assembly **1070** connected to the staple pulley **1061** and the reciprocating assembly **550** connected to the staple link assembly **1070** are moved as a whole toward the proximal end **1001g** (i.e., are moved backward). In addition, as shown in FIG. **134**, when the reciprocating assembly **550** is moved toward the proximal end **1001g** to be coupled to the ratchet member **543** of the operation member **540**, the deactivated state is switched to the activated state.

[0838] After switching to the activated state as described above, as the staple pulley **1061** performs an alternating rotational motion, the reciprocating assembly **550** is repeatedly moved forward and backward, which causes the operation member **540** to move forward.

[0839] FIGS. **135** to **139** are views illustrating a process of performing stapling and cutting motions as the end tool of the surgical instrument of FIG. **122** is switched from the deactivated state to the activated state. Here, (a) of each drawing illustrates a state in which the jaw **1003** is opened, (b) of each drawing illustrates a state in which the jaw **1003** is closed, and (c) of each drawing illustrates a state in which the staple pulley **1061** is rotated in the state in which the jaw **1003** is closed.

[0840] FIG. **135** is a view illustrating the entire process of performing the stapling and cutting motions as the end tool of the surgical instrument of FIG. **122** is switched from the deactivated state to the activated state.

[0841] In a state in which the jaw **1003** is opened as shown in FIG. **135A**, the end tool is switched to the deactivated state in which the reciprocating assembly **550** and the operation member **540** are not coupled to and separated from each other. That is, as compared to the state of FIG. **135B** in which the jaw **1003** is closed, in the state of FIG. **135A** in which the jaw **1003** is opened, the reciprocating assembly **550** is moved forward toward the distal end **1001f**. Accordingly, the reciprocating assembly **550** and the operation member **540** are in a state of being maximally spaced apart from each other.

[0842] Meanwhile, in the closed state as shown in FIG. **135B**, the recess **550b** of the reciprocating assembly **550** and the ratchet **543a** of the ratchet member **543** of the operation member **540** are in contact with each other but are not coupled to each other. That is, even in this state, the end tool remains in the deactivated state in which the reciprocating assembly **550** and the operation member **540** are not coupled to and separated from each other.

[0843] Meanwhile, in a state in which the jaw **1003** is closed as shown in FIG. **135B**, when the staple pulley **1061** is rotated in the direction of the arrow R as shown in FIG. **135C**, the reciprocating assembly **550** is moved toward the proximal end **1001g** to be coupled to the ratchet member **543** of the operation member **540**, so that the deactivated state is switched to the activated state.

[0844] After switching to the activated state as described above, as the staple pulley **1061** performs an alternating rotational motion, the reciprocating assembly **550** is repeatedly moved forward and backward, which causes the operation member **540** to move forward.

[0845] FIG. **136** is a view illustrating opened/closed states of the jaw **1003** in the end tool of the surgical instrument of FIG. **122**.

[0846] In a state in which the jaw **1003** is opened as shown in FIG. **136A**, the end tool is switched to the deactivated state in which the reciprocating assembly **550** and the operation member **540** are not coupled to and separated from each other. That is, as compared to the state of FIG. **136B** in which the jaw **1003** is closed, in the state of FIG. **136A** in which the jaw **1003** is opened, the reciprocating assembly **550** is moved forward toward the distal end **1001f**. Accordingly, the reciprocating assembly **550** and the operation member **540** are in a state of being maximally spaced apart from each other.

[0847] Meanwhile, in the closed state as shown in FIG. **136B**, the recess **550b** of the reciprocating assembly **550** and the ratchet **543a** of the ratchet member **543** of the operation member **540** are in contact with each other but are not coupled to each other. That is, even in this state, the end tool remains in the deactivated state in which the reciprocating assembly **550** and the operation member **540** are not coupled to and separated from each other.

[0848] FIG. **137** is a view illustrating a state in which the reciprocating assembly **550** and the operation member **540** are coupled to each other as the staple pulley is rotated in a state in which the jaw **1003** is closed, in the end tool of the surgical instrument of FIG. **122**.

[0849] In the state in which the jaw **1003** is closed as shown in FIG. **137B**, when the staple pulley **1061** is rotated in the direction of the arrow R as shown in FIG. **137C**, the reciprocating assembly **550** is moved toward the proximal end **1001g** to be coupled to the ratchet member **543** of the operation member **540**, so that the deactivated state is switched to the activated state.

[0850] FIG. **138** is a view illustrating a motion in which the operation member **540** is initially moved forward after the end tool of the surgical instrument of FIG. **122** is switched to the activated state, and FIG. **139** is a view illustrating a motion in which the operation member **540** is gradually moved forward as the staple pulley **161** performs an alternating rotational motion in the end tool of the surgical instrument of FIG. **122**.

[0851] After switching to the activated state as described above, as the staple pulley **1061** performs an alternating rotational motion, the reciprocating assembly **550** is repeatedly moved forward and backward, thereby performing stapling and cutting motions while the operation member **540** is moved forward.

Sixth Embodiment-Cam/Slot-Type

[0852] Hereinafter, an end tool **1100** of a surgical instrument according to a sixth embodiment of the present disclosure will be described. Here, the end tool **1100** of the surgical instrument according to the sixth embodiment of the present disclosure is different from the end tool (see **100** of FIG. **2** or the like) of the surgical instrument according to the fifth embodiment of the present disclosure described above in that a configuration of a staple pulley assembly **1160** and a staple link assembly **1170** is different. Hereinafter, the configuration that is different from that of the fifth embodiment will be described in detail.

[0853] FIG. **140** is a perspective view illustrating the end tool of the surgical instrument according to the sixth embodiment of the present disclosure, and FIG. **141** is a magnified view of the end tool of the surgical instrument of FIG. **140**. FIG. **142** is a side view illustrating the end tool of the surgical instrument of FIG. **140**. FIGS. **143** and **144** are exploded perspective views of the end tool

of the surgical instrument of FIG. 140. FIG. 145 is an exploded perspective view of a staple pulley assembly of the surgical instrument of FIG. 140. FIG. 146 is a side view illustrating operating states of a staple pulley in the end tool of FIG. 140. FIGS. 147 and 148 are plan views illustrating opening and closing motions of the end tool of the surgical instrument of FIG. 140. FIG. 149 is a perspective view illustrating a process in which the end tool of the surgical instrument of FIG. 140 is switched from a deactivated state to an activated state.

[0854] Here, FIG. 141 illustrates a state in which an end tool hub is removed. FIGS. 146 to 148 mainly describe operations of the staple pulley assembly, the staple link assembly, and a reciprocating assembly. FIG. 149 mainly describes a coupling relationship between the reciprocating member and an operation member.

[0855] Referring to FIGS. 140 to 149, the end tool 1100 of the sixth embodiment of the present disclosure includes a pair of jaws 1103 for performing a grip motion, that is, a first jaw 1101 and a second jaw 1102. Here, each of the first jaw 1101 and the second jaw 1102, or a component encompassing the first jaw 1101 and the second jaw 1102 may be referred to as the jaw.

[0856] Meanwhile, the end tool 1100 includes a plurality of pulleys including a pulley 1111 and a pulley 1112 that are related to a rotational motion of the first jaw 1101. The pulleys related to the rotational motion of the first jaw 1101 described in the present embodiment are substantially the same as the pulley 111, the pulley 112, the pulley 113, the pulley 114, the pulley 115, and the pulley 116 described with reference to FIG. 8 or the like of the first embodiment, and thus, detailed descriptions thereof will be omitted herein.

[0857] Meanwhile, the end tool 1100 includes a plurality of pulleys including a pulley 1121 and a pulley 1122 that are related to a rotational motion of the second jaw 1102. The pulleys related to the rotational motion of the second jaw 1102 described in the present embodiment are substantially the same as the pulley 121, the pulley 122, the pulley 123, the pulley 124, the pulley 125, and the pulley 126 described with reference to FIG. 8 or the like of the first embodiment, and thus, detailed descriptions thereof will be omitted herein.

[0858] In addition, the end tool 1100 of the sixth embodiment of the present disclosure may include a rotation shaft 1141, a rotation shaft 1142, a rotation shaft 1143, and a rotation shaft 1144. Here, the rotation shaft 1141 and the rotation shaft 1142 may be inserted through an end tool hub 1180, and the rotation shaft 1143 and the rotation shaft 1144 may be inserted through a pitch hub 1107. The rotation shaft 1141, the rotation shaft 1142, the rotation shaft 1143, and the rotation shaft 1144 may be arranged sequentially from a distal end 1104 of the end tool 1100 toward a proximal end 1105.

[0859] In addition, the end tool 1100 of the sixth embodiment of the present disclosure may include the end tool hub 1180 and the pitch hub 1107.

[0860] The rotation shaft 1141 and the rotation shaft 1142 may be inserted through the end tool hub 1180, and the pulley 1111 and the pulley 1121 axially coupled to the rotation shaft 1141 and at least some of the first jaw 1101 and the second jaw 1102 coupled to the pulley 1111 and the pulley 1121 may be accommodated inside the end tool hub 1180.

[0861] The rotation shaft 1143 and the rotation shaft 1144 may be inserted through the pitch hub 1107, and the pitch hub 1107 may be axially coupled to the end tool hub 1180 by the rotation shaft 1143.

[0862] Accordingly, the end tool hub 1180 may be formed to be pitch-rotatable around the rotation shaft 1143 with respect to the pitch hub 1107.

[0863] Meanwhile, the end tool 1100 of the sixth embodiment of the present disclosure may further include components, such as a staple drive assembly (see 150 of FIG. 13) including the staple pulley assembly 1160 and the staple link assembly 1170, to perform stapling and cutting motions.

[0864] The staple pulley assembly 1160 may be formed between the pulley 1111 and the pulley 1121 to be adjacent to the pulley 1111 and the pulley 1121. In the present embodiment, it is assumed that the staple pulley assembly 1160 includes one staple pulley 1161.

[0865] In the sixth embodiment of the present disclosure, by disposing the staple pulley assembly **1160** between the pulley **1111**, which is a first jaw pulley, and the pulley **1121**, which is a second jaw pulley, the end tool **1100** is allowed to perform pitch and yaw motions as well as stapling and cutting motions using a cartridge **1110**.

[0866] Hereinafter, the staple pulley assembly **1160**, the staple link assembly **1170**, and a reciprocating assembly **1150** of the end tool **1100** of the surgical instrument according to the sixth embodiment of the present disclosure will be described in more detail.

[0867] In the end tool **1100** of the surgical instrument according to the sixth embodiment of the present disclosure, the staple pulley assembly **1160** and the staple link assembly **1170** form a cam-slot structure. In addition, with such a structure, a force for moving the reciprocating assembly **550** forward may be amplified. Furthermore, a deactivated state, in which the staple link assembly **1170** and the reciprocating assembly **1150** are separated from each other, may be switched to an activated state in which the staple link assembly **1170** and the reciprocating assembly **1150** are coupled to each other.

[0868] Referring to FIGS. **122** to **139** and the like, the staple pulley assembly **1160** may include one or more staple pulleys **1161**.

[0869] A shaft pass-through part **1161a** may be formed in the staple pulley **1161**. The shaft pass-through part **1161a** may be formed in the form of a hole, and the rotation shaft **1141**, which is an end tool jaw pulley rotation shaft, may be inserted through the shaft pass-through part **1161a**.

[0870] Further, a protruding member **1161b** may be formed on the staple pulley **1161**. A link member **1171** of the staple link assembly **1170** may be coupled to the protruding member **1161b**. Here, the center of the protruding member **1161b** does not coincide with the center of the staple pulley **1161**, and the protruding member **1161b** may be formed to be eccentric to a certain extent with respect to the staple pulley **1161**. The protruding member **1161b** may be fitted into a slot **1172d** of the link member **1171** to be described later.

[0871] Meanwhile, the end tool **1100** of the sixth embodiment of the present disclosure may further include the staple link assembly **1170** connected to the staple pulley assembly **1160**, and the staple link assembly **1170** may include the link member **1171**. Here, the staple link assembly **1170** may serve to connect the staple pulley assembly **1160** to the reciprocating assembly **1150** of the cartridge **1110** to be described later.

[0872] In the present embodiment, the staple link assembly **1170** includes only one link. That is, by coupling the staple pulley assembly **1160** and the staple link assembly **1170** by a cam-slot structure, it is possible to convert a rotational motion of the staple pulley assembly **1160** into a linear motion of the staple link assembly **1170** even when the staple link assembly **1170** includes only one link.

[0873] In detail, the link member **1171** may be formed as a single link, which is a first link **1172**. In other words, the first link **1172** may be formed as a single member.

[0874] The first link **1172** is formed in the form of an elongated bar, and may include a first protrusion **1172a**, a second protrusion **1172b**, a coupling part **1172c**, and the slot **1172d**.

[0875] The first protrusion **1172a** and the second protrusion **1172b** may be formed in one region of a central portion of the first link **1172**. The first protrusion **1172a** and the second protrusion **1172b** may be fitted into a guide groove **1101b** of the first jaw **1101**.

[0876] As described above, as the first protrusion **1172a** and the second protrusion **1172b** are moved along the guide groove **1101b** in a state in which the first protrusion **1172a** and the second protrusion **1172b** of the first link **1172** formed in a protruding shape are fitted into the groove-shaped guide groove **1101b**, the link member **1171** is moved with respect to the first jaw **1101** (and the cartridge **500** therein). This will be described in more detail later.

[0877] Meanwhile, the coupling part **1172c** may be formed at one end portion of the first link **1172**. The coupling part **1172c** may be coupled to a coupling part **551a** of a reciprocating member **551** of the cartridge **500**, which will be described later.

[0878] Meanwhile, the slot **1172d** may be formed at one end portion of the first link **1172**. Here, the

slot **1172d** may be formed in the form of an elongated hole, into which the protruding member **1161b** may be fitted. In detail, the slot **1172d** may be formed to have a predetermined curvature, and may be formed in an approximately elliptical shape. Here, the slot **1172d** may be formed to be greater than the protruding member **1161b** by a certain extent. Accordingly, in a state in which the protruding member **1161b** of the staple pulley **1161** is fitted into the slot **1172d** of the first link **1172**, the protruding member **1161b** is moved to a certain extent in the slot **1172d**.

[0879] As described above, the protruding member **1161b** may be formed to be eccentric with respect to the staple pulley **1161** by a certain extent. Accordingly, when the staple pulley **1161** is rotated, the protruding member **1161b**, while in contact with the slot **1172d**, may push the slot **1172d** to move the first link **1172**. That is, when the staple pulley **1161** is rotated, the protruding member **1161b** is moved in the slot **1172d** while coming into contact with the slot **1172d**, which causes the first link **1172** to be linearly moved along the guide groove **1101b** of the first jaw **1101**. (Displacement of Staple Link Assembly According to Rotation of Staple Pulley)

[0880] Hereinafter, the displacement of the staple link assembly **1170** according to the rotation of the staple pulley **1161** will be described.

[0881] Referring to FIG. **146**, in the sixth embodiment of the present disclosure, the staple pulley **1161** and the staple link assembly **1170** are coupled in the form of a cam-slot. That is, the protruding member **1161b** formed on the staple pulley **1161** is coupled to the slot **1172d** formed in the form of a slot in the staple link assembly **1170**. Accordingly, when the staple pulley **1161** is rotated in the direction of an arrow R, the displacement of the protruding member **1161b** of the staple pulley **1161** in the X-axis direction becomes D1. In addition, the displacement of the staple link assembly **1170** in the X-axis direction becomes D2.

[0882] Meanwhile, referring to FIG. **38** or the like, in the first embodiment of the present disclosure, the staple pulley **161** and the staple link assembly **170** are axially coupled to each other and thus the displacement of the protruding member **1161b** of the staple pulley **1161** in the X-axis direction becomes much greater than that of the fourth embodiment.

[0883] In other words, as compared with the case in which the staple pulley and the staple link assembly are axially coupled as in the first embodiment of the present disclosure or the like, when the staple pulley and the staple link assembly are cam-slot coupled with each other as in the present embodiment, the displacement of the staple link assembly in the X-axis direction is reduced even when the staple pulley is rotated by the same distance.

[0884] Meanwhile, since work is the product of force and displacement, assuming that the work for rotating the staple pulley is the same, the displacement and the force are inversely proportional to each other. Accordingly, when the displacement is reduced, the force is increased in inverse proportion to the displacement.

[0885] As a result, in the sixth embodiment of the present disclosure, the staple pulley **1161** and the staple link assembly **1170** are coupled in the form of a cam-slot, and the displacement of the staple link assembly **1170** in the X-axis direction caused by the rotation of the staple pulley **1161** is relatively reduced compared to the other embodiments, and thus the force received by the staple link assembly **1170** in the X-axis direction is relatively increased compared to the other embodiments.

[0886] According to the sixth embodiment of the present disclosure described above, a force for moving the staple link assembly **1170** and the reciprocating assembly **550** connected thereto forward is amplified, and thus a stapling motion may be performed more robustly.

(Activated State and Deactivated State)

[0887] Hereinafter, the activated state and the deactivated state according to the present embodiment will be described.

[0888] In the sixth embodiment of the present disclosure, there is a deactivated state in which the reciprocating assembly **550** and an operation member **540** are not coupled to and separated from each other, and an activated state in which the reciprocating assembly **550** and the operation

member **540** are coupled. In addition, the deactivated state may be maintained in a jaw open state, and the deactivated state may be switched to the activated state in a jaw close state. In addition, after switching to the activated state, as the reciprocating assembly is repeatedly moved forward and backward according to an alternating rotational motion of the staple pulley, the operation member is moved forward.

[0889] This distinction between the activated and deactivated states may be made possible by the cam-slot structure described above. Hereinafter, this will be described in more detail.

[0890] Referring to FIGS. **147** to **149** and the like, in a state in which the jaw **1103** is closed as shown in FIG. **147**, when the first jaw **1101** is rotated in the direction of an arrow R of FIG. **148** and the second jaw **1102** is rotated in the opposite direction of the arrow R, the jaw **1103** is in an opened state. Here, it is assumed that the position of a jaw rotation shaft **1145** in the X-axis direction is fixed, while the other components move.

[0891] When the jaw **1103** is in the opened state, the first jaw pulley **1111** and the staple pulley **1161** are rotated in the same direction.

[0892] When the first jaw pulley **1111** is rotated in the direction of the arrow R of FIG. **148**, the pulley **1111**, the staple pulley **1161**, the staple link assembly **1170**, and the reciprocating assembly **550** are moved together toward a distal end **1101f** (i.e., are moved forward) in a state in which the jaw shaft **1145** is fixed. Accordingly, A1 of FIG. **147**, which is a distance between the rotation shaft **1141** and the jaw rotation shaft **1145** in a state in which the jaw **1103** is closed, is greater than A2 of FIG. **148**, which is a distance between the rotation shaft **1141** and the jaw rotation shaft **1145** in a state in which the jaw **1103** is opened (i.e., $A1 > A2$).

[0893] At the same time, when the staple pulley **1161** is rotated in the direction of the arrow R of FIG. **148**, the staple link assembly **1170** connected to the staple pulley **1161** and the reciprocating assembly **550** connected to the staple link assembly **1170** are moved as a whole (i.e., are moved backward) toward a proximal end **1101g**. Accordingly, B1 of FIG. **147**, which is a distance between the rotation shaft **1141** and the staple link assembly **1170** (the second protrusion **1172b** thereof) in a state in which the jaw **1103** is closed, is greater than B2 of FIG. **148**, which is a distance between the rotation shaft **1141** and the staple link assembly **1170** (the second protrusion **1172b** thereof) in a state in which the jaw **1103** is opened (i.e., $B1 > B2$).

[0894] However, here, due to the cam-slot structure as described above, the displacement of the staple link assembly **1170** in the x-axis direction when the staple pulley **1161** is rotated is relatively small (compared to other embodiments).

[0895] That is, in the sixth embodiment of the present disclosure, a relationship of $(A1 - A2) > (B1 - B2)$ is established.

[0896] In other words, when the jaw **1103** is opened, the movement amount $(A1 - A2)$ of the pulley **1111**, the staple pulley **1161**, the staple link assembly **1170**, and the reciprocating assembly **550** toward the distal end is greater than the movement amount $(B1 - B2)$ of the staple link assembly **1170** and the reciprocating assembly **550** connected thereto toward the proximal end.

[0897] As a result, when the two movement components of the reciprocating assembly **550** described above are synthesized, the reciprocating assembly **550** is moved toward the distal end **1101f** when the jaw **1103** is opened. That is, the reciprocating assembly **550** is moved forward.

[0898] That is, in the first embodiment, the reciprocating member is moved backward when the moving-backward amount of the reciprocating assembly due to the rotation of the staple pulley is larger than the moving-forward amount of the reciprocating assembly due to the rotation of the jaw pulley, whereas in the present embodiment, the reciprocating member is moved forward when the moving-forward amount of the reciprocating assembly due to the rotation of the jaw pulley is larger than the moving-backward amount of the reciprocating assembly due to the rotation of the staple pulley.

[0899] In addition, when the reciprocating assembly **550** is moved forward as described above, the reciprocating assembly **550** and the operation member **540** are further spaced apart from each other,

and thus, when the jaw is opened, the deactivated state in which the reciprocating assembly **550** and the operation member **540** are not coupled is maintained.

[0900] In contrast, in a state in which the jaw **1103** is opened as shown in FIG. **148**, the first jaw **1101** is rotated in the opposite direction of the arrow R of FIG. **148**, and when the second jaw **1102** is rotated in the direction of the arrow R, the jaw **1103** is in a closed state as shown in FIG. **131**.

[0901] When the jaw **1103** is closed as described above, the reciprocating assembly **550** is moved toward the proximal end **1101g**. That is, the reciprocating assembly **550** is moved backward.

[0902] In the closed state as shown in FIG. **147**, a recess **550b** of the reciprocating assembly **550** and a ratchet **543a** of a ratchet member **543** of the operation member **540** are in contact with each other, but are not coupled to each other.

[0903] Accordingly, the deactivated state in which the reciprocating assembly **550** and the operation member **540** are not coupled to and separated from each other is maintained between the state of FIG. **147**, in which the jaw **1103** is fully opened, and the state of FIG. **148**, in which the jaw **1103** is closed. That is, in this state, even when the staple pulley **1161** is rotated, the operation member **540** is not moved, and thus stapling and cutting motions are not performed.

[0904] Next, for stapling and cutting motions, the staple pulley **1161** is rotated in the direction of the arrow R of FIG. **149B** in a state in which the jaw **1103** is closed as shown in FIG. **149A**.

[0905] When the staple pulley **1161** is rotated in the direction of the arrow R, the staple link assembly **1170** connected to the staple pulley **1161** and the reciprocating assembly **550** connected to the staple link assembly **1170** are moved as a whole toward the proximal end **1101g** (i.e., are moved backward). In addition, as shown in FIG. **149B**, when the reciprocating assembly **550** is moved toward the proximal end **1101g** to be coupled to the ratchet member **543** of the operation member **540**, the deactivated state is switched to the activated state.

[0906] After switching to the activated state as described above, as the staple pulley **1161** performs an alternating rotational motion, the reciprocating assembly **550** is repeatedly moved forward and backward, thereby performing stapling and cutting motions while the operation member **540** is moved forward.

Seventh Embodiment—Adding Second Auxiliary Pulley for Staple Wire

[0907] Hereinafter, an end tool **1200** of a surgical instrument according to a seventh embodiment of the present disclosure will be described. Here, the end tool **1200** of the surgical instrument according to the seventh embodiment of the present disclosure is different from the end tool (see **100** of FIG. **2** or the like) of the surgical instrument according to the first embodiment of the present disclosure described above in that a staple second auxiliary pulley **967**, which is a staple wire auxiliary pulley, is additionally provided. Hereinafter, the configuration that is different from that of the first embodiment will be described in detail.

[0908] FIGS. **150** to **153** are perspective views illustrating the end tool of the surgical instrument according to the seventh embodiment of the present disclosure. Here, FIGS. **150** and **152** illustrate a state in which the end tool is closed, and FIGS. **151** and **153** illustrate a state in which the end tool is opened. Meanwhile, FIGS. **152** and **153** illustrate a state in which an end tool hub is removed. FIGS. **154** and **155** are side views illustrating the end tool of the surgical instrument of FIG. **150**. FIGS. **156** and **157** are plan views illustrating the end tool of the surgical instrument of FIG. **150**. FIGS. **158** to **160** are plan views illustrating a stapling motion of the end tool of the surgical instrument of FIG. **150**, and is a view illustrating a process of performing a stapling motion in a state in which jaws are yaw-rotated by +90°. FIGS. **161** to **163** are plan views illustrating a stapling motion of the end tool of the surgical instrument of FIG. **150**, and is a view illustrating a process of performing a stapling motion in a state in which the jaws are yaw-rotated by -90°.

[0909] Referring to FIGS. **150** to **163**, the end tool **1200** of the seventh embodiment of the present disclosure includes a pair of jaws **1203** for performing a grip motion, that is, a first jaw **1201** and a second jaw **1202**. Here, each of the first jaw **1201** and the second jaw **1202**, or a component encompassing the first jaw **1201** and the second jaw **1202** may be referred to as the jaw.

[0910] Meanwhile, the end tool **1200** includes a plurality of pulleys including a pulley **1211** and a pulley **1212** that are related to a rotational motion of the first jaw **1201**. The pulleys related to the rotational motion of the first jaw **1201** described in the present embodiment are substantially the same as the pulley **111**, the pulley **112**, the pulley **113**, the pulley **114**, the pulley **115**, and the pulley **116** described with reference to FIG. 8 or the like of the first embodiment, and thus, detailed descriptions thereof will be omitted herein.

[0911] Meanwhile, the end tool **1200** includes a plurality of pulleys including a pulley **1221** and a pulley **1222** that are related to a rotational motion of the second jaw **1202**. The pulleys related to the rotational motion of the second jaw **1202** described in the present embodiment are substantially the same as the pulley **121**, the pulley **122**, the pulley **123**, the pulley **124**, the pulley **125**, and the pulley **126** described with reference to FIG. 8 or the like of the first embodiment, and thus, detailed descriptions thereof will be omitted herein.

[0912] In addition, the end tool **1200** of the seventh embodiment of the present disclosure may include a rotation shaft **1241**, a rotation shaft **1242**, a rotation shaft **1243**, and a rotation shaft **1244**. Here, the rotation shaft **1241** and the rotation shaft **1242** may be inserted through an end tool hub **1280**, and the rotation shaft **1243** and the rotation shaft **1244** may be inserted through a pitch hub **1207**. The rotation shaft **1241**, the rotation shaft **1242**, the rotation shaft **1243**, and the rotation shaft **1244** may be arranged sequentially from a distal end **1204** of the end tool **1200** toward a proximal end **1205**.

[0913] In addition, the end tool **1200** of the seventh embodiment of the present disclosure may include the end tool hub **1280** and the pitch hub **1207**.

[0914] The rotation shaft **1241** and the rotation shaft **1242** may be inserted through the end tool hub **1280**, and the pulley **1211** and the pulley **1221** axially coupled to the rotation shaft **1241** and at least some of the first jaw **1201** and the second jaw **1202** coupled to the pulley **1211** and the pulley **1221** may be accommodated inside the end tool hub **1280**.

[0915] The rotation shaft **1243** and the rotation shaft **1244** may be inserted through the pitch hub **1207**, and the pitch hub **1207** may be axially coupled to the end tool hub **1280** by the rotation shaft **1243**. Accordingly, the end tool hub **1280** may be formed to be pitch-rotatable around the rotation shaft **1243** with respect to the pitch hub **1207**.

[0916] Meanwhile, the end tool **1200** of the seventh embodiment of the present disclosure may further include components, such as a staple drive assembly (see **150** of FIG. 13) including a staple pulley assembly **1260** and a staple link assembly **1270**, to perform stapling and cutting motions.

[0917] The staple pulley assembly **1260** may be formed between the pulley **1211** and the pulley **1221** to be adjacent to the pulley **1211** and the pulley **1221**. In the present embodiment, it is assumed that the staple pulley assembly **1260** includes one staple pulley **1261**.

[0918] In the seventh embodiment of the present disclosure, by disposing the staple pulley assembly **1260** between the pulley **1211**, which is a first jaw pulley, and the pulley **1221**, which is a second jaw pulley, the end tool **1200** is allowed to perform pitch and yaw motions as well as stapling and cutting motions using a cartridge **1210**.

[0919] Hereinafter, a pulley configuration of the end tool **1200** of the surgical instrument according to the seventh embodiment of the present disclosure will be described in more detail.

[0920] The end tool **1200** of the seventh embodiment of the present disclosure may include the staple pulley **1261**, a staple auxiliary pulley **1262**, a pulley **1263**, a pulley **1264**, a pulley **1265**, and a pulley **1266** that are related to a linear motion/rotational motion of the staples. In addition, the end tool **1200** of the seventh embodiment of the present disclosure may further include a pulley **1267** and a pulley **1268**, which are staple second auxiliary pulleys.

[0921] The staple pulley **1261** is formed to face each of a pulley (see **111** of FIG. 6) and a pulley (see **121** of FIG. 6), which are end tool jaw pulleys, and are formed to be rotatable independently of each other around the rotation shaft **1241** that is an end tool jaw pulley rotation shaft.

[0922] Here, in the present disclosure, the staple pulley **1261**, the pulley (see **111** of FIG. 6), and

the pulley (see **121** of FIG. **6**) are formed to rotate around the same shaft. As described above, as the staple pulley **1261**, the pulley (see **111** of FIG. **6**), and the pulley (see **121** of FIG. **6**) are formed to rotate around the same shaft, it is possible to perform a pitch motion/yaw motion/actuation motion as well as a cutting motion using the staple.

[0923] The staple auxiliary pulley **1262** may be further provided on one side of the staple pulley **1261**. In other words, the staple auxiliary pulley **1262** may be disposed between the pulley **1261** and the pulley **1263**/the pulley **1264**. The staple auxiliary pulley **1262** may be formed to be rotatable independently of each other around the rotation shaft **1242** together with the pulley (see **112** of FIG. **6**) and the pulley (see **122** of FIG. **6**).

[0924] The pulleys **1263** and **1264** function as staple pitch main pulleys, and the pulleys **1265** and **1266** function as staple pitch sub-pulleys.

[0925] The pulleys **1267** and **1268**, which are staple second auxiliary pulleys, may be disposed between the staple auxiliary pulley **1262** and the pulley **1263**/the pulley **1264**. The staple auxiliary pulley **1262** may be formed to be rotatable around a predetermined axis parallel to the rotation shaft **1243** which is a central axis of the pulleys **1263** and **1264**.

[0926] Hereinafter, components related to the rotation of the staple pulley **1261** will be described.

[0927] The pulleys **1263** and **1264** function as staple pitch main pulleys. Here, the wire **307**, which is a staple wire, is wound around the pulley **1263**, and the wire **308**, which is a staple wire, is wound around the pulley **1264**.

[0928] The pulleys **1265** and **1266** function as staple pitch sub-pulleys. Here, the wire **307**, which is a staple wire, is wound around the pulley **1265**, and the wire **308**, which is a staple wire, is wound around the pulley **1266**.

[0929] Here, the pulleys **1263** and **1264** are disposed on one side of the staple pulley **1261** and the staple auxiliary pulley **1262** to face each other. Here, the pulleys **1263** and **1264** are formed to be rotatable independently of each other around the rotation shaft **1243** that is an end tool pitch rotating shaft. In addition, the pulleys **1265** and **1266** are respectively disposed on one sides of the pulleys **1263** and **1264** to face each other. Here, the pulleys **1265** and **1266** are formed to be rotatable independently of each other around the rotation shaft **1244**, which is an end tool pitch auxiliary rotating shaft. Here, in the drawings, it is illustrated that all of the pulley **1263**, the pulley **1265**, the pulley **1264**, and the pulley **1266** are formed to be rotatable around the Y-axis direction, but the concept of the present disclosure is not limited thereto, and the rotating axes of the respective pulleys may be formed in various directions according to configurations thereof.

[0930] The wire **307**, which is a staple wire, is sequentially wound to make contact with at least portions of the pulley **1265**, the pulley **1263**, the staple auxiliary pulley **1262**, and the staple pulley **1261**. In addition, the wire **308** connected to the wire **307** by the coupling member **329** is sequentially wound to make contact with at least portions of the staple pulley **1261**, the staple auxiliary pulley **1262**, the pulley **1268**, the pulley **1264**, and the pulley **1266**.

[0931] In other words, the wires **307** and **308**, which are staple wires, sequentially wound to make contact with at least portions of the pulley **1265**, the pulley **1263**, the pulley **1267**, the staple auxiliary pulley **1262**, the staple pulley **1261**, the staple auxiliary pulley **1262**, the pulley **1268**, the pulley **1264**, and the pulley **1266**, and the wires **307** and **308** are formed to move along the above pulleys while rotating the above pulleys.

[0932] Accordingly, when the wire **307** is pulled, the coupling member **329**, to which the wire **307** is coupled, and the staple pulley **1261** coupled to the coupling member **329** are rotated in one direction. In contrast, when the wire **308** is pulled, the coupling member **329**, to which the wire **308** is coupled, and the staple pulley **1261** coupled to the coupling member **329** are rotated in a direction opposite to the one direction.

[0933] Here, in the end tool **1200** of the surgical instrument according to the seventh embodiment of the present disclosure, the pulleys **1267** and **1268**, which are staple second auxiliary pulleys, are additionally provided to serve to prevent the wires **307** and **308**, which are staple wires, from being

separated.

[0934] That is, the staple second auxiliary pulley **1267** is disposed between the staple auxiliary pulley **1262** and the pulley **1263** to change the path of the wire **307** entering the end tool hub **1280** to a certain extent. In addition, the staple second auxiliary pulley **1268** is disposed between the staple auxiliary pulley **1262** and the pulley **1264** to change the path of the wire **308** entering the end tool hub **1280** to a certain extent.

[0935] In detail, a height of the wire **307** in the Z-axis direction coming from the staple auxiliary pulley **1262** and a height of the wire **307** in the Z-axis direction entering the pulley **1263** are different from each other. Here, when the pulley **1267**, which is a staple second auxiliary pulley, is not present, the path of the wire **307** becomes oblique, and thus there is a risk that the wire **307** is removed from the pulley. Similarly, a height of the wire **308** in the Z-axis direction coming from the staple auxiliary pulley **1262** and a height of the wire **308** in the Z-axis direction entering the pulley **1264** are different from each other. Here, when the pulley **1268**, which is a staple second auxiliary pulley, is not present, the path of the wire **308** becomes oblique, and thus there is a risk that the wire **308** is removed from the pulley.

[0936] Accordingly, in the present embodiment, the staple second auxiliary pulley **1267** is disposed between the staple auxiliary pulley **1262** and the pulley **1263** to change the path of the wire **307** to a certain extent so that the wire **307** coming out of the staple auxiliary pulley **1262** maintains a parallel path. Similarly, the staple second auxiliary pulley **1268** is disposed between the staple auxiliary pulley **1262** and the pulley **1264** to change the path of the wire **308** to a certain extent so that the wire **308** coming out of the staple auxiliary pulley **1262** maintains a parallel path.

[0937] According to the present disclosure, the wires **307** and **308**, which are staple wires, are prevented from being separated from the pulleys, thereby more smoothly performing the cutting motion.

[0938] FIGS. **158** to **160** are plan views illustrating stapling and cutting motions of the end tool of the surgical instrument of FIG. **150**, and is a view illustrating a process of performing the stapling and cutting motions in a state in which the jaws are yaw-rotated by $+90^\circ$. As shown in FIGS. **158** to **160**, the end tool of the surgical instrument according to the seventh embodiment of the present disclosure is formed to normally perform the stapling and cutting motions even when the jaws are yaw-rotated by $+90^\circ$.

[0939] In detail, in a state in which the pulley **1211**, the pulley **1221**, and the staple pulley **1261** are rotated by $+90^\circ$ around the rotation shaft **1241**, when the staple pulley **1261** is alternately rotated in the clockwise/counterclockwise directions, a link member **1271** and a reciprocating member **551** connected thereto are repeatedly moved forward and backward. In addition, when the reciprocating member **551** is moved forward, an operation member **540** may be moved forward together with the reciprocating member **551**, and when the reciprocating member **551** is moved backward, only the reciprocating member **551** is moved backward and the operation member **540** remains stationary. By repeating this process, the stapling and cutting motions are performed while the operation member **540** is moved toward a distal end **502**.

[0940] FIGS. **161** to **163** are plan views illustrating stapling and cutting motions of the end tool of the surgical instrument of FIG. **150**, and are views illustrating a process of performing the stapling and cutting motions in a state in which the jaws are yaw-rotated by -90° . As shown in FIG. **63**, the end tool of the surgical instrument according to the first embodiment of the present disclosure is formed to normally perform the stapling and cutting motions even when the jaws are yaw-rotated by -90° .

[0941] In detail, in a state in which the pulley **1211**, the pulley **1221**, and the staple pulley **1261** are rotated by -90° around the rotation shaft **1241**, when the staple pulley **1261** is alternately rotated in the clockwise/counterclockwise directions, the link member **1271** and the reciprocating member **551** connected thereto are repeatedly moved forward and backward. In addition, when the reciprocating member **551** is moved forward, the operation member **540** may be moved forward

together with the reciprocating member **551**, and when the reciprocating member **551** is moved backward, only the reciprocating member **551** is moved backward and the operation member **540** remains stationary in place. By repeating this process, the stapling and cutting motions are performed while the operation member **540** is moved toward the distal end **502**.

Eighth Embodiment—Dual Rack & Spring

[0942] Hereinafter, an end tool **1300** of a surgical instrument according to an eighth embodiment of the present disclosure will be described. Here, the end tool **1300** of the surgical instrument according to the eighth embodiment of the present disclosure is different from the end tool (see **900** of FIG. **105** or the like) of the surgical instrument according to the fourth embodiment of the present disclosure described above in that a staple link assembly **1370** and a reciprocating assembly **1350** of a cartridge **1310** are different. Hereinafter, the configuration that is different from that of the fourth embodiment will be described in detail.

[0943] FIGS. **164** to **166** are perspective views illustrating the end tool of the surgical instrument according to the eighth embodiment of the present disclosure. Here, FIGS. **164** to **166** illustrate a state in which a first jaw, a second jaw, an end tool hub, and the like are removed. FIGS. **167** and **168** are perspective views illustrating the cartridge of the surgical instrument of FIG. **164**. FIG. **169** is a bottom view of the cartridge of FIG. **167**. FIGS. **170** and **171** are perspective views illustrating respective operating states of the end tool of FIG. **164**. Here, FIGS. **170** and **171** mainly illustrate operations of the reciprocating assembly and an operation member.

[0944] Referring to FIGS. **164** to **171**, the end tool **1300** of the eighth embodiment of the present disclosure includes a pair of jaws for performing a grip motion, that is, a first jaw (see **901** of FIG. **105**) and a second jaw (see **902** in FIG. **105**).

[0945] Meanwhile, the end tool **1300** includes a plurality of pulleys including a pulley **1311** and a pulley **1312** that are related to a rotational motion of the first jaw (see **901** of FIG. **105**). The pulleys related to the rotational motion of the first jaw (see **901** of FIG. **105**) described in the present embodiment are substantially the same as the pulley **111**, the pulley **112**, the pulley **113**, the pulley **114**, the pulley **115**, and the pulley **116** described with reference to FIG. **105** or the like of the fourth embodiment, and thus, detailed descriptions thereof will be omitted herein.

[0946] Meanwhile, the end tool **1300** includes a plurality of pulleys including a pulley **1321** and a pulley **1322** that are related to a rotational motion of second jaw (see **902** in FIG. **105**) The pulleys related to the rotational motion of the second jaw (see **902** in FIG. **105**) described in the present embodiment are substantially the same as the pulley **121**, the pulley **122**, the pulley **123**, the pulley **124**, the pulley **125**, and the pulley **126** described with reference to FIG. **105** or the like of the fourth embodiment, and thus, detailed descriptions thereof will be omitted herein.

[0947] Meanwhile, the end tool **1300** of the eighth embodiment of the present disclosure may further include components, such as a staple drive assembly (see **150** of FIG. **13**) including a staple pulley assembly **1360** and the staple link assembly **1370**, to perform stapling and cutting motions.

[0948] The staple pulley assembly **1360** may be formed between the pulley **1311** and the pulley **1321** to be adjacent to the pulley **1311** and the pulley **1321**. In the present embodiment, it is assumed that the staple pulley assembly **1360** includes one staple pulley **1361**.

[0949] In the eighth embodiment of the present disclosure, by disposing the staple pulley assembly **1360** between the pulley **1311**, which is a first jaw pulley, and the pulley **1321**, which is a second jaw pulley, the end tool **1300** is allowed to perform pitch and yaw motions as well as stapling and cutting motions using the cartridge **1310**.

[0950] Hereinafter, the staple pulley assembly **1360**, the staple link assembly **1370**, and the reciprocating assembly **1350** of the cartridge **1310** of the end tool **1300** of the surgical instrument according to the eighth embodiment of the present disclosure will be described in more detail.

[0951] In the end tool **1300** of the surgical instrument according to the eighth embodiment of the present disclosure, the staple link assembly **1370** includes a first link member **1371** and a second link member **1376**, and the reciprocating assembly **1350** of the cartridge **1310** includes a first

reciprocating member **1351** and a second reciprocating member **1352**, so that a kind of dual rack structure is formed in the end tool **1300**.

[0952] Referring to FIGS. **105** to **121** and the like, the staple pulley assembly **1360** may include one or more staple pulleys **1361**.

[0953] A shaft pass-through part **1361a** may be formed in the staple pulley **1361**. The shaft pass-through part **1361a** may be formed in the form of a hole, and a rotation shaft **1341**, which is an end tool jaw pulley rotation shaft, may be inserted through the shaft pass-through part **1361a**.

[0954] In addition, a first link coupling part **1361b** and a second link coupling part **1361c** may be formed in the staple pulley **1361**. The first link member **1371** of the staple link assembly **1370** may be coupled to the first link coupling part **1361b**, and the second link member **1376** of the staple link assembly **1370** may be coupled to the second link coupling part **1361c**. Here, the first link coupling part **1361b** and the second link coupling part **1361c** may be disposed on opposite sides with respect to a central axis of the staple pulley **1361**.

[0955] Meanwhile, the end tool **1300** of the eighth embodiment of the present disclosure may further include the staple link assembly **1370** connected to the staple pulley assembly **1360**. Here, the staple link assembly **1370** may serve to connect the staple pulley assembly **1360** to the reciprocating assembly **1350** of the cartridge **1310** to be described later. In the end tool **1300** of the eighth embodiment of the present disclosure, the staple link assembly **1370** includes two pairs of link members in the first link member **1371** and the second link member **1376**.

[0956] The first link member **1371** may include a first link **1372** and a second link **1373**.

[0957] The second link member **1376** may include a third link **1377** and a fourth link **1378**.

[0958] The configuration of the first link member **1371** and the second link member **1376** is substantially the same as the configuration of the first link member (see **971** of FIG. **110**) and the second link member (see **976** of FIG. **110**) of the fourth embodiment illustrated in FIG. **105** or the like, and thus, detailed descriptions thereof will be omitted in the present embodiment.

[0959] In the state of FIG. **164**, when the staple pulley **1361** is rotated in the direction of an arrow **A1** of FIG. **164** (i.e., in the clockwise direction), the first link member **1376** connected to the staple pulley **1361** may be moved in the direction of an arrow **B1** of FIG. **164**, in other words, toward a distal end **1301f** of the first jaw **1301**. In contrast, when the staple pulley **1361** is rotated in the counterclockwise direction, the first link member **1376** connected to the staple pulley **1361** may be moved in the direction of an arrow **C1** of FIG. **164**, in other words, toward a proximal end **1301g** of the first jaw **1301**.

[0960] Meanwhile, when the staple pulley **1361** is rotated in the direction of the arrow **A1** of FIG. **164** (i.e., in the clockwise direction), the second link member **1376** connected to the staple pulley **1361** may be moved in the direction of the arrow **C1** of FIG. **164**, in other words, toward the proximal end **1301g** of the first jaw **1301**. In contrast, when the staple pulley **1361** is rotated in the counterclockwise direction, the second link member **1376** connected to the staple pulley **1361** may be moved in the direction of the arrow **B1** of FIG. **114**, in other words, toward the distal end **1301f** of the first jaw **1301**.

[0961] That is, when the staple pulley **1361** is rotated in one direction (e.g., the clockwise direction), the first link member **1371** is moved forward and the second link member **1376** is moved backward. Meanwhile, when the staple pulley **1361** is rotated in the other direction (e.g., the counterclockwise direction), the first link member **1371** is moved backward and the second link member **1376** is moved forward.

[0962] With this configuration, when the staple pulley **1361** is alternately rotated in the clockwise and counterclockwise directions, the first link member **1371** and the second link member **1376** perform a linear reciprocating motion, and in this case, the first link member **1371** and the second link member **1376** are moved in opposite directions. That is, as the first link member **1371** is moved forward (i.e., toward the distal end), the second link member **1376** is moved backward (i.e., toward the proximal end). In contrast, as the first link member **1371** is moved backward (i.e.,

toward the proximal end), the second link member **1376** is moved forward (i.e., toward the distal end).

[0963] As a result, a bidirectional rotational motion of the staple pulley assembly **1360** may cause reciprocating linear motions of the first reciprocating member **1351** and the second reciprocating member **1352** of the cartridge **1310** through the first link member **1371** and the second link member **1376** of the staple link assembly **1370**.

(Cartridge)

[0964] Hereinafter, the cartridge **1310** of the end tool **1300** of the surgical instrument according to the eighth embodiment of the present disclosure will be described in more detail.

[0965] In the cartridge **1310** of the end tool **1300** of the surgical instrument according to the eighth embodiment of the present disclosure, the reciprocating assembly **1350** includes the first reciprocating member **1351** and the second reciprocating member **1352**.

[0966] In detail, referring to FIGS. **164** to **169** and the like, the cartridge **1310** is formed to be mountable to and dismountable from the first jaw **1301** and includes a plurality of staples (see **530** of FIG. **22**) and an operation member **1340** therein to perform suturing and cutting tissue.

[0967] Here, the cartridge **1310** may include a cover (see **510** of FIG. **22**), a housing (see **520** of FIG. **22**), a staple (see **530** of FIG. **22**), the operation member **1340**, and the reciprocating assembly **1350**.

[0968] The housing (see **520** of FIG. **22**) forms an outer shape of the cartridge **1310**, and may be formed entirely in the form of a hollow box with one surface (upper surface) removed to accommodate the reciprocating assembly **1350**, the operation member **1340**, and the staple (see **530** of FIG. **22**) therein. Here, the housing (see **520** of FIG. **22**) may be formed in an approximately “U” shape in cross section.

[0969] A plurality of staples (see **530** of FIG. **22**) may be disposed inside the housing (see **520** of FIG. **22**). As the operation member **1340**, which will be described later, is linearly moved in one direction, the plurality of staples (see **530** of FIG. **22**) are sequentially pushed and raised from the inside of the housing (see **520** of FIG. **22**) to the outside, thereby performing sealing, that is, stapling.

[0970] The reciprocating assembly **1350** may be disposed at an inner lower side of the housing (see **520** of FIG. **22**). In the present embodiment, the reciprocating assembly **1350** includes the first reciprocating member **1351** and the second reciprocating member **1352**.

[0971] In the present embodiment, the first reciprocating member **1351** and the second reciprocating member **1352** may form a rack. The first reciprocating member **1351** may include a recess **1351b** and a coupling part **1351a**.

[0972] In detail, the first reciprocating member **1351** may be formed in the form of an elongated bar, and a plurality of recesses **1351b** having a sawtooth shape may be formed on one surface thereof. The recess **1351b** may be formed to be in contact with an operation member **1340** to be described later, in particular, a first ratchet member **1343** of the operation member **1340**. In other words, the first reciprocating member **1351** may include the plurality of recesses **1351b** shaped to engage with first ratchets **1343a** of the first ratchet member **1343**.

[0973] Similarly, the second reciprocating member **1352** may include a recess **1352b** and a coupling part **1352a**. That is, the second reciprocating member **1352** may include a plurality of recesses **1352b** shaped to engage with recesses **1347b** of a second ratchet member **1347**.

[0974] Here, the first reciprocating member **1351** and the second reciprocating member **1352** are not fixedly coupled to other components of the cartridge **1310**, and may be formed to be movable relative to other components of the cartridge **1310**. That is, the first reciprocating member **1351** and the second reciprocating member **1352** may perform a reciprocating linear motion with respect to the housing (see **520** of FIG. **22**) and the cover (see **510** of FIG. **22**).

[0975] Meanwhile, in the first reciprocating member **1351**, the coupling part **1351a** may be formed at the proximal end side adjacent to the pulley **1311** and the coupling part **1351a** may be fastened

and coupled to the first link member **1371** of the staple link assembly **1370** of the end tool **1300**. Thus, when the first link member **1371** performs a reciprocating linear motion in the extension direction (i.e., the Y-axis direction) of the connection part **400**, the first reciprocating member **1351** coupled thereto may also perform a reciprocating linear motion in the extension direction (i.e., the Y-axis direction) of the connection part **400**.

[0976] Similarly, in the second reciprocating member **1352**, the coupling part **1352a** may be formed at the proximal end side adjacent to the pulley **1311** and the coupling part **1352a** may be fastened and coupled to the second link member **1376** of the staple link assembly **1370** of the end tool **1300**. Thus, when the second link member **1376** performs a reciprocating linear motion in the extension direction (i.e., the Y-axis direction) of the connection part **400**, the second reciprocating member **1352** coupled thereto may also perform a reciprocating linear motion in the extension direction (i.e., the Y-axis direction) of the connection part **400**.

[0977] The operation member **1340** may be disposed inside the housing **1320**. The operation member **1340** is formed to be in contact with the first reciprocating member **1351** and the second reciprocating member **1352**, and may be formed to linearly move in one direction according to the reciprocating linear motion of the first reciprocating member **1351** and the second reciprocating member **1352**. In other words, the operation member **1340** interacts with the first reciprocating member **1351** and the second reciprocating member **1352** to perform stapling and cutting while moving in the extension direction of the connection part **400**.

[0978] The operation member **1340** may include a wedge (see **541** of FIG. 22), a blade (see **542** of FIG. 22), the first ratchet member **1343**, a first elastic member **1344**, the second ratchet member **1347**, a second elastic member **1348**, and the like. Here, in the drawings, components other than the ratchet member and the elastic member, that is, the wedge, the blade, and the like are omitted from the operation member **1340**, but it is of course possible that such components may be included in the operation member **1340**. Here, the first ratchet member **1343** and the second ratchet member **1347** may be collectively referred to as a ratchet member.

[0979] The first ratchet member **1343** may include the first ratchets **1343a**, and the second ratchet member **1347** may include second ratchets **1347a**. Here, the first ratchet **1343a** of the first ratchet member **1343** may be formed to be engageable with the first reciprocating member **1351**, and the second ratchet **1347a** of the second ratchet member **1347** may be formed to be engageable with the second reciprocating member **1372**. That is, the motion of the operation member **1340** being moved forward by the first reciprocating member **1351** as the first ratchet **1343a** of the first ratchet member **1343** is engaged with (or brought into close contact with) the first reciprocating member **1351**, and the motion of the operation member **1340** being moved forward by the second reciprocating member **1352** as the second ratchet **1347a** of the second ratchet member **1347** is engaged with (or brought into close contact with) the second reciprocating member **1352** are alternately performed.

[0980] The first ratchet member **1343** may be formed to be rotatable around a rotation shaft **1345a**. In an example, the rotation shaft **1345a** is formed to protrude from a lower portion of a body **1345**, and the first ratchet member **1343** may be fitted to the rotation shaft **1345a**. Accordingly, when the first ratchet member **1343** is pressed by the first reciprocating member **1351** or is subjected to an elastic force by the first elastic member **1344**, the first ratchet member **1343** may be rotated around the rotation shaft **1345a**.

[0981] The first elastic member **1344** is formed on one side of the first ratchet member **1343**, and serves to apply a predetermined elastic force to the first ratchet member **1343**. In an example, one region of the elastic member **1344** may be in contact with a support **1345c** formed to protrude from a lower portion of the body **1345**, and another region of the elastic member **1344** may be in contact with the first ratchet member **1343**. Here, the first elastic member **1344** may apply an elastic force in a direction in which the first ratchet member **1343** comes into close contact with the first reciprocating member **1351**. To this end, the first elastic member **1344** may be formed in the form

of a leaf spring, and may be provided in various forms capable of providing a predetermined elastic force to the first ratchet member **1343**, such as a coil spring, a dish spring, and the like.

[0982] The second ratchet member **1347** may be formed to be rotatable around a rotation shaft **1345b**. In an example, the rotation shaft **1345b** is formed to protrude from the lower portion of the body **1345**, and the second ratchet member **1347** may be fitted to the rotation shaft **1345b**. Accordingly, when the second ratchet member **1347** is pressed by the second reciprocating member **1352** or is subjected to an elastic force by the second elastic member **1348**, the second ratchet member **1347** may be rotated around the rotation shaft **1345b**.

[0983] The second elastic member **1348** is formed on one side of the second ratchet member **1347**, and serves to apply a predetermined elastic force to the second ratchet member **1347**. In an example, one region of the elastic member **1344** may be in contact with a support **1345d** formed to protrude from the lower portion of the body **1345**, and another region of the elastic member **1344** may be in contact with the second ratchet member **1347**. Here, the second elastic member **1348** may apply an elastic force in a direction in which the second ratchet member **1347** comes into close contact with the second reciprocating member **1352**. To this end, the second elastic member **1348** may be formed in the form of a leaf spring, and may be provided in various forms capable of providing a predetermined elastic force to the second ratchet member **1347**, such as a coil spring, a dish spring, and the like.

(Operations of Reciprocating Assembly and Operation Member)

[0984] Hereinafter, operations of the reciprocating assembly **1350** and the operation member **1340** will be described in more detail.

[0985] FIGS. **170** and **171** are views illustrating respective operating states of the reciprocating assembly **1350** and the operation member **1340**. Here, FIG. **170** is a view illustrating when the first reciprocating member **1351** is moved forward and the second reciprocating member **1352** is moved backward, and FIG. **171** is a view illustrating when the first reciprocating member **1351** is moved backward and the second reciprocating member **1352** is moved forward.

[0986] Here, FIG. **170** is a view illustrating when the first reciprocating member **1351** is moved forward and the second reciprocating member **1352** is moved backward as the staple pulley **1361** is rotated in the clockwise direction (in the direction of the arrow **A1** of FIG. **164**).

[0987] In addition, FIG. **171** is a view illustrating when the first reciprocating member **1351** is moved backward and the second reciprocating member **1352** is moved forward as the staple pulley **1361** is rotated in the counterclockwise direction (in the opposite direction of the arrow **A1** of FIG. **164**).

[0988] First, operations of FIGS. **170A** to **170E** will be described.

[0989] FIG. **170A** illustrates a state in which the first ratchet **1343a** of the first ratchet member **1343** and the first reciprocating member **1351** are in close contact with each other, and the second ratchet **1347a** of the second ratchet member **1347** and the second reciprocating member **1352** are in close contact with each other.

[0990] In this state, when the staple pulley **1361** is rotated in the clockwise direction (in the direction of the arrow **A1** of FIG. **164**), as shown in FIG. **170B**, the first reciprocating member **1351** is moved in the direction of an arrow **b1** (i.e., is moved forward), and the second reciprocating member **1352** is moved in the direction of an arrow **b2** (i.e., is moved backward). Then, the recess **1352b** of the second reciprocating member **1352** pushes the second ratchet **1347a** of the second ratchet member **1347** toward the first reciprocating member **1351**, which causes the second ratchet member **1347** to be rotated as a whole in the direction of an arrow **b3** around the rotation shaft **1345b**. In addition, as the second ratchet member **1347** is rotated, the second ratchet **1347a** of the second ratchet member **1347** starts to be spaced apart from the recess **1352b** of the second reciprocating member **1352**.

[0991] In this state, when the staple pulley **1361** is further rotated in the clockwise direction (in the direction of the arrow **A1** of FIG. **164**), as shown in FIG. **170C**, the first reciprocating member

1351 is further moved in the direction of an arrow **c1** (i.e., is moved forward), and the second reciprocating member **1352** is further moved in the direction of an arrow **c2** (i.e., is moved backward). Then, the recess **1352b** of the second reciprocating member **1352** further pushes the second ratchet **1347a** of the second ratchet member **1347** toward the first reciprocating member **1351**, so that the entire second ratchet member **1347** is further rotationally moved in the direction of an arrow **c3**. In addition, as the second ratchet member **1347** is rotated, the second ratchet **1347a** of the second ratchet member **1347** is further spaced apart from the recess **1352b** of the second reciprocating member **1352**.

[0992] In this state, when the staple pulley **1361** is further rotated in the clockwise direction (in the direction of the arrow **A1** of FIG. **164**), as shown in FIG. **170D**, the first reciprocating member **1351** is further moved in the direction of an arrow **d1** (i.e., is moved forward), and the second reciprocating member **1352** is further moved in the direction of an arrow **d2** (i.e., is moved backward). Then, the recess **1352b** of the second reciprocating member **1352** further pushes the second ratchet **1347a** of the second ratchet member **1347** toward the first reciprocating member **1351**, and meets the next second ratchet **1347a** on the second ratchet member **1347** when an inclined first surface (see **551b** of FIG. **36**) of the second reciprocating member **1352** is moved beyond an end of an inclined first surface (see **543a1** of FIG. **36**) of the second ratchet member **1347**. In this case, since the second elastic member **1348** applies an elastic force in a direction in which the second ratchet member **1347** comes into close contact with the second reciprocating member **1352**, front surfaces of the second reciprocating member **1352** and the second ratchet member **1347** are again brought into close contact with each other.

[0993] In this state, when the staple pulley **1361** is further rotated in the clockwise direction (in the direction of the arrow **A1** of FIG. **164**), as shown in FIG. **170E**, the first reciprocating member **1351** is further moved in the direction of an arrow **e1** (i.e., is moved forward), and the second reciprocating member **1352** is further moved in the direction of an arrow **e2** (i.e., is moved backward). Then, the first ratchet member **1343** is linearly moved in the direction of an arrow **e4** by the first reciprocating member **1351** while repeating the above-described operations.

[0994] Next, operations of FIGS. **171A** to **171E** will be described.

[0995] FIG. **171A** illustrates a state in which the first ratchet **1343a** of the first ratchet member **1343** and the first reciprocating member **1351** are in close contact with each other, and the second ratchet **1347a** of the second ratchet member **1347** and the second reciprocating member **1352** are in close contact with each other.

[0996] In this state, when the staple pulley **1361** is rotated in the counterclockwise direction (in the opposite direction of the arrow **A1** of FIG. **164**, the first reciprocating member **1351** is moved in the direction of an arrow **b1** (i.e., is moved backward), and the second reciprocating member **1352** is moved in the direction of an arrow **b2** (i.e., is moved forward) as shown in FIG. **171B**. Then, the recess **1351b** of the first reciprocating member **1351** pushes the first ratchet **1343a** of the first ratchet member **1343** toward the second reciprocating member **1352**, so that the first ratchet member **1343** is entirely rotated in the direction of an arrow **b3** around the rotation shaft **1345a**. In addition, as the first ratchet member **1343** is rotated, the first ratchet **1343a** of the first ratchet member **1343** starts to be spaced apart from the recess **1351b** of the first reciprocating member **1351**.

[0997] In this state, when the staple pulley **1361** is further rotated in the counterclockwise direction (in the opposite direction of the arrow **A1** of FIG. **164**), the first reciprocating member **1351** is further moved in the direction of an arrow **c1** (i.e., is moved backward), and the second reciprocating member **1352** is further moved in the direction of an arrow **c2** (i.e., is moved forward) as shown in FIG. **171C**. Then, the recess **1351b** of the first reciprocating member **1351** further pushes the first ratchet **1343a** of the first ratchet member **1343** toward the second reciprocating member **1352**, so that the entire first ratchet member **1343** is further rotationally moved in the direction of an arrow **c3**. In addition, as the first ratchet member **1343** is rotated, the

first ratchet **1343a** of the first ratchet member **1343** is further spaced apart from the recess **1351b** of the first reciprocating member **1351**.

[0998] In this state, when the staple pulley **1361** is further rotated in the counterclockwise direction (in the opposite direction of the arrow **A1** of FIG. **164**), the first reciprocating member **1351** is further moved in the direction of an arrow **d1** (i.e., is moved backward), and the second reciprocating member **1352** is further moved in the direction of an arrow **d2** (i.e., is moved forward) as shown in FIG. **171D**. Then, the recess **1351b** of the first reciprocating member **1351** further pushes the first ratchet **1343a** of the first ratchet member **1343** toward the second reciprocating member **1352**, and meets the next first ratchet **1343a** on the first ratchet member **1343** when the inclined first surface (see **551b** of FIG. **36**) of the first reciprocating member **1351** is moved beyond an end of the inclined first surface (see **543a1** of FIG. **36**) of the first ratchet member **1343**. In this case, since the first elastic member **1344** applies an elastic force in a direction in which the first ratchet member **1343** comes into close contact with the first reciprocating member **1351**, the first reciprocating member **1351** and the first ratchet member **1343** are brought into close contact with each other again.

[0999] In this state, when the staple pulley **1361** is further rotated in the counterclockwise direction (in the opposite direction of the arrow **A1** of FIG. **164**), the first reciprocating member **1351** is further moved in the direction of an arrow **e1** (i.e., is moved backward), and the second reciprocating member **1352** is further moved in the direction of an arrow **e2** (i.e., is moved forward) as shown in FIG. **171E**. Then, the second ratchet member **1347** is linearly moved in the direction of an arrow **e4** by the second reciprocating member **1352** while repeating the above-described operations.

[1000] In conclusion, when the staple pulley **1361** is rotated in one direction, the first reciprocating member **1351** is moved forward and the second reciprocating member **1352** is moved backward, and in this case, the backward-moving second reciprocating member **1352** pushes the operation member **1340** toward the first reciprocating member **1351** to bring the operation member **1340** into close contact with the first reciprocating member **1351**, and the forward-moving first reciprocating member **1351** moves the operation member **1340**, which is in close contact therewith, forward.

[1001] Meanwhile, when the staple pulley **1361** is rotated in the other direction, the second reciprocating member **1352** is moved forward and the first reciprocating member **1351** is moved backward, and in this case, the backward-moving first reciprocating member **1351** pushes the operation member **1340** toward the second reciprocating member **1352** to bring the operation member **1340** into close contact with the second reciprocating member **1352**, and the forward-moving second reciprocating member **1352** moves the operation member **1340**, which is in close contact therewith, forward.

[1002] As a result, when the staple pulley **1361** is rotated in one direction, the first reciprocating member **1351** moves the operation member **1340** forward, and when the staple pulley **1361** is rotated in the opposite direction, the second reciprocating member **1352** moves the operation member **1340** forward, and thus a forward-moving speed of the operation member **1340** is approximately doubled as compared to the first embodiment, thereby reducing a stapling and cutting time.

Ninth Embodiment—Anvil Clamp Structure

[1003] Hereinafter, an end tool **1500** of a surgical instrument according to a ninth embodiment of the present disclosure will be described. Here, the end tool **1500** of the surgical instrument according to the ninth embodiment of the present disclosure is different from the end tool (see **100** of FIG. **2** or the like) of the surgical instrument according to the first embodiment of the present disclosure described above in that a clamp **1545** is additionally formed on an operation member **1540** of a cartridge **1510**. Hereinafter, the configuration that is different from that of the first embodiment will be described in detail.

[1004] FIG. **172** is a perspective view illustrating the end tool of the surgical instrument according

to the ninth embodiment of the present disclosure. FIG. 173 is a side cross-sectional view illustrating the cartridge of the surgical instrument of FIG. 172. FIG. 174 is a perspective view illustrating the operation member of the cartridge of the surgical instrument of FIG. 172. FIG. 175 is a magnified perspective view illustrating the cartridge of the surgical instrument of FIG. 172. FIGS. 176 to 179 are views illustrating a process in which the operation member is moved. FIGS. 180 to 182 are views illustrating a process in which the operation member is coupled to a second jaw. FIGS. 183 to 185 are views illustrating a process in which the operation member is separated from the second jaw.

[1005] Referring to FIGS. 172 to 185, the end tool 1500 of the ninth embodiment of the present disclosure includes a pair of jaws 1503 for performing a grip motion, that is, a first jaw 1501 and a second jaw 1502. Here, each of the first jaw 1501 and the second jaw 1502, or a component encompassing the first jaw 1501 and the second jaw 1502, may be referred to as the jaw 1503. This will be described in more detail later.

[1006] Meanwhile, the end tool 1500 includes a plurality of pulleys including a first jaw pulley (see 111 in FIG. 8) and a first jaw auxiliary pulley (see 121 of FIG. 8) that are related to the rotational motion of the first jaw 1501. The pulleys related to the rotational motion of the first jaw 1501 described in the present embodiment are substantially the same as the pulley 111, the pulley 112, the pulley 113, the pulley 114, the pulley 115, and the pulley 116 described with reference to FIG. 8 or the like of the first embodiment, and thus, detailed descriptions thereof will be omitted herein.

[1007] Meanwhile, the end tool 1500 includes a plurality of pulleys including a pulley 1521 and a pulley 1522 that are related to a rotational motion of the second jaw 1502. The pulleys related to the rotational motion of the second jaw 1502 described in the present embodiment are substantially the same as the pulley 151, the pulley 152, the pulley 153, the pulley 154, the pulley 155, and the pulley 156 described with reference to FIG. 8 or the like of the first embodiment, and thus, detailed descriptions thereof will be omitted herein.

[1008] Further, the end tool 1500 of the ninth embodiment of the present disclosure may include a rotation shaft 1541, a rotation shaft 1542, a rotation shaft 1543, and a rotation shaft 1544. Here, the rotation shaft 1541 and the rotation shaft 1542 may be inserted through an end tool hub 1580, and the rotation shaft 1543 and the rotation shaft 1544 may be inserted through a pitch hub 1507. The rotation shaft 1541, the rotation shaft 1542, the rotation shaft 1543, and the rotation shaft 1544 may be arranged sequentially from a distal end 1504 of the end tool 1500 toward a proximal end 1505.

[1009] In addition, the end tool 1500 of the ninth embodiment of the present disclosure may include the end tool hub 1580 and the pitch hub 1507.

[1010] The rotation shaft 1541 and the rotation shaft 1542 may be inserted through the end tool hub 1580, and a pulley 1511 and the pulley 1521 axially coupled to the rotation shaft 1541 and at least some of the first jaw 1501 and the second jaw 1502 coupled to the pulley 1511 and the pulley 1521 may be accommodated inside the end tool hub 1580.

[1011] The rotation shaft 1543 and the rotation shaft 1544 may be inserted through the pitch hub 1507, and the pitch hub 1507 may be axially coupled to the end tool hub 1580 by the rotation shaft 1543. Accordingly, the end tool hub 1580 may be formed to be pitch-rotatable around the rotation shaft 1543 with respect to the pitch hub 1507.

[1012] Meanwhile, the end tool 1500 of the ninth embodiment of the present disclosure may further include components such as a staple drive assembly (see 150 of FIG. 13) including a staple pulley assembly (see 160 of FIG. 13) and a staple link assembly (see 170 of FIG. 13) to perform stapling and cutting motions.

[1013] The staple pulley assembly (see 160 of FIG. 13) may be formed between the pulley 1511 and the pulley 1521 to be adjacent to the pulley 1511 and the pulley 1521. In the present embodiment, it is assumed that the staple pulley assembly (see 160 of FIG. 13) includes one staple pulley 1561.

[1014] The staple link assembly (see 170 of FIG. 13) may include one or more link members 1571.

The staple link assembly (see **170** of FIG. **13**) may serve to connect a staple pulley assembly **1560** and a reciprocating assembly **1550** of the cartridge **1510**. In the present embodiment, it is assumed that the staple link assembly (see **170** of FIG. **13**) includes one link member **1571**, and the link member **1571** includes two links.

[1015] As described above, in the ninth embodiment of the present disclosure, by disposing the staple pulley assembly (see **160** of FIG. **13**) and the staple link assembly (see **170** of FIG. **13**) between the pulley **1511**, which is a first jaw pulley, and the pulley **1521**, which is a second jaw pulley, the end tool **1500** is allowed to perform pitch and yaw motions as well as stapling and cutting motions using the cartridge **1510**. In the present embodiment, components for performing the stapling and cutting motions are substantially the same as those described in the first embodiment, and thus detailed descriptions thereof will be omitted herein.

[1016] Hereinafter, the second jaw **1502** and the cartridge **1510** of the ninth embodiment of the present disclosure will be described in more detail, and in particular, the clamp **1546** formed in the operation member **1540** of the cartridge **1510** will be mainly described.

[1017] The second jaw **1502** includes an anvil **1502a**, a guide groove **1502b**, a first coupling groove **1502h**, and a second coupling groove **1502i**.

[1018] The second jaw **1502** is formed entirely in the shape of an elongated bar, the anvil **1502a** is formed in a distal end **1502f** side, and a pulley **1512** is coupled to a proximal end **1502g**, so that the second jaw **1502** is formed to be rotatable around the rotation shaft **1541**.

[1019] In detail, the anvil **1502a** is formed in the form of a flat plane, on one surface of which shapes corresponding to the shapes of staples **1530** to be described later may be formed. The above-described anvil **1502a** may serve as a support for supporting the staple **1530** on the opposite side of the operation member **1540** when the operation member **1540** pushes and raises the staple **1530** during a stapling motion, so that the staple **1530** is bent.

[1020] The guide groove **1502b** for guiding the movement of the operation member **1540** of the cartridge **1510** may be formed inside the second jaw **1502**. The guide groove **1502b** may be formed in the shape of a groove formed along a moving path of the operation member **1540**. In addition, the clamp **1546** is moved along the guide groove **1502b** in a state in which the clamp **1546** of the operation member **1540** formed in a protruding shape is fitted into the groove-shaped guide groove **1502b**, so that the operation member **1540** is moved with respect to the second jaw **1502**. That is, the operation member **1540** may be moved along the guide groove **1502b** of the second jaw **1502**.

[1021] The first coupling groove **1502h** may be formed in the proximal end **1502g** of the second jaw **1502**, and the second coupling groove **1502i** may be formed in the distal end **1502f**. Here, the first coupling groove **1502h** and the second coupling groove **1502i** may be formed at both end portions of the guide groove **1502b**. Each of the first coupling groove **1502h** and the second coupling groove **1502i** is formed to be slightly larger than the clamp **1546** of the operation member **1540** so that the clamp **1546** may be inserted into or withdrawn from the second jaw **1502**. That is, the clamp **1546** may be inserted into the second jaw **1502** through the first coupling groove **1502h**, moved along the guide groove **1502b**, and then withdrawn to the outside of the second jaw **1502** through the second coupling groove **1502i**. This will be described in more detail below.

[1022] The cartridge **1510** may include a cover (see **510** of FIG. **22**), a housing (see **520** of FIG. **22**), the staple **1530**, the operation member **1540**, and the reciprocating assembly **1550**. Here, the other components of the cartridge **1510** except for the operation member **1540** are substantially the same as those of the cartridge (see **500** of FIG. **22**) described in the first embodiment, and thus detailed descriptions thereof will be omitted.

[1023] The operation member **1540** may include a wedge **1541**, a blade **1542**, a ratchet member **1543**, an elastic member **1544**, and a body **1545**. Furthermore, the operation member **1540** of the ninth embodiment of the present disclosure further includes the clamp **1546**.

[1024] The clamp **1546** may be formed on one side of the blade **1542** and may be formed in a shape that is approximately parallel to the body **1545** or the wedge **1541**. In addition, a protrusion

1546a may be formed at one end portion of the clamp **1546**, and the protrusion **1546a** may be moved along the guide groove **1502b**.

[1025] Here, the operation member **1540** according to the ninth embodiment of the present disclosure is formed as a whole in a “C” shape by further including the clamp **1546** parallel to the wedge **1541**, and thus may better withstand a strong pressure applied to the wedge **1541** during a stapling motion.

[1026] FIGS. **176** to **179** are views illustrating a process in which the operation member is moved.

[1027] As described above, in a state in which the first and second jaws **1501** and **1502** are closed, as shown in FIG. **176**, the clamp **1546** may pass through the first coupling groove **1502h** and be located in the guide groove **1502b**.

[1028] In this state, as shown in FIG. **177**, when the operation member **1546** is moved in the direction of an arrow A, the clamp **1546** may also be moved along the guide groove **1502b**.

[1029] In this state, as shown in FIG. **178**, when the operation member **1546** continues to move in the direction of an arrow B and reaches the distal end **1502f** of the second jaw **1502**, the clamp **1546** may pass through the second coupling groove **1502i** and be withdrawn to the outside of the guide groove **1502b**.

[1030] FIGS. **180** to **182** are views illustrating a process in which the operation member is coupled to the second jaw.

[1031] In the state as shown in FIG. **180**, the second jaw **1502** is rotated in the direction of an arrow A on FIG. **181** and the direction of an arrow B of FIG. **182**, so that the first and second jaws **1501** and **1502** are closed. Then, the clamp **1546** passes through the first coupling groove **1502h** and is located in the guide groove **1502b**.

[1032] FIGS. **183** to **185** are views illustrating a process in which the operation member is separated from the second jaw.

[1033] In the state as shown in FIG. **183**, the second jaw **1502** is rotated in the direction of an arrow A of FIG. **184** and the direction of an arrow B of FIG. **185**, so that the first and second jaws **1501** and **1502** are open. Then, the clamp **1546** passes through the second coupling groove **1502i** and comes out of the guide groove **1502b**.

[1034] As described above, the operation member **1540** according to the ninth embodiment of the present disclosure may better withstand a strong pressure applied to the wedge **1541** during a stapling motion by further including the clamp **1546** parallel to the wedge **1541**.

First Modified Example of First Embodiment—Ratchet Modification: Hinge Type 1

[1035] Hereinafter, a surgical instrument according to a first modified example of the first embodiment of the present disclosure will be described. Here, the surgical instrument according to the first modified example of the first embodiment of the present disclosure is different from the surgical instrument (see **10** of FIG. **2** or the like) according to the first embodiment of the present disclosure in that the shape of an elastic member **564** of the cartridge is different.

[1036] FIGS. **76** and **77** are views illustrating the cartridge of the surgical instrument according to the first modified example of the first embodiment of the present disclosure.

[1037] Referring to FIGS. **76** and **77**, a support **566** may be formed to protrude from one surface of the body (see **545** of FIG. **27**), for example, from a lower surface of the body (see **545** of FIG. **27**). In addition, the elastic member **564** may be formed such that one end portion is coupled to the support **566** of the body (see **545** of FIG. **27**) and the other end portion of the elastic member **564** is in contact with a ratchet member **563**.

[1038] At this time, the elastic member **564** may apply an elastic force in a direction in which the ratchet member **563** comes into close contact with the reciprocating member (see **551** of FIG. **22**). To this end, the elastic member **564** may be formed in the form of a leaf spring, and may be provided in various forms capable of providing a predetermined elastic force to the ratchet member **563**, such as a coil spring, a dish spring, and the like.

[1039] Here, in the first modified example of the first embodiment of the present disclosure, the

elastic member **564** is not integrally formed with an operation member **560** but is formed as a separate member and is disposed between the support **566** of the body (see **545** of FIG. **27**) and the ratchet member **563**.

[1040] In detail, in the first embodiment of the present disclosure, the elastic member (see **544** of FIG. **28**) is integrally formed with the operation member (see **540** of FIG. **28**), whereas in the present modified example, the elastic member **564** is formed as a separate member from the operation member **560** and is interposed between the support **566** and the ratchet member **563** of the body (see **545** of FIG. **27**) of the operation member **560**.

Second Modified Example of First Embodiment—Ratchet Modification: Hinge Type 2

[1041] Hereinafter, a surgical instrument according to a second modified example of the first embodiment of the present disclosure will be described. Here, the surgical instrument according to the second modified example of the first embodiment of the present disclosure is different from the surgical instrument (see **10** of FIG. **2** or the like) according to the first embodiment of the present disclosure in that the shape of an elastic member **574** of the cartridge is different.

[1042] FIGS. **78** and **79** are views illustrating the cartridge of the surgical instrument according to the second modified example of the first embodiment of the present disclosure.

[1043] Referring to FIGS. **78** and **79**, a support **576** may be formed to protrude from one surface of a body (see **545** of FIG. **27**), for example, from a lower surface of the body (see **545** of FIG. **27**). In addition, the elastic member **574** may be formed such that one end portion is in contact with the support **576** of the body (see **545** of FIG. **27**) and the other end portion of the elastic member **574** is coupled to a ratchet member **573**.

[1044] At this time, the elastic member **574** may apply an elastic force in a direction in which the ratchet member **573** comes into close contact with the reciprocating member (see **551** of FIG. **22**). To this end, the elastic member **574** may be formed in the form of a leaf spring, and may be provided in various forms capable of providing a predetermined elastic force to the ratchet member **573**, such as a coil spring, a dish spring, and the like.

[1045] Here, in the second modified example of the first embodiment of the present disclosure, the elastic member **574** is not integrally formed with an operation member **570** but is formed as a separate member, and is disposed between the support **576** of the body (see **545** of FIG. **27**) and the ratchet member **573**.

[1046] In detail, in the first embodiment of the present disclosure, the elastic member (see **544** of FIG. **28**) is integrally formed with the operation member (see **540** of FIG. **28**), whereas in the present modified example, the elastic member **574** is formed as a separate member from the operation member **570** and is interposed between the support **576** and the ratchet member **573** of the body (see **545** of FIG. **27**) of the operation member **570**.

Third Modified Example of First Embodiment—Ratchet Modification: Hinge Type 3

[1047] Hereinafter, a surgical instrument according to a third modified example of the first embodiment of the present disclosure will be described. Here, the surgical instrument according to the third modified example of the first embodiment of the present disclosure is different from the surgical instrument (see **10** of FIG. **2** or the like) according to the first embodiment of the present disclosure in that the shape of an elastic member **584** of the cartridge is different.

[1048] FIGS. **80** and **81** are views illustrating the cartridge of the surgical instrument according to the third modified example of the first embodiment of the present disclosure.

[1049] Referring to FIGS. **80** and **81**, the elastic member **584** may be formed such that one end portion is in contact with an inner wall of the housing **520** and the other end portion of the elastic member **584** is coupled to a ratchet member **583**.

[1050] Here, the elastic member **584** may apply an elastic force in a direction in which the ratchet member **583** comes into close contact with the reciprocating member **551**. To this end, the elastic member **584** may be formed in the form of a leaf spring, and may be provided in various forms capable of providing a predetermined elastic force to the ratchet member **583**, such as a coil spring,

a dish spring, and the like.

[1051] Here, in the third modified example of the first embodiment of the present disclosure, the elastic member **584** is not integrally formed with an operation member **580** but is formed as a separate member, and is disposed between the housing **520** and the ratchet member **583**.

[1052] In detail, in the first embodiment of the present disclosure, the elastic member (see **544** of FIG. **28**) is integrally formed with the operation member (see **540** of FIG. **28**), whereas in the present modified example, the elastic member **584** is formed as a separate member from the operation member **580** and is interposed between the housing **520** and the ratchet member **583**.

Fourth Modified Example of First Embodiment—Ratchet Modification: Vertical Type

[1053] Hereinafter, a surgical instrument according to a fourth modified example of the first embodiment of the present disclosure will be described. Here, the surgical instrument according to the fourth modified example of the first embodiment of the present disclosure is different from the surgical instrument (see **10** of FIG. **2** or the like) according to the first embodiment of the present disclosure in that the shape of an elastic member **594** of the cartridge is different.

[1054] FIGS. **82** and **83** are views illustrating the cartridge of the surgical instrument according to the fourth modified example of the first embodiment of the present disclosure.

[1055] Referring to FIGS. **82** and **83**, a support **596** may be formed to protrude from one surface of the body (see **545** of FIG. **27**), for example, from a lower surface of the body (see **545** of FIG. **27**). Here, the entire support **596** may be formed in a roughly “C” shape, in which the elastic member **594** may be accommodated. Accordingly, the elastic member **594** may be formed such that both end portions are in contact with the support **596** and a central portion is in contact with a ratchet member **593**.

[1056] At this time, the elastic member **594** may apply an elastic force in a direction in which the ratchet member **593** comes into close contact with the reciprocating member (see **551** of FIG. **22**). To this end, the elastic member **594** may be formed in the form of a leaf spring, and may be provided in various forms capable of providing a predetermined elastic force to the ratchet member **593**, such as a coil spring, a dish spring, and the like.

[1057] Here, in the fourth modified example of the first embodiment of the present disclosure, the elastic member **594** is not integrally formed with an operation member **590** but is formed as a separate member, and is disposed between the support **596** of the body (see **545** of FIG. **27**) and the ratchet member **593**.

[1058] In detail, in the first embodiment of the present disclosure, the elastic member (see **544** of FIG. **28**) is integrally formed with the operation member (see **540** of FIG. **28**), whereas in the present modified example, the elastic member **594** is formed as a separate member from the operation member **590** and is interposed between the support **596** and the ratchet member **593** of the body (see **545** of FIG. **27**) of the operation member **590**.

Fifth Modified Example of First Embodiment—Moving-Backward Prevention Snap

[1059] Hereinafter, a surgical instrument according to a fifth modified example of the first embodiment of the present disclosure will be described. Here, the surgical instrument according to the fifth modified example of the first embodiment of the present disclosure is different from the surgical instrument (see **10** of FIG. **2** or the like) according to the first embodiment of the present disclosure in that a snap **646** for preventing a backward movement is further formed in a cartridge **600**.

[1060] FIGS. **84** and **85** are views illustrating the cartridge of the surgical instrument according to the fifth modified example of the first embodiment of the present disclosure. FIG. **86** is a view illustrating operating states of the cartridge of FIG. **84**.

[1061] Referring to FIGS. **84** to **86**, the cartridge **600** is formed to be mountable to and dismountable from the first jaw (see **101** of FIG. **2**), and includes a plurality of staples (see **530** of FIG. **25**) and a blade (see **642** of FIG. **27**) therein to perform suturing and cutting tissue. Here, the cartridge **600** may include a cover **610**, a housing **620**, the staples (see **530** of FIG. **25**), a

withdrawal member (see 535 of FIG. 25), an operation member 640, and a reciprocating assembly 650.

[1062] The housing 620 forms an outer shape of the cartridge 600, and may be formed entirely in the form of a hollow box with one surface (upper surface) removed to accommodate the reciprocating assembly 650, the operation member 640, and the staples (see 530 of FIG. 25) therein. Here, the housing 620 may be formed in an approximately “U” shape in cross section.

[1063] One or more protrusions 621 may be formed on an inner surface of the housing 620, more specifically, on the inner surface of the housing 620 facing the recess (see 551b of FIG. 22) of the reciprocating assembly 650. In other words, it may be said that one or more protrusions 621 are formed in a region of the inner surface of the housing 620, which is to be in contact with the operation member 640. The one or more protrusions 621 may be formed in a wedge shape. In other words, it may also be described that the one or more protrusions 621 include a slope formed to have a greater height at the distal end side of the cartridge 600 than the proximal end side. The protrusions 621 may be formed to be in contact with the snap 646 of the operation member 640 to prevent a backward movement of the operation member 640 (that is, the movement toward the proximal end). This will be described in more detail later.

[1064] The cover 610 is formed to cover an upper portion of the housing 620. Staple holes (see 511 of FIG. 23) through which the plurality of staples 630 may be ejected to the outside may be formed in the cover 610. Stapling is performed as the staples (see 530 of FIG. 25), which are accommodated in the housing 620 before a stapling operation, are pushed and raised upward by the operation member 640 during the stapling operation, and pass through the staple holes (see 511 of FIG. 23) of the cover 610 to be withdrawn to the outside of the cartridge 600.

[1065] The reciprocating assembly 650 may be disposed at an inner lower side of the housing 620. The reciprocating assembly 650 may include one or more reciprocating members 651. In this embodiment, the reciprocating member 651 may be a rack.

[1066] The operation member 640 may include a wedge 641, a blade 642, a ratchet member 643, the elastic member 644, the body 645, and the snap 646.

[1067] The snap 646 may be formed to protrude from any one side of the body 645 or the wedge 641. The snap 646 is formed in the shape of a wedge, and is formed to be in contact with the inner surface of the housing 620. Here, the snap 646 may be formed to engage with the protrusion 621 of the housing 620. When the snap 646 is engaged with the protrusion 621, the operation member 640 may be prevented from moving backward (i.e., moving toward the proximal end) due to the snap 646 and the protrusion 621.

[1068] Here, the snap 646 may be formed to be elastically deformable to a certain extent. That is, the snap 646 may serve as a kind of leaf spring. Accordingly, when the snap 646 is moved along an inclined surface of the protrusion 621, the snap 646 may be elastically deformed to a certain extent toward the body 645. In addition, when the snap 646 is brought into contact with the next protrusion 621 beyond the inclined surface of the protrusion 621, the snap 646 may be elastically restored toward the housing 620.

[1069] In summary, the cartridge 600 is accommodated in the cartridge accommodation part (see 101a of FIG. 2) of the first jaw (see 101 of FIG. 2), and in this case, the reciprocating member 651 of the cartridge 600 is coupled to the staple link assembly (see 170 of FIG. 13) of the end tool (see 100 of FIG. 2). Accordingly, a rotational motion of the staple pulley (see 161 of FIG. 13) of the end tool (see 100 of FIG. 2) is converted into a linear motion of the reciprocating member 651 through the staple link assembly (see 170 of FIG. 13).

[1070] In this case, when a coupling part 651a of the reciprocating member 651 is connected to the staple pulley 161 through the staple link assembly 170, and the staple pulley 161 is rotated alternately in the clockwise/counterclockwise directions, the reciprocating member 651 may be repeatedly moved forward and backward. In addition, when the reciprocating member 651 is moved forward, the operation member 640 is moved forward together with the reciprocating

member **651**, and when the reciprocating member **651** is moved backward, only the reciprocating member **651** may be moved backward and the operation member **640** may remain stationary in place. As the operation member **640** is moved forward while repeating this process, the staple **630** may be stapled by the wedge **641** while the blade **642** cuts the stapled tissue.

[1071] Here, when the reciprocating member **651** is moved backward, only the reciprocating member **651** may be moved backward and the operation member **640** may more securely remain stationary in place. That is, the operation member **640** may be more securely prevented from moving toward the proximal end (see **501** of FIG. **23**) of the cartridge **600** by the snap **646** and the protrusion **621**.

[1072] This will be described in more detail as follows.

[1073] First, in the state shown in FIG. **86A**, when the staple pulley (see **161** of FIG. **13**) is rotated in one direction as shown in FIG. **86B**, the staple link assembly (see **170** of FIG. **13**) connected to the staple pulley (see **161** of FIG. **13**) and the reciprocating member **651** coupled to the staple link assembly (see **170** of FIG. **13**) are moved in the direction of an arrow **b1** (i.e., toward the distal end of the cartridge). In this state, since the reciprocating member **651** and the operation member **640** are in close contact with each other by the elastic member **644**, when the reciprocating member **651** is moved in the direction of the arrow **b1**, the operation member **640** is also moved in the direction of the arrow **b1** together with the reciprocating member **651**. In addition, at this time, the snap **646** may be elastically deformed to a certain extent in the direction of an arrow **b2** (i.e., toward the body **645**) while moving along the inclined surface of the protrusion **621**.

[1074] Next, as shown in FIG. **86C**, when the staple pulley (see **161** of FIG. **13**) is further rotated in the one direction, the staple link assembly (see **170** of FIG. **13**) connected thereto, the reciprocating member **651**, and the operation member **640** are further moved in the direction of an arrow **c1**. In addition, at this time, when the snap **646** is brought into contact with the next protrusion **621** beyond the inclined surface of the protrusion **621**, the snap **646** may be elastically restored toward the housing **620**.

[1075] Next, in the state shown in FIG. **86C**, when the staple pulley (see **161** of FIG. **13**) is rotated in the opposite direction as shown in FIG. **86D**, the staple link assembly (see **170** of FIG. **13**) connected to the staple pulley (see **161** of FIG. **13**) and the reciprocating member **651** coupled to the staple link assembly (see **170** of FIG. **13**) are moved in the direction of an arrow **d1** (i.e., toward proximal end of the cartridge). In this state, due to the coupling structure of the ratchet member **643** and the reciprocating member **651**, even when the reciprocating member **651** is moved in the direction of the arrow **d1**, the ratchet member **643** is repeatedly spaced apart from the reciprocating member **651** to a certain extent as the elastic member **644** is repeatedly elastically deformed in the direction of an arrow **d2**, in a state in which the overall position (i.e., position in the X-axis direction) of the operation member **640** remains unchanged. That is, even when the reciprocating member **651** is moved in the direction of the arrow **d1**, the operation member **640** remains stationary when viewed in the X-axis direction.

[1076] At this time, since the snap **646** is engaged with the protrusion **621** of the housing **620**, the movement of the operation member in the X-axis direction is more securely prevented.

[1077] Next, as shown in FIG. **86E**, when the staple pulley (see **161** of FIG. **13**) is further rotated in the opposite direction, only the staple link assembly (see **170** of FIG. **13**) connected thereto, and the reciprocating member **651** are further moved in the direction of an arrow **e1**, and the operation member **640** remains stationary in place when viewed in the X-axis direction.

[1078] When the staple pulley (see **161** of FIG. **13**) is alternately rotated in the clockwise/counterclockwise directions while repeating the above process, the reciprocating member **651** is repeatedly moved forward and backward, and the operation member **640** repeats moving forward and stopping, and as a result, the operation member **640** is moved toward the distal end (see **502** of FIG. **23**). In addition, as the operation member **640** is moved toward the distal end (see **502** of FIG. **23**), a stapling motion by the wedge **641** and a cutting motion by the blade **642** are

simultaneously performed.

[1079] As such, the present disclosure has been described with reference to an embodiment shown in the drawings, but it will be understood that this is merely exemplary, and those of ordinary skill in the art will understand that various modifications and variations of the embodiments are possible therefrom. Accordingly, the true technical protection scope of the present disclosure should be defined by the technical spirit of the appended claims.

INDUSTRIAL USABILITY

[1080] The present disclosure relates to an end tool of a surgical instrument and a surgical instrument including the same, and more particularly, may be used to an end tool of a surgical instrument that may be mounted on a robotic arm or operable manually to be used in laparoscopic surgery or other various surgeries, wherein the end tool is rotatable in two or more directions and is moved in a way that intuitively matches a motion of a manipulation part, and a surgical instrument including the same.

Claims

1. An end tool of a surgical instrument, the end tool comprising: a first jaw capable of accommodating a cartridge; a second jaw formed to face the first jaw; a first jaw pulley coupled to the first jaw and formed to be rotatable around a first shaft; a second jaw pulley coupled to the second jaw, formed to be rotatable around a shaft substantially the same as or parallel to the first shaft, and formed to be spaced apart from the first jaw pulley by a certain extent; a staple drive assembly including one or more staple pulleys at least partially formed between the first jaw pulley and the second jaw pulley; and a staple wire at least partially in contact with the staple pulley and configured to transmit, to the staple pulley, a driving force necessary for rotating the staple pulley, wherein the staple drive assembly is connected to a reciprocating assembly of the cartridge and configured to convert a rotational motion of the staple pulley into a linear motion of the reciprocating assembly.
2. The end tool of claim 1, further comprising an end tool hub including a first jaw pulley coupling part and a second jaw pulley coupling part, which are formed to face each other, and a guide part configured to connect the first jaw pulley coupling part and the second jaw pulley coupling part, wherein the first jaw pulley is disposed adjacent to the first jaw pulley coupling part of the end tool hub, the second jaw pulley is disposed adjacent to the second jaw pulley coupling part of the end tool hub, and at least a portion of the staple drive assembly is formed between the first jaw pulley and the second jaw pulley.
3. The end tool of claim 2, wherein the first shaft is sequentially inserted through the first jaw pulley coupling part, the first jaw pulley, the staple pulley, the second jaw pulley, and the second jaw pulley coupling part.
4. The end tool of claim 2, wherein the first jaw pulley, the staple pulley, and the second jaw pulley are sequentially stacked in the end tool hub.
5. The end tool of claim 2, wherein the first jaw pulley, the staple pulley, and the second jaw pulley are formed to be rotatable independently of each other.
6. The end tool of claim 2, further comprising a staple auxiliary pulley disposed between the staple pulley and the guide part.
7. The end tool of claim 6, wherein the staple wire is located on a common internal tangent of the staple pulley and the staple auxiliary pulley, and a rotation angle of the staple pulley is increased by the staple auxiliary pulley.
8. The end tool of claim 2, wherein a region of the guide part adjacent to the first jaw pulley, the staple pulley, and the second jaw pulley is formed such that a cross section thereof is curved with a predetermined curvature.
9. The end tool of claim 8, wherein the staple wire is located on a common internal tangent of the

staple pulley and the guide part, and a rotation angle of the staple pulley is increased by the guide part.

10. The end tool of claim 1, wherein the staple drive assembly includes a staple link assembly configured to connect the staple pulley and the reciprocating assembly.

11. The end tool of claim 10, wherein the staple link assembly includes a first link coupled to the staple pulley, and a second link coupled to each of the first link and the reciprocating assembly.

12. The end tool of claim 10, wherein when the staple pulley is alternately rotated in a clockwise direction and a counterclockwise direction, the staple link assembly connected to the staple pulley is alternately moved toward a distal end and a proximal end of the end tool.

13. The end tool of claim 10, wherein a bidirectional rotational motion of the staple pulley is converted into a reciprocating linear motion of the reciprocating assembly, which is connected to the staple link assembly, by the staple link assembly.

14. The end tool of claim 10, wherein a guide groove is formed in the first jaw in a length direction of the first jaw, and the staple link assembly is moved along the guide groove.

15. The end tool of claim 1, further comprising a jaw rotation shaft inserted through the first jaw and the second jaw and being the center of rotation of the first jaw and the second jaw, wherein the first shaft is a jaw pulley rotation shaft, which is inserted through the first jaw pulley and the second jaw pulley and being the center of rotation of the first jaw pulley and the second jaw pulley, and the jaw rotation shaft is moved relative to the jaw pulley rotation shaft when the first jaw pulley and the second jaw pulley are rotated around the jaw pulley rotation shaft.

16. The end tool of claim 15, wherein when the first jaw and the second jaw are closed, the jaw rotation shaft is moved toward a distal end of the end tool, and when the first jaw and the second jaw are opened, the jaw rotation shaft is moved toward a proximal end of the end tool.

17. The end tool of claim 1, further comprising: a pair of end tool first jaw pitch main pulleys formed on one side of the first jaw pulley, and formed to be rotatable around a second shaft forming a predetermined angle with the first shaft; and a pair of end tool second jaw pitch main pulleys formed on one side of the second jaw pulley, and formed to be rotatable around a shaft that is substantially the same as or parallel to the second shaft.

18. The end tool of claim 17, wherein the end tool is formed to be yaw-rotatable around the first shaft and simultaneously pitch-rotatable around the second shaft.

19. The end tool of claim 17, further comprising: a first jaw wire of which at least a portion is wound around the first jaw pulley and the pair of end tool first jaw pitch main pulleys; and a second jaw wire of which at least a portion is wound around the second jaw pulley and the pair of end tool second jaw pitch main pulleys.
